

Statistical Inference for Linear Stochastic Approximation with Richardson-Romberg Extrapolation

Abstract

This thesis studies statistical inference for constant-stepsize linear stochastic approximation with Markovian noise. The main object is the Polyak–Ruppert averaged Richardson–Romberg estimator formed from two coupled step sizes, α and 2α . The stationary theorem is for an assembled augmented-chain comparison statistic; statements for the deterministic-start estimator are obtained only after the burn-in transfer. We prove a stationary augmented-chain Berry–Esseen assembly for scalar RR statistics, identify the asymptotic covariance target Σ_∞ , and then transfer the result to deterministic starts under mixing-scale burn-in conditions with logarithmic factors. At the balanced scale $\alpha = cn^{-1/2}$, the final burned-in statistic has a non-asymptotic normal approximation with $n^{-1/4}$ polynomial rate up to logarithmic factors.

1. Introduction

Stochastic approximation (SA) algorithms are a cornerstone of modern computational statistics, optimization, and reinforcement learning. Introduced by Robbins and Monro (1951), these iterative procedures provide a principled way to find roots of equations or optimize objectives when only noisy observations are available. A particularly important subclass is the *linear stochastic approximation* (LSA) algorithm, which arises naturally in temporal-difference (TD) learning, policy evaluation, and stochastic gradient descent for linear models.

In this work, we study the LSA recursion with a *constant step size* $\alpha > 0$:

$$\theta_k^{(\alpha)} = \theta_{k-1}^{(\alpha)} - \alpha \{A(Z_k)\theta_{k-1}^{(\alpha)} - b(Z_k)\}, \quad k \geq 1, \quad (1)$$

where $\{Z_k\}_{k \in \mathbb{N}}$ is a time-homogeneous Markov chain on a measurable space $(Z, \mathcal{Z})$ with transition kernel Q and unique invariant distribution π . The mappings $A : Z \rightarrow \mathbb{R}^{d \times d}$ and $b : Z \rightarrow \mathbb{R}^d$ are measurable functions satisfying $\bar{A} := \int_Z A(z) d\pi(z)$ and $\bar{b} := \int_Z b(z) d\pi(z)$. We use the stability convention that $-\bar{A}$ is Hurwitz, equivalently all eigenvalues of $\bar{A}$ have strictly positive real parts. Then the target parameter $\theta^* = \bar{A}^{-1}\bar{b}$ is uniquely defined.

The *Polyak–Ruppert averaging* procedure (Polyak, 1990; Ruppert, 1988) provides an effective variance reduction technique. Given a burn-in period $n_0 \geq 0$, the averaged iterate is defined as

$$\bar{\theta}_n^{(\alpha)} = \frac{1}{n - n_0} \sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}. \quad (2)$$

1.1. Bias of constant step-size iterates

The use of a constant step size $\alpha > 0$ offers several practical advantages: it enables geometrically fast forgetting of the initial condition (Dieuleveut, Durmus, and Bach, 2020) and simplifies hyperparameter tuning compared to diminishing step-size schedules. However, unlike the classical regime $\alpha_k \rightarrow 0$ with $\sum \alpha_k = \infty$ and $\sum \alpha_k^2 < \infty$, a constant step size produces iterates that converge only *in distribution* to a stationary

measure Π_α , rather than almost surely to θ^* . The stationary expectation $\mathbb{E}[\theta_\infty^{(\alpha)}]$ is generally *biased* with respect to θ^* , and this bias cannot be eliminated by Polyak–Ruppert averaging alone.

As shown in Levin, Naumov, and Samsonov (2025), the stationary bias has a leading linear term in α :

$$\lim_{n \rightarrow \infty} \mathbb{E}[\theta_n^{(\alpha)}] = \theta^* + \alpha\Delta + O(\alpha^{3/2}), \quad (3)$$

where $\Delta = \bar{A}^{-1} \sum_{k=1}^{\infty} \mathbb{E}[\{Q^k \tilde{A}(Z_\infty)\} \epsilon(Z_\infty)]$ depends on the correlation structure of the Markov chain, and $\tilde{A}(z) = A(z) - \bar{A}$ is the centered matrix-valued function. Under stronger expansion assumptions, the power-series approach of Huo, Chen, and Xie (2024) gives higher-order bias expansions in integer powers of α ; in the Levin decomposition, the first misadjustment bias component itself has an $O(\alpha^2)$ remainder after the leading $\alpha\Delta$ term.

1.2. Richardson–Romberg extrapolation

To eliminate the leading $O(\alpha)$ bias term, we employ the *Richardson–Romberg* (RR) *extrapolation* procedure. Two LSA sequences are run *on the same Markov chain trajectory* $\{Z_k\}$ with step sizes α and 2α , and the RR iterate is formed as

$$\bar{\theta}_n^{(\alpha, \text{RR})} = 2\bar{\theta}_n^{(\alpha)} - \bar{\theta}_n^{(2\alpha)}. \quad (4)$$

Since both sequences share the same noise realization, the leading bias term $\alpha\Delta$ cancels, leaving a residual bias of order $O(\alpha^{3/2})$ or higher (Levin et al., 2025).

More generally, one can consider the multi-level extrapolation with M step sizes $\mathcal{A} = \{\alpha_1, \dots, \alpha_M\}$ and coefficients $\{h_m\}$ determined by the Vandermonde system (Huo et al., 2024):

$$\sum_{m=1}^M h_m = 1, \quad \sum_{m=1}^M h_m \alpha_m^l = 0, \quad l = 1, \dots, M-1, \quad (5)$$

which cancels successive powers in settings where such a power-series expansion is available.

1.3. Problem statement and goals

The high-order moment bounds for the PR-averaged RR iterate $\bar{\theta}_n^{(\alpha, \text{RR})}$ established in Levin et al. (2025) show that the leading error term scales as $\sqrt{\text{Tr } \Sigma_\epsilon^{(M)}} \cdot n^{-1/2}$, where $\Sigma_\epsilon^{(M)}$ is the Markovian noise covariance defined below in the key quantities subsection. This is the usual parametric $n^{-1/2}$ benchmark for averaged LSA; this thesis does not prove a separate Hájek–Le Cam or minimax lower bound. Berry–Esseen type bounds and bootstrap inference procedures for the *standard* Polyak–Ruppert average $\bar{\theta}_n$ (without extrapolation) under Markovian noise have been obtained in Samsonov, Sheshukova, Moulines, and Naumov (2025).

These results do not directly give the distributional approximation needed for the PR-averaged Richardson–Romberg statistic under Markovian noise. Two distinctions are essential here. First, the stationary theorem is proved for an augmented-chain scalar statistic, where the martingale noise, Poisson boundary term, and Richardson–Romberg misadjustment are controlled separately. Second, the deterministic-start estimator requires an additional burn-in transfer, because the deterministic transient and startup remainders are absent from the stationary augmented-chain theorem.

The main goal of this work is therefore to build a non-asymptotic distributional approximation for scalar projections of the PR-averaged Richardson–Romberg statistic. The stationary result should be read as an $n_0 = 0$ augmented-chain theorem, not as a fixed- α central limit theorem centered exactly at θ^* . Its final deterministic-start consequence is obtained at the balanced triangular-array scale $\alpha_n = cn^{-1/2}$, where the residual RR terms are absorbed into explicit remainders and the covariance target is Σ_∞ .

Concretely, we establish:

1. A stationary full-window augmented-chain Berry–Esseen assembly for scalar RR statistics, including the martingale approximation, predictable-variance comparison, Poisson remainder, and stationary RR misadjustment.
2. A balanced triangular-array specialization, in particular for $\alpha_n = cn^{-1/2}$, which identifies Σ_∞ as the covariance target and gives the resulting stationary $n_0 = 0$ CLT interpretation.
3. A deterministic-start transfer theorem under mixing-scale burn-in conditions with logarithmic factors, yielding the corresponding balanced-scale bound for the main burned-in statistic.

1.4. Contribution map and notation guide

The proof combines existing non-asymptotic tools with RR-specific algebra and the deterministic-start transfer developed here. The main proof blocks are:

Block	Role in the proof	Status
RR weight algebra	Controls the deterministic kernels Q_l^{RR} and $Q_{l;n_0,n}^{\text{RR}}$ and their variation.	Derived in Section 2 and Section 4.
Poisson and martingale reduction	Turns the Markovian depth-zero weighted sum into a martingale plus a bounded Poisson remainder.	Adapted from the Samsonov et al. framework and reproved for RR weights.
Stationary misadjustment	Controls the non-martingale RR remainder $J^{(1)} + J^{(2)} + H^{(2)}$ under the stationary augmented chain.	Uses Levin et al. inputs; the stationary RR assembly is carried out here.
Burn-in transfer	Transfers the stationary theorem to finite starts by controlling the deterministic transient, random initial product, and startup discrepancy.	Developed in Section 5.

The theorem-level outputs should be read in the following order:

Output	Label	Object controlled
Stationary augmented-chain assembly	Theorem 3	$S_{n,\text{stat}}^{\text{RR}}(u)$ with the finite-window variance $\sigma_n^{\text{RR}}(u)$.
Stationary balanced triangular-array corollaries	Corollary 6 and Corollary 7	$S_{n,\text{stat}}^{\text{RR}}(u)$ at $\alpha_n = cn^{-1/2}$, with either finite-window or asymptotic normalization.
Deterministic-start finite-window transfer	Theorem 6	$\Xi_{n,n_0}^{\text{bRR}}(u)$ after burn-in, still normalized by $\sigma_{n,n_0}^{\text{bRR}}(u)$.
Deterministic-start balanced theorem	Theorem 7 and Corollary 13	$\Xi_{n,n_0}^{\text{asy,RR}}(u)$ and then the final $\sqrt{n}$ statistic $\Xi_{n,n_0}^{\text{n,RR}}(u)$.

The following notation is used throughout the later chapters:

Notation	Meaning
$\alpha, 2\alpha$	The two constant step sizes used by the two-level RR estimator.
n_0 and $m = n - n_0$	Burn-in length and effective averaging window.
$(Z_k)_{k \geq 0}$ and ξ	Base Markov chain and initial law $\xi = \mathcal{L}(Z_0)$. The recursion uses observations Z_k , $k \geq 1$; stationary covariance formulas may use Z_0 by stationarity.
$\mathcal{F}_k$	Natural filtration $\sigma(Z_0, \dots, Z_k)$. Local displays that begin at Z_1 use the same filtration with the harmless extra variable Z_0 .

$(Z_k^{\text{stat}})_{k \in \mathbb{Z}}$	Two-sided stationary copy of the base chain. Under the stationary augmented-chain convention the superscript is usually omitted.
$B_w = I - w\bar{A}$	Deterministic linearized contraction at step size w .
Q_l^{RR}	Full-window RR PR weight in the stationary $n_0 = 0$ chapter.
$Q_{l;n_0,n}^{\text{RR}}$ or Q_l^{bRR}	Burned-in RR PR weight for the window $k = n_0, \dots, n - 1$.
$S_{n,\text{stat}}^{\text{RR}}(u)$	Stationary augmented-chain scalar statistic.
$T_{n,n_0}^{\text{RR}}(u)$	Finite-start burned-in scalar statistic with $\sqrt{m}$ normalization.
$\Xi_{n,n_0}^{\text{bRR}}(u)$	Finite-window normalized burned-in statistic.
$\sigma(u)^2 = u^\top \Sigma_\infty u$	Asymptotic scalar variance target.
$\text{polylog}(n)$	A generic fixed power of $\log n$, possibly changing from line to line. It never hides polynomial dependence on n or any dependence on α , p , or q that is rate-relevant in the surrounding display.

1.5. Setting and assumptions

We now formalize the setting and state the assumptions that will be used throughout this work.

Let $\{Z_k\}_{k \in \mathbb{N}}$ be a Markov chain on a complete separable metric space $(Z, \mathcal{Z})$ with transition kernel Q .

Assumption 1 (Uniform geometric ergodicity). The kernel Q admits a unique invariant distribution π and is *uniformly geometrically ergodic*: there exists $t_{\text{mix}} \in \mathbb{N}^*$ such that for all $k \in \mathbb{N}^*$,

$$\Delta(Q^k) := \sup_{z, z' \in Z} \frac{1}{2} \|Q^k(z, \cdot) - Q^k(z', \cdot)\|_{\text{TV}} \leq (1/4)^{\lfloor k/t_{\text{mix}} \rfloor}. \quad (6)$$

Equivalently, there exist constants $\zeta > 0$ and $\rho \in (0, 1)$ such that $\sup_z \|Q^k(z, \cdot) - \pi\|_{\text{TV}} \leq \zeta \rho^k$ for all $k \geq 1$.

Assumption 2 (Hurwitz condition and boundedness). The matrix $-\bar{A}$ is Hurwitz; equivalently, all eigenvalues of $\bar{A}$ have strictly positive real parts. Moreover,

$$C_A := \max \left(\sup_{z \in Z} \|A(z)\|, \sup_{z \in Z} \|\tilde{A}(z)\| \right) < \infty, \quad (7)$$

where $\tilde{A}(z) := A(z) - \bar{A}$.

Assumption 3 (Noise regularity). The noise function $\epsilon(z) = \tilde{A}(z)\theta^* - \tilde{b}(z)$, where $\tilde{b}(z) = b(z) - \bar{b}$, satisfies

$$\|\epsilon\|_\infty := \sup_{z \in Z} \|\epsilon(z)\| < +\infty. \quad (8)$$

By construction, $\pi(\tilde{A}) = 0$, $\pi(\tilde{b}) = 0$, and hence $\pi(\epsilon) = 0$.

Under Assumptions 1–3, the error $\theta_k^{(\alpha)} - \theta^*$ satisfies the recursion

$$\theta_k^{(\alpha)} - \theta^* = (I - \alpha A(Z_k))(\theta_{k-1}^{(\alpha)} - \theta^*) - \alpha \epsilon(Z_k). \quad (9)$$

1.6. Key quantities

The *Markovian noise covariance matrix* captures both the marginal variance and the temporal correlations of the noise:

$$\Sigma_\epsilon^{(\text{M})} = \mathbb{E}_\pi [\epsilon(Z_0)\epsilon(Z_0)^\top] + \sum_{\ell=1}^{\infty} \left(\mathbb{E}_\pi [\epsilon(Z_0)\epsilon(Z_\ell)^\top] + \mathbb{E}_\pi [\epsilon(Z_\ell)\epsilon(Z_0)^\top] \right). \quad (10)$$

The series is absolutely convergent under Assumptions 1 and 3. Indeed, for centered bounded ϵ , $\|Q^\ell \epsilon\|_\infty \leq 2 \|\epsilon\|_\infty (1/4)^{\lfloor \ell/t_{\text{mix}} \rfloor}$, and hence

$$\|\mathbb{E}_\pi [\epsilon(Z_0)\epsilon(Z_\ell)^\top]\| \leq \|\epsilon\|_\infty \|Q^\ell \epsilon\|_\infty \leq 2 \|\epsilon\|_\infty^2 (1/4)^{\lfloor \ell/t_{\text{mix}} \rfloor}. \quad (11)$$

The same bound applies to the transposed covariance term, so the covariance series converges absolutely in operator norm. This matrix is the limiting covariance in the Markov chain CLT for the partial sums $n^{-1/2} \sum_{t=0}^{n-1} \epsilon(Z_t)$ (cf. Douc et al., 2018, Theorem 21.2.10).

The *asymptotically optimal covariance matrix* is given by

$$\Sigma_\infty = \bar{A}^{-1} \Sigma_\epsilon^{(M)} (\bar{A}^{-1})^\top. \quad (12)$$

This is the covariance target attained by the averaged linearized recursion. We call it optimal in the usual averaged-SA sense; a full Hájek–Le Cam optimality statement would require an additional local-asymptotic experiment argument, which is not part of this thesis.

The *Lyapunov equation* plays a central role in the contraction analysis. For any $P = P^\top \succ 0$, there exists a unique $Q = Q^\top \succ 0$ satisfying $\bar{A}^\top Q + Q \bar{A} = P$. Defining $a = \lambda_{\min}(P)/(2 \|Q\|)$ and $\kappa_Q = \lambda_{\max}(Q)/\lambda_{\min}(Q)$, the key contraction property holds: for all $\alpha \in [0, \alpha_\infty]$,

$$\|I - \alpha \bar{A}\|_Q^2 \leq 1 - \alpha a. \quad (13)$$

Since the iterates $\theta_k^{(\alpha)}$ alone are generally not Markovian (due to the Markovian noise), we consider the *joint process* $(\theta_k^{(\alpha)}, Z_{k+1})$ with kernel

$$\bar{P}_\alpha f(\theta, z) = \int_Z Q(z, dz') f(F_z(\theta), z'), \quad (14)$$

where $F_z(\theta) = (I - \alpha A(z))\theta + \alpha b(z)$. Thus the current second coordinate z is the observation used to update θ , and the next coordinate z' is carried forward to the following step. Under Assumptions 1–3, this joint chain admits a unique invariant distribution Π_α for sufficiently small $\alpha > 0$ (Levin et al., 2025).

2. Zeroth-Order Richardson–Romberg Difference

2.1. LSA Error Decomposition

This chapter is a preliminary deterministic-product calculation for the zero-order RR difference. The final Berry–Esseen assembly uses the full PR-weight notation and hypotheses introduced in Chapter 4; the present chapter mainly fixes notation and explains the cancellation mechanism in a simpler last-iterate setting.

We consider the recursion

$$\theta_k = \theta_{k-1} - \alpha_k (A(Z_k)\theta_{k-1} - b(Z_k)), \quad \alpha_k = \alpha = \text{const.} \quad (15)$$

Define the transition products

$$\Gamma_{m:k} = \prod_{l=m}^k (I - \alpha A(Z_l)). \quad (16)$$

Write $B_\alpha := I - \alpha \bar{A}$ and introduce the deterministic-product linearized term

$$J_k^{(0,\alpha)} = -\alpha \sum_{l=1}^k B_\alpha^{k-l} \epsilon(Z_l), \quad J_0^{(0,\alpha)} = 0. \quad (17)$$

The difference between the exact random-product expansion and this deterministic-product term is denoted by $R_k^{(\alpha)}$:

$$\theta_k^{(\alpha)} - \theta^* = J_k^{(0,\alpha)} + B_\alpha^k (\theta_0 - \theta^*) + R_k^{(\alpha)}. \quad (18)$$

This convention matches the weight decomposition in Section 4.1.

A standard PR-averaged decomposition (cf. Chapter 4 for the full derivation) yields

$$\sqrt{n}(\bar{\theta}_n^{(\alpha)} - \theta^*) = W + D_1, \quad (19)$$

where the leading martingale-like term is

$$W = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l \epsilon(Z_l), \quad Q_l = \alpha \sum_{k=l}^{n-1} B_\alpha^{k-l}, \quad (20)$$

and the residual term is

$$D_1 = \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} B_\alpha^k (\theta_0 - \theta^*) + \frac{1}{\sqrt{n}} \sum_{k=1}^{n-1} R_k^{(\alpha)}. \quad (21)$$

2.2. Last-Iterate RR Combination and the $\tilde{J}_{n, \text{last}}^{(0,\alpha)}$ Term

Applying the deterministic-product decomposition of the previous subsection separately to step sizes α and 2α , and writing $B_{2\alpha} := I - 2\alpha \bar{A}$, define the chapter-local last-iterate Richardson–Romberg object

$$\theta_{n, \text{last}}^{(\text{RR}, \alpha)} := 2\theta_n^{(\alpha)} - \theta_n^{(2\alpha)}. \quad (22)$$

It is not the PR-averaged RR estimator $\bar{\theta}_n^{(\text{RR}, \alpha)}$ studied in Chapters 4–5. Its deterministic-product decomposition has the form

$$\begin{aligned} \theta_{n, \text{last}}^{(\text{RR}, \alpha)} - \theta^* &= [2B_\alpha^n - B_{2\alpha}^n](\theta_0 - \theta^*) \\ &\quad + [2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)}] + [2R_n^{(\alpha)} - R_n^{(2\alpha)}]. \end{aligned} \quad (23)$$

The first bracket is the deterministic transient in this convention. The second bracket is the linearized stochastic RR difference, and the last one is the higher-order random-product remainder. In this subsection we focus on the zeroth-order RR difference, which we denote

$$\tilde{J}_{n, \text{last}}^{(0, \alpha)} = 2J_n^{(0, \alpha)} - J_n^{(0, 2\alpha)}, \quad (24)$$

where, by definition,

$$J_n^{(0, \alpha)} = -\alpha \sum_{j=1}^n (I - \alpha \bar{A})^{n-j} \epsilon(Z_j). \quad (25)$$

Substituting and using the elementary identity $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$ with $X = I - \alpha \bar{A}$, $Y = I - 2\alpha \bar{A}$, $X - Y = \alpha \bar{A}$, we get

$$\begin{aligned} \tilde{J}_{n, \text{last}}^{(0, \alpha)} &= -2\alpha \sum_{j=1}^n \left[(I - \alpha \bar{A})^{n-j} - (I - 2\alpha \bar{A})^{n-j} \right] \epsilon(Z_j) \\ &= -2\alpha^2 \bar{A} \sum_{j=1}^n \underbrace{\sum_{i=1}^{n-j} (I - \alpha \bar{A})^{i-1} (I - 2\alpha \bar{A})^{n-j-i}}_{=: H_j^{(n)}} \epsilon(Z_j) \\ &= -2\alpha^2 \bar{A} \sum_{j=1}^{n-1} H_j^{(n)} \epsilon(Z_j), \end{aligned} \quad (26)$$

where the last equality uses $H_n^{(n)} = 0$ because the inner sum is empty. The left-factored matrix-power identity is legitimate here because $I - \alpha \bar{A}$, $I - 2\alpha \bar{A}$, and $\bar{A}$ are polynomials in the same matrix $\bar{A}$ and therefore commute. The extra factor $\alpha \bar{A}$ pulled out front is the source of the additional α -decay of the RR difference compared to a single LSA trajectory.

2.3. Norm Estimate for $H_j^{(n)}$

To bound $\|H_j^{(n)}\|$ we use the following standard result.

Lemma 1. (Lyapunov contraction.) Let $-\bar{A}$ be a Hurwitz matrix. Then for any $P = P^\top \succ 0$ there exists a unique $Q = Q^\top \succ 0$ satisfying

$$\bar{A}^\top Q + Q \bar{A} = P. \quad (27)$$

Moreover, letting

$$a := \frac{\lambda_{\min}(P)}{2 \|Q\|}, \quad \alpha_\infty := \min \left(\frac{\lambda_{\min}(P)}{2\kappa_Q \|\bar{A}\|_Q^2}, \frac{\|Q\|}{\lambda_{\min}(P)} \right), \quad (28)$$

with $\kappa_Q := \lambda_{\max}(Q)/\lambda_{\min}(Q)$, one has for all $\alpha \in [0, \alpha_\infty]$:

$$\alpha a \leq 1/2, \quad \|I - \alpha \bar{A}\|_Q^2 \leq 1 - \alpha a. \quad (29)$$

In the RR estimates below we assume explicitly that $0 < \alpha$ and $2\alpha \leq \alpha_\infty$. Hence the Lyapunov contraction applies at both step sizes α and 2α .

For $1 \leq j \leq n-1$, we estimate $\|H_j^{(n)}\|$ by combining the triangle inequality, submultiplicativity, the equivalence $\|X\| \leq \kappa_Q^{1/2} \|X\|_Q$ (applied to **each** operator-norm factor, hence two factors of $\kappa_Q^{1/2}$ multiplying to κ_Q), and the Lyapunov contraction at step sizes α and 2α :

$$\begin{aligned}
\|H_j^{(n)}\| &\leq \sum_{i=1}^{n-j} \|I - \alpha \bar{A}\|^{i-1} \|I - 2\alpha \bar{A}\|^{n-j-i} && \text{(triangle + submult.)} \\
&\leq \kappa_Q \sum_{i=1}^{n-j} (1 - \alpha a)^{(i-1)/2} (1 - 2\alpha a)^{(n-j-i)/2} && \text{(equiv. + Lyapunov)} \\
&= \kappa_Q (1 - \alpha a)^{(n-j-1)/2} \sum_{k=0}^{n-j-1} \left(\frac{1 - 2\alpha a}{1 - \alpha a} \right)^{k/2} && \text{(reindex } k = n - j - i) \\
&\leq \kappa_Q (1 - \alpha a)^{(n-j-1)/2} \frac{1}{1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}}} && \text{(geometric series).}
\end{aligned} \tag{30}$$

For $\alpha a \leq 1/2$ the geometric rate is bounded below by an **elementary** inequality: combining $\sqrt{(1-2\alpha a)/(1-\alpha a)} \leq \sqrt{1-\alpha a}$ (since $(1-2\alpha a) \leq (1-\alpha a)^2$) with $1 - \sqrt{1-x} \geq x/2$ on $[0, 1]$ gives

$$1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}} \geq 1 - \sqrt{1-\alpha a} \geq \frac{\alpha a}{2}, \quad \text{i.e.} \quad \frac{1}{1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}}} \leq \frac{2}{\alpha a}. \tag{31}$$

Defining the chapter-local norm-equivalence constant

$$K_Q := \kappa_Q \tag{32}$$

(distinct from the assumption-2 sup-norm constant C_A , which enters separately via $\|\bar{A}\|$ in the next subsection), we arrive at the final estimate

$$\|H_j^{(n)}\| \leq K_Q (1 - \alpha a)^{(n-j-1)/2} \frac{2}{\alpha a}. \tag{33}$$

The kernel decays geometrically in $n-j$ at rate $\sqrt{1-\alpha a}$ but its summed weight is of order $1/(\alpha a)$. The next subsection keeps the remaining powers of a explicit when this kernel is multiplied by the prefactor α^2 in $\tilde{J}_{n, \text{last}}^{(0, \alpha)}$.

2.4. Scalar L^p Bound for the Zeroth-Order Term

The expression

$$\tilde{J}_{n, \text{last}}^{(0, \alpha)} = - \sum_{j=1}^{n-1} 2\alpha^2 \bar{A} H_j^{(n)} \epsilon(Z_j) \tag{34}$$

is a weighted sum of values of the centered noise function ϵ along the Markov chain. In the Berry–Esseen argument below only fixed scalar projections are used, so we state the concentration input in scalar form.

Lemma 2. (Weighted Markov concentration.) Assume **UGE 1**. Let $\{g_i\}_{i=1}^n$ be a family of measurable functions $g_i : Z \rightarrow \mathbb{R}$ such that

$$c_i = \|g_i\|_\infty < \infty \quad \text{for all } i \geq 1, \quad \pi(g_i) = 0 \quad \text{for all } i \in \{1, \dots, n\}. \tag{35}$$

Then, for any initial distribution ξ on $(Z, \mathcal{Z})$, any $n \in \mathbb{N}$, and any $p \geq 2$,

$$\left\| \sum_{i=1}^n g_i(Z_i) \right\|_{L_{p(\xi)}} \leq C_{\text{MC}, 0} \sqrt{p t_{\text{mix}} \sum_{i=1}^n c_i^2}. \tag{36}$$

Proof. This is the local scalar specialization of the imported Markov concentration statement Lemma 27. The only dependence on the coefficient sequence is through $\sqrt{p t_{\text{mix}} \sum_i c_i^2}$. Their result assumes the centering condition $\pi(g_i) = 0$ and is stated for arbitrary initial law ξ , so no additional initial-bias term is introduced here.

Fix a deterministic direction $u \in \mathbb{R}^d$. We apply the lemma with $g_j^{u(z)} = -2\alpha^2 u^\top \bar{A} H_j^{(n)} \epsilon(z)$ for $1 \leq j \leq n-1$ and $g_n^u = 0$. Each g_j^u is centered under π since $\pi(\epsilon) = 0$, so the centering hypothesis holds. Combining the bound $\|\bar{A}\| \leq C_A$ (Assumption 2) with the previous subsection, define

$$K_{A,Q} := C_A K_Q = C_A \kappa_Q. \quad (37)$$

Then the per-summand bound is

$$\begin{aligned} \|g_j^u\|_\infty &\leq 2\alpha^2 \|u\| K_{A,Q} (1-\alpha a)^{(n-j-1)/2} \frac{2}{\alpha a} \|\epsilon\|_\infty \\ &= \frac{4\alpha \|u\| K_{A,Q} \|\epsilon\|_\infty}{a} (1-\alpha a)^{(n-j-1)/2}, \quad 1 \leq j \leq n-1. \end{aligned} \quad (38)$$

Note the prefactor α^2 collapsing to α/a : the $1/(\alpha a)$ blow-up in $H_j^{(n)}$ is absorbed only by one factor of α . Squaring and summing over j :

$$\begin{aligned} \sum_{j=1}^n \|g_j^u\|_\infty^2 &= \sum_{j=1}^{n-1} \|g_j^u\|_\infty^2 \\ &\leq \frac{16\alpha^2 \|u\|^2 K_{A,Q}^2 \|\epsilon\|_\infty^2}{a^2} \sum_{j=1}^{n-1} (1-\alpha a)^{n-j-1} \\ &\leq \frac{16\alpha^2 \|u\|^2 K_{A,Q}^2 \|\epsilon\|_\infty^2}{a^2} \frac{1}{\alpha a} \\ &= \frac{16\alpha \|u\|^2 K_{A,Q}^2 \|\epsilon\|_\infty^2}{a^3}. \end{aligned} \quad (39)$$

Plugging this into the Markov concentration input Eq. 36 and defining

$$K_{\text{last},0} := 32 K_{A,Q} \|\epsilon\|_\infty \frac{\sqrt{t_{\text{mix}}}}{a^{3/2}}, \quad (40)$$

gives

$$\sqrt{p t_{\text{mix}} \sum_{j=1}^n \|g_j^u\|_\infty^2} \leq \sqrt{p} \|u\| K_{\text{last},0} \sqrt{\alpha}. \quad (41)$$

Applying Eq. 36 with $X = u^\top \tilde{J}_{n,\text{last}}^{(0,\alpha)}$ gives, for any $p \geq 2$,

$$\mathbb{E}^{1/p} \left[\left| u^\top \tilde{J}_{n,\text{last}}^{(0,\alpha)} \right|^p \right] \leq C_{\text{MC},0} \sqrt{p} \|u\| K_{\text{last},0} \sqrt{\alpha}, \quad (42)$$

or equivalently

$$\frac{1}{\sqrt{\alpha}} \|u^\top \tilde{J}_{n,\text{last}}^{(0,\alpha)}\|_{L_p} \leq C_{\text{MC},0} \sqrt{p} \|u\| K_{\text{last},0}. \quad (43)$$

Thus every fixed scalar projection of the last-iterate zeroth-order RR difference is $O(\sqrt{\alpha})$ in L^p , uniformly in n . A Euclidean-norm bound can be recovered in fixed dimension by applying the scalar estimate to a coordinate basis, but no dimension-free vector concentration is claimed here.

3. Last Iterate Analysis

3.1. Centered Bound for the Shifted First-Order Perturbation

This chapter is partly preliminary. The last-iterate RR discussion at the end is motivational and is not used in the final Berry–Esseen assembly; the downstream input retained by Chapter 4 is the centered shifted first-order bound proved below. The proof uses the deterministic-product perturbation expansion from Samsonov et al. (2025, Proposition 9), specialized to the constant-stepsizes setting, and isolates the future-centered bilinear concentration input used in that argument.

Let

$$B = I - \alpha \bar{A}. \quad (44)$$

Throughout this section we assume $0 < \alpha \leq \alpha_\infty$, so that

$$\|B^m\|_Q \leq (1 - \alpha a)^{m/2}, \quad m \geq 0. \quad (45)$$

When the shifted estimate is transferred back from $T_n^{(1,\alpha)}$ to $J_n^{(1,\alpha)}$, we also use the elementary inverse-admissibility ceiling

$$\alpha_{\text{inv}} := \frac{1}{2 \|\bar{A}\|}. \quad (46)$$

Indeed, if $w \leq \alpha_{\text{inv}}$, then the Neumann series gives $\|(I - w\bar{A})^{-1}\| \leq 2$. The single-stepsizes estimates below are stated for α . Whenever the Richardson–Romberg combination is formed, the same estimates are applied at both $w = \alpha$ and $w = 2\alpha$; the local assumptions are then $2\alpha \leq \alpha_\infty$ and $2\alpha \leq \alpha_{\text{inv}}$. All constants denoted by C may depend on fixed problem parameters, including fixed powers of a^{-1} , but not on α , n , p , or q . Dependence on the displayed variables, and in particular on t_{mix} when it is tracked in the rate, is written explicitly.

The deterministic-product components are defined by

$$J_n^{(0,\alpha)} = B J_{n-1}^{(0,\alpha)} - \alpha \epsilon(Z_n), \quad J_0^{(0,\alpha)} = 0, \quad (47)$$

and

$$J_n^{(1,\alpha)} = B J_{n-1}^{(1,\alpha)} - \alpha \tilde{A}(Z_n) J_{n-1}^{(0,\alpha)}, \quad J_0^{(1,\alpha)} = 0. \quad (48)$$

Hence

$$J_n^{(0,\alpha)} = -\alpha \sum_{k=1}^n B^{n-k} \epsilon(Z_k), \quad (49)$$

and

$$\begin{aligned} J_n^{(1,\alpha)} &= -\alpha \sum_{j=1}^n B^{n-j} \tilde{A}(Z_j) J_{j-1}^{(0,\alpha)} \\ &= \alpha^2 \sum_{1 \leq k < j \leq n} B^{n-j} \tilde{A}(Z_j) B^{j-1-k} \epsilon(Z_k) \\ &= \alpha^2 \sum_{k=1}^{n-1} \sum_{r=1}^{n-k} B^{n-k-r} \tilde{A}(Z_{k+r}) B^{r-1} \epsilon(Z_k). \end{aligned} \quad (50)$$

The sign in the second line is positive because $J_{j-1}^{(0,\alpha)}$ already contains the minus sign.

The last display is for $J_n^{(1,\alpha)}$ itself. The last-iteration estimate below operates on a **shifted** version, obtained by inserting one additional left factor B into the recursion:

$$S_n = \sum_{t=0}^{n-1} B^{n-t} \tilde{A}(Z_{t+1}) J_t^{(0,\alpha)}. \quad (51)$$

Reindexing $j = t + 1$ rewrites S_n in the same form as $J_n^{(1,\alpha)}$ but with one extra power of B :

$$S_n = \sum_{j=1}^n B^{n-j+1} \tilde{A}(Z_j) J_{j-1}^{(0,\alpha)} = -\frac{1}{\alpha} B J_n^{(1,\alpha)}. \quad (52)$$

Thus the corresponding shifted first-order contribution is

$$T_n^{(1,\alpha)} = -\alpha S_n = B J_n^{(1,\alpha)}, \quad (53)$$

i.e. $T_n^{(1,\alpha)}$ is exactly $J_n^{(1,\alpha)}$ pre-multiplied by one additional B . Transferring a bound from $T_n^{(1,\alpha)}$ back to $J_n^{(1,\alpha)}$ therefore requires a local inverse bound on $B^{-1} = (I - \alpha \bar{A})^{-1}$; this inverse-bound step is used explicitly when the shifted estimate is applied below.

Lemma 3. (Imported future-centered bilinear estimate.) Assume UGE 1 and stationarity. Let $\mathcal{F}_k = \sigma(Z_0, \dots, Z_k)$. For $1 \leq k \leq n-1$ and $1 \leq l \leq n-k$, let $g_{k,l} : \mathcal{Z} \rightarrow \mathbb{R}^d$ be deterministic functions satisfying $\pi(g_{k,l}) = 0$ and $\|g_{k,l}\|_\infty \leq \beta_{k,l}$. Let ξ_k be $\mathcal{F}_k$ -measurable vectors with $\sup_k \|\xi_k\|_\infty \leq M_\xi$. Then, for every $p \geq 2$,

$$\begin{aligned} & \left\| \sum_{k=1}^{n-1} \left\{ \sum_{l=1}^{n-k} g_{k,l}(Z_{k+l})^\top \xi_k - \mathbb{E} \left[\sum_{l=1}^{n-k} g_{k,l}(Z_{k+l})^\top \xi_k \mid \mathcal{F}_k \right] \right\} \right\|_{L_p} \\ & \leq C p^{3/2} t_{\text{mix}}^{1/2} M_\xi \left(\sum_{k=1}^{n-1} \sum_{l=1}^{n-k} \beta_{k,l}^2 \right)^{1/2}. \end{aligned} \quad (54)$$

This is the scalar, constant-stepsizes specialization of the block-decomposition and Berbee-coupling estimate used in Samsonov et al. (2025, Appendix D.2, Proposition 9; see in particular their Lemma 11 and the treatment of the coupled term $T_{21} + T_{22}$). The centering in the display is the conditional centering with respect to $\mathcal{F}_k$; no stationary-centering inequality is applied to a future chain started from Z_k .

Lemma 4. (Centered shifted first-order bound.) Assume the Markov chain is started from stationarity, that is, the law of Z_1 is π , and $\pi(\tilde{A}) = 0$. For every deterministic direction $u \in \mathbb{R}^d$ and every $p \geq 2$,

$$\|u^\top (S_n - \mathbb{E} S_n)\|_{L_p} \leq C \|u\| \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right). \quad (55)$$

Consequently,

$$\|u^\top (T_n^{(1,\alpha)} - \mathbb{E} T_n^{(1,\alpha)})\|_{L_p} \leq C \alpha \|u\| \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right). \quad (56)$$

Proof. Since $J_0^{(0,\alpha)} = 0$, the term $t = 0$ in S_n vanishes. Substituting the explicit formula for $J_t^{(0,\alpha)}$ gives

$$\begin{aligned} S_n &= -\alpha \sum_{t=1}^{n-1} \sum_{k=1}^t B^{n-t} \tilde{A}(Z_{t+1}) B^{t-k} \epsilon(Z_k) \\ &= -\alpha \sum_{k=1}^{n-1} H_{k+1}^{(w)} \epsilon(Z_k), \end{aligned} \quad (57)$$

where, after the change of summation index $l = t - k + 1$,

$$H_{k+1}^{(w)} = \sum_{l=1}^{n-k} B^{n-k-l+1} \tilde{A}(Z_{k+l}) B^{l-1}. \quad (58)$$

The kernel $H_{k+1}^{(w)}$ acts on the past noise $\epsilon(Z_k)$ through future states Z_{k+l} , so $H_{k+1}^{(w)}\epsilon(Z_k)$ is a future-weighted bilinear functional of the trajectory.

Define

$$\mu_k^{(w)} = \mathbb{E}_\pi \left[H_{k+1}^{(w)} \epsilon(Z_k) \right]. \quad (59)$$

Then

$$S_n - \mathbb{E}S_n = -\alpha \sum_{k=1}^{n-1} \left\{ H_{k+1}^{(w)} \epsilon(Z_k) - \mu_k^{(w)} \right\}. \quad (60)$$

Let $\mathcal{F}_k = \sigma(Z_0, \dots, Z_k)$. The Markov property gives

$$\mathbb{E} \left[H_{k+1}^{(w)} \epsilon(Z_k) \mid \mathcal{F}_k \right] = v_k^{(w, \epsilon)}(Z_k), \quad (61)$$

where

$$v_k^{(w)}(z) = \sum_{l=1}^{n-k} B^{n-k-l+1} (Q^l \tilde{A})(z) B^{l-1}, \quad v_k^{(w, \epsilon)}(z) = v_k^{(w)}(z) \epsilon(z). \quad (62)$$

Here Q denotes the one-step Markov transition kernel of $(Z_k)_{k \geq 1}$, acting on bounded matrix-valued functions by integration against the conditional law:

$$(Q\tilde{A})(z) = \int \tilde{A}(u) Q(z, du) = \mathbb{E}[\tilde{A}(Z_{k+1}) \mid Z_k = z], \quad (63)$$

and Q^l is its l -fold iterate, the l -step kernel $Q^l(z, du) = \mathbb{P}(Z_{k+l} \in du \mid Z_k = z)$, so that

$$(Q^l \tilde{A})(z) = \int \tilde{A}(u) Q^l(z, du) = \mathbb{E}[\tilde{A}(Z_{k+l}) \mid Z_k = z]. \quad (64)$$

In particular $(Q^l \tilde{A})(z) \rightarrow \pi(\tilde{A}) = 0$ at the geometric rate dictated by UGE, which is the only fact about Q^l used below.

Under stationarity, $\mu_k^{(w)} = \pi(v_k^{(w, \epsilon)})$. Therefore

$$S_n - \mathbb{E}S_n = -\alpha(U_M + U_R), \quad (65)$$

with

$$U_M = \sum_{k=1}^{n-1} \left\{ H_{k+1}^{(w)} \epsilon(Z_k) - v_k^{(w, \epsilon)}(Z_k) \right\} \quad (66)$$

and

$$U_R = \sum_{k=1}^{n-1} \left\{ v_k^{(w, \epsilon)}(Z_k) - \pi(v_k^{(w, \epsilon)}) \right\}. \quad (67)$$

This decomposition splits the centered statistic into two structurally different parts: U_R is an ordinary centered additive functional of the original Markov chain, while U_M is a future-centered bilinear term — the conditional expectation given $\mathcal{F}_k$ vanishes summand-wise, but the summands are not forward martingale differences. We bound each piece separately.

Fix a deterministic direction u . The case $u = 0$ is trivial. By homogeneity, it is enough to prove the estimate for $\|u\| = 1$; the general statement follows by multiplying the right-hand side by $\|u\|$.

Step 1: bound on $u^\top U_R$. Because $\pi(\tilde{A}) = 0$, applying Q^l followed by integration against π replaces $\tilde{A}$ by its l -step propagation away from stationarity. UGE then gives the geometric Dobrushin bound

$$\|(Q^l \tilde{A})(z)\| = \left\| \int \tilde{A}(y) \{Q^l(z, dy) - \pi(dy)\} \right\| \leq 2C_A \Delta(Q^l) \leq 2C_A (1/4)^{\lfloor l/t_{\text{mix}} \rfloor}. \quad (68)$$

Inserting this into $v_k^{(w)}$ and factoring out the slow rate $(1 - \alpha a)^{(n-k)/2}$ from the two B -powers, the inner sum becomes a fast-decaying l -series:

$$\begin{aligned} \|v_k^{(w)}\|_\infty &\leq C \sum_{l=1}^{n-k} (1 - \alpha a)^{(n-k-l+1)/2} \Delta(Q^l) (1 - \alpha a)^{(l-1)/2} \text{ (triangle + Lyapunov)} \\ &\leq C (1 - \alpha a)^{(n-k)/2} \sum_{l=1}^{\infty} (1/4)^{\lfloor l/t_{\text{mix}} \rfloor} \text{ (extend to } l = \infty) \\ &\leq C t_{\text{mix}} (1 - \alpha a)^{(n-k)/2} \text{ (geometric block sum)}. \end{aligned} \quad (69)$$

Multiplying by $\|\epsilon\|_\infty$ gives

$$\|v_k^{(w, \epsilon)}\|_\infty \leq C t_{\text{mix}} \|\epsilon\|_\infty (1 - \alpha a)^{(n-k)/2}. \quad (70)$$

The functions $v_k^{(w, \epsilon)}(Z_k) - \pi(v_k^{(w, \epsilon)})$ are centered under π and uniformly bounded by the previous display. Applying the Markov concentration input in the form $C \sqrt{p t_{\text{mix}} \sum_k c_k^2}$ gives

$$\begin{aligned} \|u^\top U_R\|_{L_p} &\leq C p^{1/2} t_{\text{mix}}^{1/2} \left(\sum_{k=1}^{n-1} \|v_k^{(w, \epsilon)}\|_\infty^2 \right)^{1/2} \text{ (Markov concentration)} \\ &\leq C p^{1/2} t_{\text{mix}}^{3/2} \|\epsilon\|_\infty \left(\sum_{k=1}^{n-1} (1 - \alpha a)^{n-k} \right)^{1/2} \text{ (plug previous bound)} \\ &\leq C p^{1/2} t_{\text{mix}}^{3/2} \|\epsilon\|_\infty \frac{1}{\sqrt{\alpha a}} \text{ (geometric series)}. \end{aligned} \quad (71)$$

Step 2: bound on $u^\top U_M$. Unfold the projected matrix kernel through $(H_{k+1}^{(w)})^\top u$:

$$(H_{k+1}^{(w)})^\top u = \sum_{l=1}^{n-k} g_{k,l}(Z_{k+l}), \quad g_{k,l}(z) = (B^{l-1})^\top \tilde{A}(z)^\top (B^{n-k-l+1})^\top u. \quad (72)$$

Each $g_{k,l}$ is centered under π (since $\pi(\tilde{A}) = 0$), and the two B -powers give the uniform bound

$$\|g_{k,l}\|_\infty \leq C (1 - \alpha a)^{(n-k)/2}, \quad (73)$$

which is independent of l . Squaring and summing over l produces only linear growth in $n - k$:

$$\sum_{l=1}^{n-k} \|g_{k,l}\|_\infty^2 \leq C(n-k)(1 - \alpha a)^{n-k}. \quad (74)$$

Note the contrast with the U_R analysis: there, UGE folded the l -sum into a single t_{mix} factor, whereas here the same l -independent bound is applied $(n - k)$ times — the price of conditional centering being weaker than π -centering.

The conditional expectation in the imported future-centered estimate is exactly

$$\mathbb{E} \left[\sum_{l=1}^{n-k} g_{k,l}(Z_{k+l})^\top \epsilon(Z_k) \mid \mathcal{F}_k \right] = u^\top v_k^{(w,\varepsilon)}(Z_k). \quad (75)$$

Therefore the whole future-centered sum $u^\top U_M$ can be bounded in one step by applying the imported estimate with $\xi_k = \epsilon(Z_k)$ and $\beta_{k,l} = C(1 - \alpha a)^{(n-k)/2}$:

$$\begin{aligned} \|u^\top U_M\|_{L_p} &\leq C p^{3/2} t_{\text{mix}}^{1/2} \|\epsilon\|_\infty \left(\sum_{k=1}^{n-1} \sum_{l=1}^{n-k} (1 - \alpha a)^{n-k} \right)^{1/2} \\ &= C p^{3/2} t_{\text{mix}}^{1/2} \|\epsilon\|_\infty \left(\sum_{k=1}^{n-1} (n-k)(1 - \alpha a)^{n-k} \right)^{1/2} \\ &\leq C p^{3/2} t_{\text{mix}}^{1/2} \|\epsilon\|_\infty \frac{1}{\alpha a} \quad \left(\text{use } \sum_m m r^m \leq C(1-r)^{-2}, r = 1 - \alpha a \right). \end{aligned} \quad (76)$$

The extra factor $1/\sqrt{\alpha a}$ relative to U_R is precisely the cost of using conditional rather than stationary centering.

Step 3: assembly. Combining the two pieces via $S_n - \mathbb{E}S_n = -\alpha(U_M + U_R)$ and the triangle inequality,

$$\begin{aligned} \|u^\top (S_n - \mathbb{E}S_n)\|_{L_p} &\leq \alpha (\|u^\top U_M\|_{L_p} + \|u^\top U_R\|_{L_p}) \\ &\leq C \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right). \end{aligned} \quad (77)$$

The first term (from U_M) is the leading contribution for small α ; the second (from U_R) carries the heavier t_{mix} dependence but vanishes as $\alpha \rightarrow 0$. Restoring the factor $\|u\|$ and multiplying by α gives the asserted bound for $T_n^{(1,\alpha)} = -\alpha S_n$. $\square$

3.2. A Depth-One RR Misadjustment Bound and Its Limitation

This subsection records the natural depth-one attempt and explains why it is not used in the final Berry–Esseen assembly. The actual stationary and burned-in theorems use the depth-two Levin transfer developed in the next chapter.

The PR-averaged Richardson–Romberg expansion produces, after applying the first deterministic-product perturbation step underlying Samsonov et al. (2025, Proposition 9) separately at step sizes α and 2α , a depth-one “misadjustment” remainder

$$D_1^{\text{mis, RR}} = \frac{\sqrt{n}}{n - n_0} \sum_{k=n_0}^{n-1} \left(2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)} \right), \quad (78)$$

whose centered part must be controlled to feed into a Berry–Esseen statement by this route. We write

$$D_{1,c}^{\text{mis, RR}} := D_1^{\text{mis, RR}} - \mathbb{E}D_1^{\text{mis, RR}} \quad (79)$$

for this centered statistic.

In this subsection assume explicitly that $2\alpha \leq \alpha_\infty$ and $2\alpha \leq \alpha_{\text{inv}}$. The first condition makes the one-stepsize last-iterate lemma admissible at both RR levels; the second makes the shifted-to-unshifted transfer $T_k^{(1,w)} = (I - w\bar{A})J_k^{(1,w)}$ uniformly invertible for $w \in \{\alpha, 2\alpha\}$.

The stationary bias is smaller than the fluctuation term. By the working form Lemma 29,

$$\mathbb{E}_\pi [J_\infty^{(1,\alpha)}] = \alpha \Delta + O(\alpha^2), \quad (80)$$

so the linear term $\alpha\Delta$ cancels in the RR-combination and the per-iterate stationary RR bias is $O(\alpha^2)$. Therefore the PR-scaled stationary bias satisfies

$$\|\mathbb{E}D_1^{\text{mis, RR}}\| \leq C\sqrt{n}\alpha^2. \quad (81)$$

What remains is the centered fluctuation.

Define

$$\Phi(p, \alpha) := p^{3/2} \frac{t_{\text{mix}}^{1/2}}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha/a}. \quad (82)$$

Fix a deterministic direction u . The lemma applied at α and at 2α gives, separately,

$$\|u^\top (T_n^{(1,\alpha)} - \mathbb{E}T_n^{(1,\alpha)})\|_{L_p} \leq C \|u\| \alpha \Phi(p, \alpha), \quad \|u^\top (T_n^{(1,2\alpha)} - \mathbb{E}T_n^{(1,2\alpha)})\|_{L_p} \leq C \|u\| \alpha \Phi(p, 2\alpha) \leq C' \|u\| \alpha \Phi(p, \alpha),$$

where $C' = \sqrt{2}C$ absorbs the $\sqrt{2}$ -factor coming from the 2α scaling. Combining the two by the triangle inequality and using the index-shift identity $T_k^{(1,w)} = (I - w\bar{A}) J_k^{(1,w)}$ gives

$$u^\top J_k^{(1,w)} = \left((I - w\bar{A})^{-\top} u \right)^\top T_k^{(1,w)}. \quad (84)$$

The local inverse bound $\| (I - w\bar{A})^{-1} \| \leq 2$ therefore holds for $w \in \{\alpha, 2\alpha\}$ and yields

$$\|u^\top (2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)}) - \mathbb{E}(2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)})\|_{L_p} \leq C \|u\| \alpha \Phi(p, \alpha), \quad (85)$$

uniformly in k . PR-averaging through $\frac{\sqrt{n}}{n-n_0}$ and absorbing the constant therefore yields

$$\|u^\top D_{1,c}^{\text{mis, RR}}\|_{L_p} \leq C \|u\| \sqrt{n} \alpha \Phi(p, \alpha) = O(\sqrt{n} \alpha). \quad (86)$$

Together with the bias estimate,

$$\|u^\top D_1^{\text{mis, RR}}\|_{L_p} \leq C \|u\| \sqrt{n} \alpha \Phi(p, \alpha) + C \|u\| \sqrt{n} \alpha^2. \quad (87)$$

At the optimal scale $\alpha \asymp n^{-1/2}$ the centered-fluctuation term is $O(1)$, whereas the stationary bias is $O(n^{-1/2})$. Hence this depth-one route still does not yield a useful Berry–Esseen remainder of order $n^{-1/4}$: the centered misadjustment must be controlled more sharply to be subleading.

4. Richardson–Romberg PR Weight Bounds

Throughout this chapter, unnamed constants C may change from line to line. They may depend on fixed problem parameters and may absorb fixed powers of a^{-1} , but they never hide dependence on n , α , p , or q . Any dependence on these four variables is displayed explicitly or appears as an argument of a named quantity. Named constants follow the same convention: fixed problem factors may be absorbed only when they are not being tracked explicitly in the surrounding display.

4.1. Admissibility Thresholds

The exact external working forms are collected in Section 7. This section records only the admissibility ceilings used by the stationary and burned-in theorems. The cited papers sometimes use the plus-form SA convention; throughout this thesis we use $\theta_{k+1} = \theta_k - w(A(Z_{k+1})\theta_k - b(Z_{k+1}))$, so their stability assumptions are read after this sign conversion.

For stationary depth-two arguments define

$$\alpha_*(q, t_{\text{mix}}) := \min(\alpha_{\text{L,P2}}(q, t_{\text{mix}}), \alpha_{\text{L,C6}}(q, t_{\text{mix}}), \alpha_{\text{L,P5}}(q, t_{\text{mix}}), \alpha_{\text{L,P8}}(q, t_{\text{mix}}), \alpha_{\text{L,P9}}(q, t_{\text{mix}}), \alpha_{\text{L,inv}}(q, t_{\text{mix}})). \quad (88)$$

Here the six ceilings are the positive restrictions needed for Lemma 29, Lemma 30, Lemma 33, Lemma 31, Lemma 32, and Lemma 34. Thus Input C is exactly the minimum of the Levin stationary depth-two ceilings whose working forms are stated in Section 7.1.

For deterministic-start burn-in arguments define

$$\alpha_{\text{st}}(p) := \min(\alpha_{\text{prod}}(2p), \alpha_{\text{rand-prod}}(2p), \alpha_{\text{full-start}}(2p)). \quad (89)$$

The first ceiling belongs to the deterministic product-stability working form Lemma 17, including its conditional form Eq. 271. The second and third belong to the local extensions Lemma 22 and Lemma 23. These two random-coupling and full-state startup statements are proved in the thesis and are not direct citations from Levin et al. Thus Input D is exactly the local threshold Eq. 89 plus the product-stability technical input and the local statements named above, not an additional external theorem. This separation is recorded in Section 7.2.

For the shifted-to-unshifted first-order transfer we also use the local inverse ceiling

$$\alpha_{\text{inv}} := \frac{1}{2 \| \bar{A} \|}. \quad (90)$$

If $w \leq \alpha_{\text{inv}}$, the Neumann series yields $\| (I - w\bar{A})^{-1} \| \leq 2$.

We collect the small-step ceilings used in the stationary and deterministic burn-in results into two admissibility thresholds:

$$\alpha_{\text{stat}}(q) := \min(\alpha_{\infty}, \alpha_{\text{inv}}, \alpha_*(q, t_{\text{mix}})), \quad \alpha_{\text{burn}}(p, q) := \min\left(\alpha_{\text{stat}}(q), \frac{1}{2a}, \alpha_{\text{st}}(p)\right). \quad (91)$$

Thus $2\alpha \leq \alpha_{\text{stat}}(q)$ makes both RR step sizes admissible for the Lyapunov, inverse, and Levin stationary depth-two inputs. The stronger condition $2\alpha \leq \alpha_{\text{burn}}(p, q)$ additionally gives $\alpha a \leq 1/4$ and the startup/product-stability ceilings used in the burn-in transfer. In the full-state startup proof the $H^{(2)}$ comparison uses Levin moment bounds at order $2p$; the burned-in theorems therefore choose q so that $p \leq q/4$.

4.2. PR-Averaged Error Decomposition and the RR Weight

The deterministic weight estimates start from the finite-start representation of $\sqrt{n}(\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^*)$ as a noise-weighted sum with a deterministic kernel. The stationary $n_0 = 0$ Berry–Esseen bound proved below is stated for the augmented-chain assembly built from the same full-window weights.

Depth-one expansion. Unfolding the error recursion of Chapter 1 gives, for every $k \geq 1$,

$$\theta_k^{(\alpha)} - \theta^* = -\alpha \sum_{l=1}^k \Gamma_{l+1:k}^{(\alpha)} \epsilon(Z_l) + \Gamma_{1:k}^{(\alpha)} (\theta_0 - \theta^*), \quad \Gamma_{l+1:k}^{(\alpha)} := \prod_{j=l+1}^k (I - \alpha A(Z_j)). \quad (92)$$

Replacing the random products in the noise sum by their deterministic counterparts $B_\alpha^{k-l} := (I - \alpha \bar{A})^{k-l}$ and keeping the initial-product discrepancy separate yields the **depth-one decomposition** (Samsonov et al., 2025, Proposition 9):

$$R_{k,\text{init}}^{(\alpha)} := (\Gamma_{1:k}^{(\alpha)} - B_\alpha^k) (\theta_0 - \theta^*), \quad (93)$$

$$\theta_k^{(\alpha)} - \theta^* = -\alpha \sum_{l=1}^k B_\alpha^{k-l} \epsilon(Z_l) + B_\alpha^k (\theta_0 - \theta^*) + R_{k,\text{init}}^{(\alpha)} + R_k^{(\alpha)}, \quad (94)$$

where $R_k^{(\alpha)} := J_k^{(1,\alpha)} + H_k^{(1,\alpha)}$ is the first noise-driven misadjustment remainder. The term $R_{k,\text{init}}^{(\alpha)}$ vanishes when $\theta_0 = \theta^*$; otherwise it is a finite-start transient and is handled in the burn-in transfer theorem, not by the stationary augmented-chain misadjustment bound. The leading component $J_k^{(1,\alpha)}$ has a stationary bias of order α , so it is not treated as an $\alpha^{3/2}$ term. The Berry–Esseen proof below controls this noise-driven remainder by refining it to depth two, where $J^{(2)} + H^{(2)}$ carries the $\alpha^{3/2}$ moment scale. The first sum is the **depth-zero** term and is the only piece that carries the limiting Gaussian.

PR averaging produces $Q_l^{(\alpha)}$. Recall the PR average $\bar{\theta}_n^{(\alpha)} = (n - n_0)^{-1} \sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}$. For notational clarity, and for the stationary result proved in this chapter, we set $n_0 = 0$ from this point on. A burned-in average has a different deterministic weight,

$$Q_{l,n_0}^{(\alpha)} = \frac{n}{n - n_0} \alpha \sum_{k=\max(n_0,l)}^{n-1} B_\alpha^{k-l}, \quad (95)$$

when the statistic is normalized as $-n^{-1/2} \sum_l Q_{l,n_0}^{(\alpha)} \epsilon(Z_l)$. The burned-in non-stationary theorem therefore requires separate weight, Poisson, and variance-comparison arguments. Subtracting θ^* , substituting the depth-one decomposition above, and **exchanging the order of summation** in the depth-zero piece,

$$\sum_{k=0}^{n-1} \sum_{l=1}^k B_\alpha^{k-l} \epsilon(Z_l) = \sum_{l=1}^{n-1} \left(\sum_{k=l}^{n-1} B_\alpha^{k-l} \right) \epsilon(Z_l), \quad (96)$$

isolates each noise sample $\epsilon(Z_l)$ together with the deterministic **PR weight**

$$Q_l^{(\alpha)} := \alpha \sum_{k=l}^{n-1} B_\alpha^{k-l} = \alpha \sum_{j=0}^{n-l-1} B_\alpha^j. \quad (97)$$

Multiplying the resulting identity by $\sqrt{n}$ gives the PR-averaged decomposition

$$\sqrt{n} (\bar{\theta}_n^{(\alpha)} - \theta^*) = W^{(\alpha)} + D^{(\alpha)}, \quad (98)$$

where the **leading martingale-like sum** is

$$W^{(\alpha)} := -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l^{(\alpha)} \epsilon(Z_l), \quad (99)$$

and the remainder $D^{(\alpha)}$ contains the deterministic transient $D_{\text{tr}}^{(\alpha)}$, the random initial-product transient $D_{\text{init}}^{(\alpha)}$, and the higher-order noise-driven stochastic part $D_R^{(\alpha)}$:

$$D^{(\alpha)} := D_{\text{tr}}^{(\alpha)} + D_{\text{init}}^{(\alpha)} + D_R^{(\alpha)}, \quad D_{\text{tr}}^{(\alpha)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} B_\alpha^k (\theta_0 - \theta^*), \quad D_{\text{init}}^{(\alpha)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} R_{k, \text{init}}^{(\alpha)}, \quad D_R^{(\alpha)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} R_k^{(\alpha)}$$

The first sum is the deterministic initial-condition transient. Under the full-average convention it is only $O((\sqrt{n}\alpha)^{-1})$ in general, so it must either be retained explicitly or removed by a centered initialization $\theta_0 = \theta^*$. The second sum is stochastic but still purely finite-start: it is controlled after burn-in by random-product stability. The third is the source of the leading non-Gaussian correction in the Berry–Esseen rate. Its RR-combination $2D_R^{(\alpha)} - D_R^{(2\alpha)}$ is the **misadjustment** $D_1^{\text{mis}, \text{RR}}$ controlled below by the Levin depth-two transfer.

RR combination produces $\mathcal{Q}_l^{\text{RR}}$. The Richardson–Romberg iterate $\bar{\theta}_n^{(\text{RR}, \alpha)} := 2\bar{\theta}_n^{(\alpha)} - \bar{\theta}_n^{(2\alpha)}$ inherits the PR decomposition **by linearity**: applying the previous display at step sizes α and 2α separately and combining,

$$\sqrt{n} (\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^*) = W^{\text{RR}} + D^{\text{RR}}, \quad (101)$$

$$W^{\text{RR}} := 2W^{(\alpha)} - W^{(2\alpha)} = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} \mathcal{Q}_l^{\text{RR}} \epsilon(Z_l), \quad (102)$$

where the **RR weight** is

$$\mathcal{Q}_l^{\text{RR}} := 2\mathcal{Q}_l^{(\alpha)} - \mathcal{Q}_l^{(2\alpha)}. \quad (103)$$

Since both PR averages share the **same** noise realization $\{Z_k\}$, the bracket $\mathcal{Q}_l^{\text{RR}}$ is a single deterministic matrix kernel: all the stochastic content of W^{RR} now lives in $\epsilon(Z_l)$ alone.

Two deterministic estimates. The rest of the martingale approximation uses two quantities determined by $\mathcal{Q}_l^{\text{RR}}$:

1. **Variance comparison.** The target CLT covariance of W^{RR} is $\Sigma_\infty = \bar{A}^{-1} \Sigma_\epsilon^{(\text{M})} \bar{A}^{-\top}$ (the Markov-chain CLT covariance). Replacing each $\mathcal{Q}_l^{\text{RR}}$ by the asymptotic weight $\bar{A}^{-1}$ gives this covariance, so the finite- n deviation is controlled by $\sum_l \|\mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1}\|^2$ — see Section 4.5.
2. **Poisson-equation / Abel-summation remainder.** The standard Berry–Esseen route for Markov-chain noise solves the Poisson equation $\hat{\epsilon} - Q\hat{\epsilon} = \epsilon$, replaces $\epsilon(Z_l)$ by $\hat{\epsilon}(Z_l) - \hat{\epsilon}(Z_{l+1})$ (up to a martingale increment), and Abel-sums against the weight sequence $\mathcal{Q}_l^{\text{RR}}$. The resulting remainder has norm bounded by the **total variation** $\sum_l \|\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}}\|$.

Both quantities are estimated in Sections 4.3–4.4 below. The closed-form identities of Section 4.2 reduce them to elementary operator-norm calculations on $B_\alpha^m - B_{2\alpha}^m$.

4.3. Closed-Form Identities

Throughout this chapter write

$$B_\alpha := I - \alpha \bar{A}, \quad B_{2\alpha} := I - 2\alpha \bar{A}, \quad k := n - l. \quad (104)$$

We assume $0 < \alpha$ and $2\alpha \leq \alpha_\infty$, so that the Lyapunov contraction $\|B_\alpha^m\|_Q^2 \leq (1 - \alpha\alpha)^m$ and $\|B_{2\alpha}^m\|_Q^2 \leq (1 - 2\alpha\alpha)^m$ hold (and we use freely $1 - 2\alpha\alpha \leq 1 - \alpha\alpha$).

The geometric-series identity $\sum_{j=0}^{m-1} B_\alpha^j = (\alpha \bar{A})^{-1} (I - B_\alpha^m)$ converts Eq. 97 into the closed form

$$\mathcal{Q}_l^{(\alpha)} = \alpha \sum_{j=0}^{n-l-1} B_\alpha^j = \alpha (\alpha \bar{A})^{-1} (I - B_\alpha^{n-l}) = \bar{A}^{-1} (I - B_\alpha^{n-l}). \quad (105)$$

This makes the convergence $Q_l^{(\alpha)} \rightarrow \bar{A}^{-1}$ as $k = n - l \rightarrow \infty$ explicit: the only obstruction is the geometric Lyapunov tail B_α^k . Two immediate consequences are

$$Q_l^{(\alpha)} - \bar{A}^{-1} = -\bar{A}^{-1} B_\alpha^{n-l}, \quad Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)} = -\alpha B_\alpha^{n-l-1}. \quad (106)$$

The first is the **asymptotic-weight error** and decays geometrically in k . The second is the **discrete derivative** used in Abel summation; note the explicit factor α in front.

Applying both identities at α and 2α and combining yields the basic **RR identities**:

$$Q_l^{\text{RR}} - \bar{A}^{-1} = -\bar{A}^{-1} (2B_\alpha^k - B_{2\alpha}^k), \quad (107)$$

$$Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}} = -2\alpha (B_\alpha^{k-1} - B_{2\alpha}^{k-1}). \quad (108)$$

These are the exact expressions bounded in the next subsection. Note the structural asymmetry between them:

- The asymptotic-weight error $2B_\alpha^k - B_{2\alpha}^k$ is at the **single-trajectory** rate $(1 - \alpha a)^{k/2}$ — the triangle inequality $|2X - Y| \leq 2|X| + |Y|$ does not see the RR coupling, and indeed the leading $-\bar{A}^{-1}$ in Q_l^{RR} is the same as in the single-step weight $Q_l^{(\alpha)}$.
- The discrete derivative $B_\alpha^{k-1} - B_{2\alpha}^{k-1}$, however, is a **true** difference of contractions evaluated at the same exponent. The elementary identity $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$ with $X = B_\alpha$, $Y = B_{2\alpha}$, $X - Y = \alpha \bar{A}$ extracts an extra factor of α , gaining one full power of α over the single-step case $Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)} = -\alpha B_\alpha^{k-1}$. The left-factored form is valid because B_α and $B_{2\alpha}$ are polynomials in the same matrix $\bar{A}$, hence commute with each other and with $\bar{A}$.

This local α -gain in the discrete derivative is the **only** manifestation of Richardson–Romberg cancellation visible at the level of the PR weights.

4.4. Pointwise Bounds for the RR Weights

Lemma 5. (Pointwise RR weight bounds.) Let $0 < \alpha$ and $2\alpha \leq \alpha_\infty$, set $C_Q := \kappa_Q^{1/2} \|\bar{A}^{-1}\|$ and $\tilde{C}_A := \kappa_Q \|\bar{A}\|$, and write $k = n - l$.

(i) For every $1 \leq l \leq n - 1$,

$$\|Q_l^{\text{RR}} - \bar{A}^{-1}\| \leq 3C_Q (1 - \alpha a)^{k/2}. \quad (109)$$

(ii) For every $1 \leq l \leq n - 2$,

$$\|Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}\| \leq 2\tilde{C}_A \alpha^2 (k - 1) (1 - \alpha a)^{(k-2)/2}. \quad (110)$$

Proof of (i). Take norms in the basic RR identity above. Using the equivalence $\|\cdot\| \leq \kappa_Q^{1/2} \|\cdot\|_Q$ and the Lyapunov contraction at both step sizes,

$$\|2B_\alpha^k - B_{2\alpha}^k\|_Q \leq 2\|B_\alpha^k\|_Q + \|B_{2\alpha}^k\|_Q \leq 2(1 - \alpha a)^{k/2} + (1 - 2\alpha a)^{k/2} \leq 3(1 - \alpha a)^{k/2}. \quad (111)$$

Submultiplicativity in $\|\cdot\|_Q$ followed by norm equivalence then gives

$$\|Q_l^{\text{RR}} - \bar{A}^{-1}\| \leq \kappa_Q^{1/2} \|\bar{A}^{-1}\| \cdot \|2B_\alpha^k - B_{2\alpha}^k\|_Q \leq 3C_Q (1 - \alpha a)^{k/2}. \quad (112)$$

Proof of (ii). Apply the elementary identity $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$ with $X = B_\alpha$, $Y = B_{2\alpha}$, $X - Y = \alpha \bar{A}$, and $m = k - 1$. The matrices commute because they are polynomials in $\bar{A}$, so the factor $\alpha \bar{A}$ may be placed on the left:

$$B_\alpha^{k-1} - B_{2\alpha}^{k-1} = \alpha \bar{A} \sum_{i=1}^{k-1} B_\alpha^{i-1} B_{2\alpha}^{k-1-i}. \quad (113)$$

Each summand obeys, by submultiplicativity in $\|\cdot\|_Q$ and Lyapunov contraction,

$$\begin{aligned} \|B_\alpha^{i-1} B_{2\alpha}^{k-1-i}\| &\leq \kappa_Q^{1/2} (1-\alpha a)^{(i-1)/2} (1-2\alpha a)^{(k-1-i)/2} \\ &\leq \kappa_Q^{1/2} (1-\alpha a)^{(k-2)/2}, \end{aligned} \quad (114)$$

where in the last step we used $1-2\alpha a \leq 1-\alpha a$. Summing over $i \in \{1, \dots, k-1\}$ and absorbing constants,

$$\|B_\alpha^{k-1} - B_{2\alpha}^{k-1}\| \leq \alpha \kappa_Q \|\bar{A}\| (k-1)(1-\alpha a)^{(k-2)/2}. \quad (115)$$

Inserting this bound into the discrete-difference identity finishes the proof. $\square$

4.5. Summed Bounds and Comparison with the Single-Step Case

Corollary 1. (Summed RR weight bounds.) Under the assumptions of the previous lemma, uniformly in $n \geq 2$,

$$\sum_{l=1}^{n-1} \|Q_l^{\text{RR}} - \bar{A}^{-1}\|^2 \leq \frac{C_1}{\alpha a}, \quad \sum_{l=1}^{n-2} \|Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}\| \leq \frac{C_2}{a^2}, \quad (116)$$

with $C_1 = 9C_Q^2$ and $C_2 = 32\tilde{C}_A$.

Proof. For the first sum apply (i) and the geometric series $\sum_{k \geq 1} (1-\alpha a)^k \leq \frac{1}{\alpha a}$. For the second, apply (ii) and

$$\sum_{k \geq 2} (k-1) (1-\alpha a)^{(k-2)/2} = \sum_{m \geq 0} (m+1) (1-\alpha a)^{m/2} \leq \frac{1}{(1-\sqrt{1-\alpha a})^2} \leq \frac{4}{(\alpha a)^2}, \quad (117)$$

where the last step uses $1-\sqrt{1-\alpha a} \geq \alpha \frac{a}{2}$ for $\alpha a \leq 1/2$. Multiplying by $2\tilde{C}_A \alpha^2$ gives the claim, up to the stated universal constant. $\square$

4.6. Variance Comparison

After the Poisson decomposition the $l=1$ noise sample is absorbed into the Poisson boundary remainder, while the martingale increments are indexed by $l \in \{2, \dots, n-1\}$. The deterministic variance proxy used in the martingale Berry–Esseen step must therefore use the same index set:

$$\Sigma_n^{\text{RR}} := \frac{1}{n} \sum_{l=2}^{n-1} Q_l^{\text{RR}} \Sigma_\epsilon^{(\text{M})} (Q_l^{\text{RR}})^\top, \quad \sigma_n^{2, \text{RR}}(u) := u^\top \Sigma_n^{\text{RR}} u, \quad (118)$$

with $\Sigma_\epsilon^{(\text{M})}$ the symmetric long-run noise covariance of the Markov chain,

$$\Sigma_\epsilon^{(\text{M})} = \mathbb{E}_\pi [\epsilon(Z_0) \epsilon(Z_0)^\top] + \sum_{j \geq 1} \left(\mathbb{E}_\pi [\epsilon(Z_0) \epsilon(Z_j)^\top] + \mathbb{E}_\pi [\epsilon(Z_j) \epsilon(Z_0)^\top] \right). \quad (119)$$

The asymptotic covariance is

$$\Sigma_\infty = \bar{A}^{-1} \Sigma_\epsilon^{(\text{M})} \bar{A}^{-\top}, \quad \sigma^2(u) = u^\top \Sigma_\infty u. \quad (120)$$

The bounds of the previous section give a quantitative comparison between Σ_n^{RR} and Σ_∞ .

Lemma 6. (Stationary finite-window variance comparison.) Let $0 < \alpha$ and $2\alpha \leq \alpha_\infty$, set $\Sigma := \Sigma_\epsilon^{(M)}$, and assume $\|\Sigma\| < \infty$. Then

$$\|\Sigma_n^{\text{RR}} - \Sigma_\infty\| \leq \frac{C_3}{n\alpha a}, \quad (121)$$

with $C_3 = 12 C_Q \|\bar{A}^{-1}\| \|\Sigma\| + 9 C_Q^2 \|\Sigma\| + 2 \|\Sigma_\infty\|$. Consequently, for every $u \in \mathbb{R}^d$,

$$|\sigma_n^{2, \text{RR}}(u) - \sigma^2(u)| \leq \frac{C_3 \|u\|^2}{n\alpha a}. \quad (122)$$

At the working scale $\alpha = c n^{-1/2}$ this is $O(n^{-1/2})$.

Proof. Write $\Delta_l := \mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1}$ and expand

$$\mathcal{Q}_l^{\text{RR}} \Sigma (\mathcal{Q}_l^{\text{RR}})^\top - \Sigma_\infty = \underbrace{\Delta_l \Sigma \bar{A}^{-\top} + \bar{A}^{-1} \Sigma \Delta_l^\top}_{R_{1,l}} + \underbrace{\Delta_l \Sigma \Delta_l^\top}_{R_{2,l}}. \quad (123)$$

Submultiplicativity of the operator norm yields the pointwise bounds

$$\|R_{1,l}\| \leq 2 \|\bar{A}^{-1}\| \|\Sigma\| \|\Delta_l\|, \quad \|R_{2,l}\| \leq \|\Sigma\| \|\Delta_l\|^2. \quad (124)$$

Summing the linear part with the bound from part (i) of the previous lemma and the geometric series $\sum_{k \geq 1} (1 - \alpha a)^{k/2} \leq \frac{1}{1 - \sqrt{1 - \alpha a}} \leq \frac{2}{\alpha a}$ (valid for $\alpha a \leq 1/2$),

$$\sum_{l=1}^{n-1} \|\Delta_l\| \leq 3 C_Q \sum_{k=1}^{n-1} (1 - \alpha a)^{k/2} \leq \frac{6 C_Q}{\alpha a}. \quad (125)$$

The quadratic part is bounded directly by the previous corollary,

$$\sum_{l=1}^{n-1} \|\Delta_l\|^2 \leq \frac{9 C_Q^2}{\alpha a}. \quad (126)$$

Since the sum starts at $l = 2$, replacing every weight by $\bar{A}^{-1}$ gives $((n-2)/n)\Sigma_\infty$, not Σ_∞ . Hence there is an additional deterministic finite-sum boundary term $2\Sigma_\infty/n$. Combining,

$$\begin{aligned} \|\Sigma_n^{\text{RR}} - \Sigma_\infty\| &\leq \frac{2 \|\Sigma_\infty\|}{n} + \frac{1}{n} \sum_{l=2}^{n-1} (\|R_{1,l}\| + \|R_{2,l}\|) \\ &\leq \frac{1}{n} \left(2 \|\bar{A}^{-1}\| \|\Sigma\| \cdot \frac{6 C_Q}{\alpha a} + \|\Sigma\| \cdot \frac{9 C_Q^2}{\alpha a} + 2 \|\Sigma_\infty\| / (\alpha a) \right) = \frac{C_3}{n\alpha a}, \end{aligned} \quad (127)$$

where we used $\alpha a \leq 1$. This proves the operator-norm bound. The scalar bound on $|\sigma_n^{2, \text{RR}}(u) - \sigma^2(u)| = |u^\top (\Sigma_n^{\text{RR}} - \Sigma_\infty) u|$ follows from the Cauchy–Schwarz inequality. $\square$

4.7. Poisson Martingale Approximation

The variance comparison of the previous section identifies the limiting covariance of W^{RR} , but it does not by itself produce a martingale. This subsection converts W^{RR} into a martingale plus a quantitatively small remainder via the Poisson equation for the Markov chain $\{Z_l\}$. The deterministic kernel bounds of Sections 4.3–4.4 enter once and for all in the boundary/Abel control of the remainder.

Poisson kernel. Let Q denote the one-step Markov transition kernel of $(Z_k)_{k \geq 0}$, acting on bounded measurable functions $f : Z \rightarrow \mathbb{R}^d$ by

$$(Qf)(z) = \mathbb{E}[f(Z_{k+1}) \mid Z_k = z]. \quad (128)$$

Under UGE 1 with mixing time t_{mix} , the geometric Dobrushin bound $\|Q^k \epsilon\|_\infty \leq 2 \|\epsilon\|_\infty (1/4)^{\lfloor k/t_{\text{mix}} \rfloor}$ (valid for centered ϵ , $\pi(\epsilon) = 0$) makes the Poisson series

$$\hat{\epsilon} := \sum_{k=0}^{\infty} Q^k \epsilon \quad (129)$$

absolutely convergent in sup-norm, with

$$\|\hat{\epsilon}\|_\infty \leq 2 \|\epsilon\|_\infty \sum_{k=0}^{\infty} (1/4)^{\lfloor k/t_{\text{mix}} \rfloor} \leq 3 t_{\text{mix}} \|\epsilon\|_\infty, \quad \|Q\hat{\epsilon}\|_\infty \leq \|\hat{\epsilon}\|_\infty, \quad (130)$$

the second inequality because Q is a Markov kernel and contracts the sup-norm. By construction $\hat{\epsilon}$ solves the **Poisson equation**

$$\hat{\epsilon} - Q\hat{\epsilon} = \epsilon. \quad (131)$$

Conditional centering. Let $\mathcal{F}_l := \sigma(Z_0, \dots, Z_l)$. Substituting Eq. 131 into the noise sample and adding and subtracting the conditional mean,

$$\epsilon(Z_l) = \underbrace{[\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})]}_{\text{martingale increment}} + \underbrace{[Q\hat{\epsilon}(Z_{l-1}) - Q\hat{\epsilon}(Z_l)]}_{\text{telescope}}, \quad l \geq 2, \quad (132)$$

where the first bracket is centered conditionally on $\mathcal{F}_{l-1}$ since the Markov property gives $\mathbb{E}[\hat{\epsilon}(Z_l) | \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$. The $l = 1$ term is kept as the left boundary of the Abel transform rather than included in M_n^{RR} ; we treat it directly via Eq. 131 as $\epsilon(Z_1) = \hat{\epsilon}(Z_1) - Q\hat{\epsilon}(Z_1)$. The decomposition therefore separates W^{RR} into a martingale piece and a deterministic-coefficient telescope; Abel summation against the kernel sequence $\{Q_l^{\text{RR}}\}$ converts the telescope into boundary terms plus a sum against the discrete derivative $Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}$, which is exactly the object bounded in the Corollary of Section 4.4.

Lemma 7. (Stationary Poisson martingale decomposition.) Assume **UGE 1** and $\pi(\epsilon) = 0$. Set

$$\Delta M_l^{\text{RR}} := Q_l^{\text{RR}} (\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})), \quad 2 \leq l \leq n-1, \quad (133)$$

and let $M_n^{\text{RR}} := \sum_{l=2}^{n-1} \Delta M_l^{\text{RR}}$. Then $\{\Delta M_l^{\text{RR}}\}_{l=2}^{n-1}$ is a sequence of $\mathcal{F}_l$ -martingale differences, and

$$W^{\text{RR}} = -\frac{1}{\sqrt{n}} M_n^{\text{RR}} + D_{2,n}^{\text{RR}}, \quad (134)$$

with the **Poisson boundary/Abel remainder**

$$D_{2,n}^{\text{RR}} := -\frac{1}{\sqrt{n}} \left[Q_1^{\text{RR}} \hat{\epsilon}(Z_1) + \sum_{l=1}^{n-2} (Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}) Q\hat{\epsilon}(Z_l) \right]. \quad (135)$$

(The right boundary $Q_{n-1}^{\text{RR}} Q\hat{\epsilon}(Z_{n-1})$ vanishes identically because $Q_{n-1}^{\text{RR}} = 0$.) Moreover, with $C_Q := \|\bar{A}^{-1}\| + 3C_Q$ a uniform bound on $\|Q_l^{\text{RR}}\|$, and C_2 the constant from the Corollary of Section 4.4,

$$\|D_{2,n}^{\text{RR}}\|_\infty \leq \frac{3 t_{\text{mix}} \|\epsilon\|_\infty}{\sqrt{n}} \left(C_Q + \frac{C_2}{a^2} \right). \quad (136)$$

Consequently, for every $p \geq 1$,

$$\|D_{2,n}^{\text{RR}}\|_{L_p} \leq \frac{C t_{\text{mix}} \|\epsilon\|_\infty}{a^2 \sqrt{n}}, \quad (137)$$

with a constant C depending only on $\|\bar{A}^{-1}\|$, κ_Q , and $\|\bar{A}\|$.

Proof. The increments ΔM_l^{RR} are $\mathcal{F}_l$ -martingale differences because Q_l^{RR} is deterministic and the Markov property gives $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$.

For the decomposition, substitute Eq. 131 into the definition in Eq. 102:

$$W^{\text{RR}} = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l^{\text{RR}} (\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_l)). \quad (138)$$

Peel off $l = 1$ and apply the conditional-centering identity above to the remaining $l \in \{2, \dots, n-1\}$:

$$W^{\text{RR}} = -\frac{1}{\sqrt{n}} Q_1^{\text{RR}} \hat{\epsilon}(Z_1) + \frac{1}{\sqrt{n}} Q_1^{\text{RR}} Q\hat{\epsilon}(Z_1) - \frac{1}{\sqrt{n}} M_n^{\text{RR}} - \frac{1}{\sqrt{n}} \sum_{l=2}^{n-1} Q_l^{\text{RR}} (Q\hat{\epsilon}(Z_{l-1}) - Q\hat{\epsilon}(Z_l)). \quad (139)$$

Set $g_l := Q\hat{\epsilon}(Z_l)$ and Abel-sum the last sum:

$$\sum_{l=2}^{n-1} Q_l^{\text{RR}} (g_{l-1} - g_l) = Q_2^{\text{RR}} g_1 - Q_{n-1}^{\text{RR}} g_{n-1} + \sum_{l=2}^{n-2} (Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}) g_l. \quad (140)$$

Adding the contribution $Q_1^{\text{RR}} g_1$ from the peeled $l = 1$ term combines with $-Q_2^{\text{RR}} g_1$ to yield $-(Q_2^{\text{RR}} - Q_1^{\text{RR}}) g_1$, which folds the boundary $l = 1$ summand back into the discrete-derivative sum:

$$\begin{aligned} & -Q_1^{\text{RR}} \hat{\epsilon}(Z_1) + Q_1^{\text{RR}} g_1 - Q_2^{\text{RR}} g_1 + Q_{n-1}^{\text{RR}} g_{n-1} - \sum_{l=2}^{n-2} (Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}) g_l \\ &= -Q_1^{\text{RR}} \hat{\epsilon}(Z_1) - \sum_{l=1}^{n-2} (Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}) g_l + Q_{n-1}^{\text{RR}} g_{n-1}. \end{aligned} \quad (141)$$

The rightmost boundary $Q_{n-1}^{\text{RR}} g_{n-1}$ vanishes because $Q_{n-1}^{\text{RR}} = 2\alpha I - 2\alpha I = 0$ identically (from $Q_{n-1}^{(\alpha)} = \alpha I$ and $Q_{n-1}^{(2\alpha)} = 2\alpha I$). Multiplying by $1/\sqrt{n}$ and absorbing the sign into $D_{2,n}^{\text{RR}}$ produces exactly the stated formula.

For the sup-norm bound Eq. 136, use $\|g_l\|_\infty \leq \|\hat{\epsilon}\|_\infty \leq 3t_{\text{mix}} \|\epsilon\|_\infty$, the uniform bound $\|Q_1^{\text{RR}}\| \leq C_Q$ on the left boundary, and the summed-total-variation bound $\sum_{l=1}^{n-2} \|Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}\| \leq C_2/a^2$ on the Abel sum:

$$\|\sqrt{n} D_{2,n}^{\text{RR}}\| \leq 3t_{\text{mix}} \|\epsilon\|_\infty (C_Q + C_2/a^2). \quad (142)$$

The L_p bound is immediate from the deterministic sup-norm bound. $\square$

4.8. Predictable Quadratic Variation Concentration

The Berry–Esseen step for the martingale M_n^{RR} requires a quantitative control of its predictable quadratic variation $\langle M^{\text{RR}} \rangle_n$ around its deterministic asymptotic counterpart $n\sigma_n^{2,\text{RR}}(u)$. This subsection produces such a control by applying the imported Markov concentration form Eq. 397 to a suitable centered quadratic functional of the chain. The result is the RR analogue of Lemmas 22–23 of Samsonov et al. (2025).

Predictable quadratic variation. By the Poisson Martingale Approximation lemma above, the increments $\Delta M_l^{\text{RR}} = Q_l^{\text{RR}} (\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}))$ are $\mathcal{F}_l$ -martingale differences. The Markov property $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$ gives, by direct computation,

$$\begin{aligned} \mathbb{E}[\Delta M_l^{\text{RR}} (\Delta M_l^{\text{RR}})^\top \mid \mathcal{F}_{l-1}] &= Q_l^{\text{RR}} \mathbb{E}[\hat{\epsilon}(Z_l) \hat{\epsilon}(Z_l)^\top \mid \mathcal{F}_{l-1}] (Q_l^{\text{RR}})^\top \\ &\quad - Q_l^{\text{RR}} Q\hat{\epsilon}(Z_{l-1}) (Q\hat{\epsilon}(Z_{l-1}))^\top (Q_l^{\text{RR}})^\top \\ &= Q_l^{\text{RR}} \mathcal{V}_\epsilon(Z_{l-1}) (Q_l^{\text{RR}})^\top, \end{aligned} \quad (143)$$

where the cross-terms in the expansion of $(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}))(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}))^\top$ cancel by the Markov property and we have set

$$\mathcal{V}_\epsilon(z) := Q(\hat{\epsilon}\hat{\epsilon}^\top)(z) - (Q\hat{\epsilon})(z)(Q\hat{\epsilon})(z)^\top. \quad (144)$$

This is a matrix-valued conditional covariance of the Poisson martingale increment; the vector noise remains denoted by ϵ .

Lemma 8. (Poisson covariance identity and absolute convergence.) Assume **UGE 1**, $\pi(\epsilon) = 0$, and $\|\epsilon\|_\infty < \infty$. Then the long-run covariance series defining $\Sigma_\epsilon^{(M)}$ is absolutely convergent in operator norm, and

$$\pi(\mathcal{V}_\epsilon) = \Sigma_\epsilon^{(M)} =: \Sigma. \quad (145)$$

Proof. First, boundedness and UGE make the covariance series absolutely convergent. For $j \geq 1$,

$$\|\mathbb{E}_\pi[\epsilon(Z_0)\epsilon(Z_j)^\top]\| = \|\pi(\epsilon(Q^j\epsilon)^\top)\| \leq \|\epsilon\|_\infty \|Q^j\epsilon\|_\infty \leq 2\|\epsilon\|_\infty^2 (1/4)^{\lfloor j/t_{\text{mix}} \rfloor}, \quad (146)$$

and the same estimate applies to $\mathbb{E}_\pi[\epsilon(Z_j)\epsilon(Z_0)^\top]$. Hence $\Sigma_\epsilon^{(M)}$ is absolutely convergent in operator norm.

Set $h := \hat{\epsilon}$ and $r := Qh = h - \epsilon$. Since $\pi Q = \pi$,

$$\begin{aligned} \pi(\mathcal{V}_\epsilon) &= \pi(hh^\top) - \pi(rr^\top) \\ &= \pi(\epsilon\epsilon^\top) + \pi(\epsilon r^\top) + \pi(r\epsilon^\top). \end{aligned} \quad (147)$$

The Poisson series gives $r = \sum_{j \geq 1} Q^j \epsilon$ in sup-norm, so the absolute convergence above justifies termwise integration:

$$\begin{aligned} \pi(\epsilon r^\top) &= \sum_{j \geq 1} \mathbb{E}_\pi[\epsilon(Z_0)\epsilon(Z_j)^\top], \\ \pi(r\epsilon^\top) &= \sum_{j \geq 1} \mathbb{E}_\pi[\epsilon(Z_j)\epsilon(Z_0)^\top]. \end{aligned} \quad (148)$$

Substituting these two series proves Eq. 145. This is the Poisson-solution covariance identity used by Samsonov et al. (2025, Eq. (10)) and by the Markov-chain CLT formulation of Douc–Moulines–Priouret–Soulier (2018, Theorem 21.2.5). $\square$

Summing over $l \in \{2, \dots, n-1\}$ and tracking that $\langle M^{\text{RR}} \rangle_n := \sum_{l=2}^{n-1} \mathbb{E}[\Delta M_l^{\text{RR}} (\Delta M_l^{\text{RR}})^\top \mid \mathcal{F}_{l-1}]$,

$$\langle M^{\text{RR}} \rangle_n = \sum_{l=2}^{n-1} Q_l^{\text{RR}} \mathcal{V}_\epsilon(Z_{l-1}) (Q_l^{\text{RR}})^\top. \quad (149)$$

By Lemma 8, $\pi(\mathcal{V}_\epsilon) = \Sigma$.

Sup-norm bound on $\mathcal{V}_\epsilon$. Since $\|\hat{\epsilon}\|_\infty \leq 3t_{\text{mix}}\|\epsilon\|_\infty$ and Q is a Markov kernel (so Qf inherits the sup-norm of f),

$$\begin{aligned} \|\mathcal{V}_\epsilon\|_\infty &\leq \|Q(\hat{\epsilon}\hat{\epsilon}^\top)\|_\infty + \|(Q\hat{\epsilon})(Q\hat{\epsilon})^\top\|_\infty \\ &\leq 2\|\hat{\epsilon}\|_\infty^2 \leq 18t_{\text{mix}}^2\|\epsilon\|_\infty^2. \end{aligned} \quad (150)$$

Lemma 9. (Stationary predictable-variation concentration.) Assume **UGE 1** and $\pi(\epsilon) = 0$, with $\|\epsilon\|_\infty < \infty$. Let C_Q be the uniform bound on $\|Q_l^{\text{RR}}\|$ from the previous lemma. There exists a universal constant $C_4 > 0$ such that, for every $u \in \mathbb{R}^d$, every $p \geq 2$, every initial distribution ξ , and every $n \geq 2$,

$$\mathbb{E}_\xi^{1/p}[\|u^\top \langle M^{\text{RR}} \rangle_n u - n\sigma_n^2(u)\|^p] \leq C_4 C_Q^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2} \sqrt{pn}. \quad (151)$$

Proof. Write $h_{l(z)} := u^\top Q_l^{\text{RR}} \mathcal{V}_\epsilon(z) (Q_l^{\text{RR}})^\top u$ and $g_{l(z)} := h_{l(z)} - \pi(h_l)$. Submultiplicativity of the operator norm and Eq. 150 gives

$$|h_{l(z)}| \leq C_Q^2 \|u\|^2 \|\mathcal{V}_\epsilon\|_\infty \leq 18 C_Q^2 \|u\|^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2, \quad \|g_l\|_\infty \leq 2 |h_{l(z)}| \leq 36 C_Q^2 \|u\|^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2. \quad (152)$$

By Eq. 149 and the variance definition $n \sigma_n^{2, \text{RR}}(u) = \sum_{l=2}^{n-1} \pi(h_l)$,

$$u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma_n^{2, \text{RR}}(u) = \sum_{l=2}^{n-1} g_{l(Z_{l-1})}. \quad (153)$$

For the centered sum, set $\tilde{g}_i := g_{i+1}$ for $i \in \{1, \dots, n-2\}$. By construction $\pi(\tilde{g}_i) = 0$ and $\|\tilde{g}_i\|_\infty \leq c := 36 C_Q^2 \|u\|^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2$. Apply the Markov concentration input Eq. 397 to $\sum_{i=1}^{n-2} \tilde{g}_{i(Z_i)}$ with $c_i = c$. For every $p \geq 2$,

$$\mathbb{E}_\xi^{1/p} \left[\left| \sum_{l=2}^{n-1} g_{l(Z_{l-1})} \right|^p \right] \leq C_{\text{MC}} \sqrt{p t_{\text{mix}} (n-2) c^2}. \quad (154)$$

Substituting c and bounding $\sqrt{n-2} \leq \sqrt{n}$,

$$\mathbb{E}_\xi^{1/p} \left[\left| \sum_{l=2}^{n-1} g_{l(Z_{l-1})} \right|^p \right] \leq C_{\text{MC}} \cdot 36 C_Q^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2} \sqrt{p n}. \quad (155)$$

Absorb the universal prefactor into C_4 to get the stated bound. $\square$

Corollary 2. (Stationary asymptotic-variance bracket comparison.) Under the assumptions of the previous lemma, for every $u \in \mathbb{R}^d$, every $p \geq 2$, and every $n \geq 2$,

$$\mathbb{E}_\xi^{1/p} [|u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma^2(u)|^p] \leq C_4 C_Q^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2} \sqrt{p n} + \frac{C_3 \|u\|^2}{\alpha a}, \quad (156)$$

with C_3 the variance-comparison constant of Section 4.5.

Proof. The triangle inequality and the variance-comparison lemma give

$$|u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma^2(u)| \leq |u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma_n^{2, \text{RR}}(u)| + n |\sigma_n^{2, \text{RR}}(u) - \sigma^2(u)|. \quad (157)$$

The first piece is bounded in L_p by Eq. 151; the second is the deterministic bound $n C_3 \|u\|^2 / (n \alpha a) = C_3 \|u\|^2 / (\alpha a)$. $\square$

4.9. Martingale Berry–Esseen Step

The Poisson decomposition (Section 4.6) writes $W^{\text{RR}} = -n^{-1/2} M_n^{\text{RR}} + D_{2,n}^{\text{RR}}$, with the remainder $D_{2,n}^{\text{RR}}$ controlled in sup-norm at the order $n^{-1/2}$. The quadratic variation concentration (Section 4.7) bounds $|u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma_n^{2, \text{RR}}(u)|$ in L_p by applying the Markov concentration form $C \sqrt{p t_{\text{mix}} \sum c_i^2}$, which after the bracket coefficient bound becomes $B(u) \sqrt{p n}$. This subsection assembles those two inputs into a Berry–Esseen rate for the **scalar** martingale $u^\top M_n^{\text{RR}}$, the leading contribution to the final RR Berry–Esseen bound.

Bounded increments. Fix $u \in \mathbb{R}^d$ and set $X_l := u^\top \Delta M_l^{\text{RR}}$ for $2 \leq l \leq n-1$. The X_l are $\mathcal{F}_l$ -martingale differences (Section 4.6), and bounding the three factors $\|u\|$, $\|Q_l^{\text{RR}}\| \leq C_Q$, and $\|\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})\| \leq 2 \|\hat{\epsilon}\|_\infty \leq 6 t_{\text{mix}} \|\epsilon\|_\infty$ yields the deterministic bound

$$|X_l| \leq \kappa(u), \quad \kappa(u) := 6 t_{\text{mix}} C_Q \|\epsilon\|_\infty \|u\|. \quad (158)$$

The conditional variances $\sigma_l^2 := \mathbb{E}[X_l^2 \mid \mathcal{F}_{l-1}]$ sum to the predictable quadratic variation along u ,

$$V_n^2 := \sum_{l=2}^{n-1} \sigma_l^2 = u^\top \langle M^{\text{RR}} \rangle_n u. \quad (159)$$

Variance lower bound. Set $s_n^2 := n \sigma_n^{2, \text{RR}}(u)$. The variance comparison lemma (Section 4.5) and assumption $\sigma^2(u) > 0$ give $|\sigma_n^{2, \text{RR}}(u) - \sigma^2(u)| \leq C_3 \|u\|^2 / (n \alpha a)$, so whenever

$$n \alpha a \geq \frac{2 C_3 \|u\|^2}{\sigma^2(u)} \quad (160)$$

one has $\sigma_n^{2, \text{RR}}(u) \geq \sigma^2(u)/2$, i.e. $s_n^2 \geq n\sigma^2(u)/2$. At the working scale $\alpha = c n^{-1/2}$, the variance lower-bound condition Eq. 160 is satisfied for $n \geq (2C_3 \|u\|^2 / (c a \sigma^2(u)))^2$. The trivial upper bound $\sigma_n^{2, \text{RR}}(u) \leq C_Q^2 \|\Sigma_\epsilon^{(M)}\| \|u\|^2$ gives $s_n^2 \leq K^2(u) n$ with $K(u) := C_Q \|\Sigma_\epsilon^{(M)}\|^{1/2} \|u\|$.

Bolthausen–Fan inequality. We apply the martingale Berry–Esseen input Eq. 398 with $N = n$ and bounded-increment parameter $\varkappa(u)$.

Theorem 1. (Stationary martingale Berry–Esseen bound.) Assume **UGE 1**, $\pi(\epsilon) = 0$, $\|\epsilon\|_\infty < \infty$, $\sigma^2(u) > 0$, $0 < \alpha$, and $2\alpha \leq \alpha_\infty$. There exist constants $C_{K,1}(u), C_{K,2}(u) > 0$ depending only on $\|u\|, \sigma(u), C_Q, t_{\text{mix}}, \|\epsilon\|_\infty, \|\Sigma_\epsilon^{(M)}\|$, and the constants $L_{B(\varkappa(u))}, C_1, C_2$ of Eq. 398, such that for every $n \geq 3$ satisfying the variance lower-bound condition Eq. 160,

$$d_K \left(\frac{u^\top M_n^{\text{RR}}}{\sqrt{n} \sigma_n^{\text{RR}}(u)}, \mathcal{N}(0, 1) \right) \leq \frac{C_{K,1}(u) \log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u) \log n}{\sqrt{n}}. \quad (161)$$

Proof. Apply Eq. 398 with $S_n = u^\top M_n^{\text{RR}}$, the increment bound $\varkappa = \varkappa(u)$ from Eq. 158, $s_n^2 = n \sigma_n^{2, \text{RR}}(u)$, and $p = \lceil \log n \rceil$ (so $\log n \leq p \leq \log n + 1$). Bound the three terms in turn, using $s_n^2 \in [n \sigma^2(u)/2, K^2(u) n]$ from the variance lower and upper bounds.

Term I (classical Bolthausen). From $s_n^3 \geq (n \sigma^2(u)/2)^{3/2}$ and $(2n+1) \log(2n+1) \leq 3 n \log n$ for $n \geq 3$,

$$L_{B(\varkappa(u))} \frac{(2n+1) \log(2n+1)}{s_n^3} \leq \frac{6\sqrt{2} L_{B(\varkappa(u))} \log n}{\sigma^3(u) \sqrt{n}} =: \frac{C^{(\text{I})}(u) \log n}{\sqrt{n}}. \quad (162)$$

Term III (Lindeberg). Write $a_p := 2p/(2p+1) = 1 - 1/(2p+1)$. Using $s_n^{-a_p} = s_n^{-1} s_n^{1/(2p+1)}$ and $s_n \leq K(u) \sqrt{n}$,

$$s_n^{1/(2p+1)} \leq \max(1, K(u))^{1/(2p+1)} n^{1/(2(2p+1))}. \quad (163)$$

For $p \geq \log n$ one has $n^{1/(2(2p+1))} \leq n^{1/(4 \log n)} = e^{1/4}$; similarly $\max(1, K(u))^{1/(2p+1)} \leq e^{1/2}$ once $n \geq \max(1, K(u))$, which is absorbed into $C^{(\text{III})}(u)$. Bounding $\varkappa(u)^{a_p} \leq \max(1, \varkappa(u))$ and $s_n^{-1} \leq \sqrt{2}/(\sigma(u) \sqrt{n})$,

$$C_2 s_n^{-a_p} p \varkappa(u)^{a_p} \leq \sqrt{2} e^{3/4} \frac{C_2 \max(1, \varkappa(u))}{\sigma(u)} \frac{p}{\sqrt{n}} \leq \frac{C^{(\text{III})}(u) \log n}{\sqrt{n}}, \quad (164)$$

with $C^{(\text{III})}(u)$ depending on $\|u\|, \sigma(u), C_Q, \|\Sigma_\epsilon^{(M)}\|, t_{\text{mix}}, \|\epsilon\|_\infty$ and on C_2 .

Term II (conditional-variance concentration). By the bracket concentration bound above, for every $p \geq 2$,

$$(\mathbb{E} | V_n^2 - s_n^2 |^p)^{1/p} \leq B(u) \sqrt{p n}, \quad B(u) := C_4 C_Q^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2}, \quad (165)$$

Using $s_n^{-a_p} = s_n^{-1} s_n^{1/(2p+1)}$, the variance lower bound $s_n^2 \geq n \sigma^2(u)/2$, and the upper bound $s_n \leq K(u) \sqrt{n}$, the second Bolthausen–Fan term is bounded by

$$\begin{aligned} & C_1 \sqrt{p} s_n^{-a_p} (\mathbb{E} | V_n^2 - s_n^2 |^p)^{1/(2p+1)} \\ & \leq C C_1 \sigma^{-1}(u) \max(1, K(u))^{1/(2p+1)} \max(1, B(u))^{p/(2p+1)} \\ & \quad \times p^{(3p+1)/(2(2p+1))} n^{-p/(2(2p+1))}. \end{aligned} \quad (166)$$

For $p = \lceil \log n \rceil$ and n large enough depending on u , $p^{(3p+1)/(2(2p+1))} \leq C \log^{3/4} n$ and $n^{-p/(2(2p+1))} \leq C n^{-1/4}$; the remaining $K(u)$ - and $B(u)$ -factors are absorbed into the direction-dependent constant. Thus

$$\text{Term II} \leq \frac{C^{(\text{II})}(u) \log^{3/4} n}{n^{1/4}}. \quad (167)$$

Adding the three term bounds and setting $C_{K,1}(u) := C^{(\text{II})}(u)$, $C_{K,2}(u) := C^{(\text{I})}(u) + C^{(\text{III})}(u)$ proves the martingale Berry–Esseen bound. $\square$

Corollary 3. (Stationary martingale asymptotic-normalization bound.) Under the hypotheses of the previous theorem,

$$d_K\left(\frac{u^\top M_n^{\text{RR}}}{\sqrt{n} \sigma(u)}, \mathcal{N}(0, 1)\right) \leq \frac{C_{K,1}(u) \log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u) \log n}{\sqrt{n}} + \frac{C_3 \|u\|^2}{n \alpha a \sigma^2(u)}. \quad (168)$$

At $\alpha = c n^{-1/2}$ the last term is $O(n^{-1/2})$, hence absorbed into $C_{K,2}(u) \log n / \sqrt{n}$ up to a constant.

Proof. Set $r := \sigma_n^{\text{RR}}(u) / \sigma(u)$ and $W := u^\top M_n^{\text{RR}} / (\sqrt{n} \sigma_n^{\text{RR}}(u))$. Then the statistic normalised by $\sigma(u)$ is Wr . Under the variance lower-bound condition Eq. 160, $r \geq 1/\sqrt{2}$, while the trivial upper bound on $\sigma_n^2, \text{RR}(u)$ gives $r \leq r_{\max}(u) < \infty$. On this compact interval the standard normal cdf satisfies

$$\sup_x |\Phi(x/r) - \Phi(x)| \leq C_\Phi |r - 1|, \quad C_\Phi := \sqrt{2}/\sqrt{\pi e}. \quad (169)$$

Therefore

$$d_{K(Wr, \mathcal{N}(0,1))} \leq d_{K(W, \mathcal{N}(0,1))} + C_\Phi |r - 1|. \quad (170)$$

The variance comparison of Section 4.5 gives

$$|r - 1| = \frac{|\sigma_n^2, \text{RR}(u) - \sigma^2(u)|}{\sigma(u) (\sigma_n^{\text{RR}}(u) + \sigma(u))} \leq \frac{C_3 \|u\|^2}{n \alpha a \sigma^2(u)}. \quad (171)$$

Adding this perturbation to the martingale Berry–Esseen bound proves the stated claim after absorbing C_Φ into the constants. $\square$

This theorem concerns only the martingale term; the stationary assembly below adds the Poisson remainder and the Levin depth-two misadjustment.

4.10. Misadjustment via Levin Depth-Two

The Berry–Esseen assembly so far controls the leading martingale piece (Section 4.8) and the Poisson boundary remainder $D_{2,n}^{\text{RR}}$ (Section 4.6). The remaining non-martingale contribution to $\sqrt{n} u^\top (\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^*)$ is the **PR-averaged misadjustment**

$$R_n^{\text{mis, RR}} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} (2R_k^{(\alpha)} - R_k^{(2\alpha)}), \quad (172)$$

where $R_k^{(\alpha)} := J_k^{(1, \alpha)} + H_k^{(1, \alpha)}$ is the depth-one remainder of Eq. 94. A direct kernel-difference route only bounds $R_n^{\text{mis, RR}}$ at order $O(\sqrt{n} \alpha) = O(1)$ at the working scale $\alpha \asymp n^{-1/2}$ — too crude for a Berry–Esseen rate $n^{-1/4}$. The depth-two route below uses the imported working forms Lemma 29, Lemma 30, Lemma 31, and Lemma 32 to recover the target $n^{-1/4} \text{polylog}(n)$ rate.

Depth-two refinement. The deterministic-product expansion from the depth-two setup extends the depth-one decomposition by one more level. For $\ell \geq 1$ define

$$J_n^{(\ell, \alpha)} := (I - \alpha \bar{A}) J_{n-1}^{(\ell, \alpha)} - \alpha \tilde{A}(Z_n) J_{n-1}^{(\ell-1, \alpha)}, \quad (173)$$

$$H_n^{(\ell, \alpha)} := (I - \alpha A(Z_n)) H_{n-1}^{(\ell, \alpha)} - \alpha \tilde{A}(Z_n) J_{n-1}^{(\ell, \alpha)}, \quad (174)$$

with $J_0^{(\ell, \alpha)} = H_0^{(\ell, \alpha)} = 0$ and the $\ell = 0$ processes as in Section 3. Substituting $A(Z_n) = \bar{A} + \tilde{A}(Z_n)$ and grouping terms gives, by induction on n , the recursive identity

$$H_n^{(\ell, \alpha)} = J_n^{(\ell+1, \alpha)} + H_n^{(\ell+1, \alpha)}, \quad \ell \geq 0. \quad (175)$$

Applying Eq. 175 with $\ell = 1$ refines $R_k^{(\alpha)}$ to

$$R_k^{(\alpha)} = J_k^{(1, \alpha)} + J_k^{(2, \alpha)} + H_k^{(2, \alpha)}, \quad (176)$$

which splits the misadjustment into three structurally different pieces:

$$R_n^{\text{mis, RR}} = T_n^{(1)} + T_n^{(2)} + T_n^{(H)}, \quad (177)$$

$$T_n^{(j)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \left(2J_k^{(j, \alpha)} - J_k^{(j, 2\alpha)} \right), \quad j \in \{1, 2\}, \quad (178)$$

with $T_n^{(H)}$ defined identically with $H^{(2)}$ in place of $J^{(j)}$. Each piece is bounded separately below. The direct Levin working forms used in this section are collected in Section 7.1: Lemma 29, Lemma 30, Lemma 31, and Lemma 32.

Lemma 10. (Finite-past construction of the stationary full depth-two augmented state.) Fix $q \geq 2$, $2 \leq p \leq q/2$, and an admissible step size $0 < w \leq \alpha_{\text{stat}}(q)$. Let $(Z_t)_{t \in \mathbb{Z}}$ be a two-sided stationary version of the driving Markov chain, and write $B_w := I - w\bar{A}$. For $m \geq 1$ define the finite-past truncation by starting at time $-m$ from

$$J_{-m,m}^{(0,w)} = J_{-m,m}^{(1,w)} = J_{-m,m}^{(2,w)} = H_{-m,m}^{(2,w)} = 0 \quad (179)$$

and evolving, for $r = -m + 1, \dots, 0$,

$$J_{r,m}^{(0,w)} = B_w J_{r-1,m}^{(0,w)} - w\epsilon(Z_r), \quad (180)$$

$$J_{r,m}^{(\ell,w)} = B_w J_{r-1,m}^{(\ell,w)} - w\tilde{A}(Z_r) J_{r-1,m}^{(\ell-1,w)}, \quad \ell \in \{1, 2\}, \quad (181)$$

$$H_{r,m}^{(2,w)} = (I - wA(Z_r)) H_{r-1,m}^{(2,w)} - w\tilde{A}(Z_r) J_{r-1,m}^{(2,w)}. \quad (182)$$

Then the vector

$$\left(J_{0,m}^{(0,w)}, J_{0,m}^{(1,w)}, J_{0,m}^{(2,w)}, H_{0,m}^{(2,w)} \right) \quad (183)$$

is Cauchy in L_p . More precisely, for $m' \geq m \geq 1$,

$$\| H_{0,m'}^{(2,w)} - H_{0,m}^{(2,w)} \|_{L_p} \leq A_{\text{fp}}(p, q, w) \exp(-c_{\text{fp}} w a m / p), \quad (184)$$

where

$$A_{\text{fp}}(p, q, w) := C_{\text{fp}}(1 + d^{1/q}) p^8 t_{\text{mix}}^5 \frac{1}{a} \sqrt{w/a} \log^3(1/(wa)), \quad (185)$$

and $C_{\text{fp}}, c_{\text{fp}} > 0$ depend only on fixed problem constants and the Levin constants. The J coordinates converge to the invariant law $\Pi_{J,2,w}$ of Lemma 34, and the L_p limit

$$H_0^{(2,w)} := \lim_{m \rightarrow \infty} H_{0,m}^{(2,w)} \quad (186)$$

satisfies the one-trajectory bound of Lemma 32. Shifting the construction in time defines a two-sided stationary process

$$\left(Z_{k+1}, J_k^{(0,w)}, J_k^{(1,w)}, J_k^{(2,w)}, H_k^{(2,w)} \right)_{k \in \mathbb{Z}} \quad (187)$$

which solves the full depth-two recursions and whose law is the stationary full augmented-chain law used below.

Proof. The J -coordinate construction is Lemma 34 together with the component contraction Lemma 33 for $Y_k^{(w)} = \left(Z_{k+1}, J_k^{(0,w)}, J_k^{(1,w)}, J_k^{(2,w)} \right)$. On the same two-sided driving chain, the finite-past truncations started at $-m'$ and $-m$ agree in the driving coordinate after time $-m$, while their J -coordinates at time $-m$ differ by the older finite-past state. The componentwise contraction Eq. 408 therefore gives, for $r \geq -m$ and $\ell \in \{0, 1, 2\}$,

$$\| J_{r,m'}^{(\ell,w)} - J_{r,m}^{(\ell,w)} \|_{L_p} \leq A_{J(p,q,w)} \exp(-c w a (r + m) / p), \quad (188)$$

with

$$A_{J(p,q,w)} := C(1 + d^{1/q}) p^7 t_{\text{mix}}^5 \sqrt{w/a} \log^3(1/(wa)). \quad (189)$$

Here the initial finite-past cost is controlled by the elementary $J^{(0,w)}$, $J^{(1,w)}$ moment bounds and Lemma 31 for $J^{(2,w)}$; the constants are uniform in m, m' .

It remains to justify the extra $H^{(2,w)}$ coordinate. Subtract the two finite-past versions of the representation Eq. 405 from time $-m$ to time 0. With $\Delta H_r := H_{r,m'}^{(2,w)} - H_{r,m}^{(2,w)}$ and similarly for $\Delta J_r^{(2,w)}$,

$$\Delta H_0 = \Gamma_{-m+1:0}^{(w)} H_{-m,m'}^{(2,w)} - w \sum_{l=-m+1}^0 \Gamma_{l+1:0}^{(w)} \tilde{A}(Z_l) \Delta J_{l-1}^{(2,w)}. \quad (190)$$

The product-stability working form Lemma 17, applied at the deterministic time $-m$, and the one-trajectory bound Lemma 32 give

$$\| \Gamma_{-m+1:0}^{(w)} H_{-m,m'}^{(2,w)} \|_{L_p} \leq C B_{H(p,q,w)} \exp(-cwam/p), \quad (191)$$

where

$$B_{H(p,q,w)} := (1 + d^{1/q}) p^{7/2} t_{\text{mix}}^{5/2} w^{3/2} \log^{3/2}(1/(wa)). \quad (192)$$

For the convolution term, combine product stability with Eq. 188:

$$w \sum_{l=-m+1}^0 \| \Gamma_{l+1:0}^{(w)} \tilde{A}(Z_l) \Delta J_{l-1}^{(2,w)} \|_{L_p} \leq Cw \sum_{l=-m+1}^0 e^{-cwa(-l)/p} A_{J(p,q,w)} e^{-cwa(l+m)/p}. \quad (193)$$

The last convolution is bounded by $C(p/a) A_{J(p,q,w)} \exp(-cwam/(2p))$. Enlarging the constant and decreasing the exponential rate gives Eq. 184. This proves the Cauchy property for the full vector.

The limits are measurable functions of the two-sided stationary driving chain. Passing to the limit in the finite-past recursions is justified in L_p , so the limiting process solves the same recursions. Stationarity follows because the finite-past construction is shift-covariant on the two-sided stationary chain. The uniform one-trajectory bound for the limiting $H^{(2,w)}$ follows from Lemma 32 and Fatou's lemma. $\square$

Joint RR stationary construction. Apply Lemma 10 simultaneously at $w = \alpha$ and $w = 2\alpha$ on the same two-sided stationary base chain $(Z_t)_{t \in \mathbb{Z}}$. The two finite-past systems are measurable functions of the same driving path, and the pair of full augmented vectors is Cauchy in the product L_p norm because each coordinate is Cauchy at its own step size. The joint limit

$$\left(Z_{k+1}, J_k^{(0,\alpha)}, J_k^{(1,\alpha)}, J_k^{(2,\alpha)}, H_k^{(2,\alpha)}, J_k^{(0,2\alpha)}, J_k^{(1,2\alpha)}, J_k^{(2,2\alpha)}, H_k^{(2,2\alpha)} \right)_{k \in \mathbb{Z}} \quad (194)$$

is therefore a stationary RR augmented state. All RR differences below, such as $2J_k^{(j,\alpha)} - J_k^{(j,2\alpha)}$ and $2H_k^{(2,\alpha)} - H_k^{(2,2\alpha)}$, are read under this joint construction.

Stationary augmented-chain convention. The recursions above are displayed in finite-time notation with $J_0^{(\ell,\alpha)} = H_0^{(\ell,\alpha)} = 0$, while Lemma 29 is a statement about the stationary augmented chain. The misadjustment estimates below use the stationary versions constructed in Lemma 10: $(Z_{t+1}, J_t^{(0,w)}, J_t^{(1,w)}, J_t^{(2,w)}, H_t^{(2,w)})$, $w \in \{\alpha, 2\alpha\}$, as deterministic centers and should be read under this stationary augmented-chain convention. The one-step shift is intentional: J_t is the perturbation state after the observation Z_{t+1} has updated the recursion, while covariance formulas for the base chain may still use Z_0 by stationarity. We keep the same symbols $J_t^{(\ell,w)}$ and $H_t^{(\ell,w)}$ to avoid duplicating notation. For a zero-start full average, the transfer from finite-start sums to stationary centered sums is not a single terminal ρ^n term: summing pointwise contractions over $k = 0, \dots, n-1$ produces a startup contribution of order

$$\frac{1}{\sqrt{n}} \sum_{k \geq 0} \rho_\alpha^k \asymp \frac{1}{\sqrt{n} \alpha a} \quad (195)$$

when $\rho_\alpha \approx \exp(-c\alpha a)$. This term is not part of the theorem proved here. A practical arbitrary-start theorem requires either a genuine burn-in average with the burned-in weights $Q_{l,n_0}^{(\alpha)}$ of Section 4.1 or a separate transfer lemma with the correct accumulated startup dependence.

For the shifted-to-unshifted first-order transfer we use the local inverse ceiling α_{inv} defined in Eq. 90.

Lemma 11. (Stationary-limit transfer for the centered first-order iterate.) Fix $w \in (0, \alpha_\infty]$ and set $B_w := I - w\bar{A}$. Assume the stationary augmented-chain convention above, **UGE 1**, $\pi(\tilde{A}) = 0$, and $\|\epsilon\|_\infty < \infty$. Let $(J_t^{(0,w)}, J_t^{(1,w)})_{t \in \mathbb{Z}}$ be the stationary two-sided solution of the first two perturbation recursions, and set $T_t^{(1,w)} := B_w J_t^{(1,w)}$. Then, with

$$\Phi_+(p, w) := 1 + p^{3/2} t_{\text{mix}}^{1/2}/a + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{w/a}, \quad (196)$$

there exists a constant $C_{\text{stat},1}$, depending only on the constants in the zero-start centered shifted first-order bound in Lemma 4, such that for every $p \geq 2$ and every $t \in \mathbb{Z}$,

$$\|T_t^{(1,w)} - \mathbb{E}_\pi T_t^{(1,w)}\|_{L_p} \leq C_{\text{stat},1} w \Phi_+(p, w). \quad (197)$$

Consequently, whenever $w \leq \alpha_{\text{inv}}$,

$$\|J_t^{(1,w)} - \mathbb{E}_\pi J_t^{(1,w)}\|_{L_p} \leq 2C_{\text{stat},1} w \Phi_+(p, w). \quad (198)$$

Proof. For $m \geq 1$, run the same recursions on the stationary driving chain over the finite window $\{t - m + 1, \dots, t\}$ with zero initial conditions at time $t - m$; denote the resulting variables by $J_{t,m}^{(0,w)}$ and $J_{t,m}^{(1,w)}$. Stationarity of the driving chain implies that $B_w J_{t,m}^{(1,w)}$ has the same law as the zero-start $T_m^{(1,w)}$ in Lemma 4. Therefore the centered last-iterate bound gives, uniformly in m and t ,

$$\|B_w J_{t,m}^{(1,w)} - \mathbb{E} B_w J_{t,m}^{(1,w)}\|_{L_p} \leq C_{\text{stat},1} w \Phi_+(p, w). \quad (199)$$

The deterministic-product expansions give

$$J_{t,m}^{(0,w)} = -w \sum_{r=0}^{m-1} B_w^r \epsilon(Z_{t-r}), \quad (200)$$

and express $J_{t,m}^{(1,w)}$ as a bounded double series whose summands are dominated by $Cw^2 s(1 - wa)^{s/2} \|\epsilon\|_\infty$ at total lag s . For the fixed admissible w in the lemma, $\sum_{s \geq 0} s(1 - wa)^{s/2} < \infty$, so the finite-past approximations are Cauchy in every L_p and converge to the stationary two-sided solution. This domination is not asserted uniformly as $w \rightarrow 0$; the later estimates retain the resulting step-size dependence through $\Phi_+(p, w)$. Passing to the limit in the preceding uniform bound yields the displayed estimate for $T_t^{(1,w)}$. The estimate for $J_t^{(1,w)}$ then follows from $J_t^{(1,w)} - \mathbb{E}_\pi J_t^{(1,w)} = B_w^{-1} (T_t^{(1,w)} - \mathbb{E}_\pi T_t^{(1,w)})$. $\square$

The lemma is therefore compatible with the triangular-array substitution $w = \alpha_n$: the finite-past domination is fixed- w , while every later bound keeps the displayed w -dependence through $\Phi_+(p, w)$ and uses constants independent of n .

Telescoping identity for $J^{(1)}$. Summing the recursion $J_k^{(1,\alpha)} = (I - \alpha\bar{A}) J_{k-1}^{(1,\alpha)} - \alpha\tilde{A}(Z_k) J_{k-1}^{(0,\alpha)}$ from $k = 1$ to n and rearranging gives, for an arbitrary initial value,

$$\bar{A} \sum_{k=0}^{n-1} J_k^{(1,\alpha)} = - \sum_{k=1}^n \tilde{A}(Z_k) J_{k-1}^{(0,\alpha)} + \frac{1}{\alpha} (J_0^{(1,\alpha)} - J_n^{(1,\alpha)}). \quad (201)$$

The stationary version of the same recursion yields $\mathbb{E}_\pi [\tilde{A}(Z_1) J_0^{(0,\alpha)}] = -\bar{A} \mathbb{E}_\pi [J_\infty^{(1,\alpha)}]$. Under the stationary augmented-chain convention, for every k the pair $(J_{k-1}^{(0,w)}, Z_k)$ has the same law as $(J_0^{(0,w)}, Z_1)$. Thus the centered function $\bar{\psi}_w$ in Lemma 30 is centered for each summand in the telescoping sum below; the index

shift is only notational. Subtracting $n \mathbb{E}_\pi [J_\infty^{(1,\alpha)}]$ from both sides and applying $\bar{A}^{-1}$ gives the **centered telescoping identity**

$$\sum_{k=0}^{n-1} \left(J_k^{(1,\alpha)} - \mathbb{E}_\pi [J_\infty^{(1,\alpha)}] \right) = -\bar{A}^{-1} \sum_{k=1}^n \bar{\psi}_\alpha \left(J_{k-1}^{(0,\alpha)}, Z_k \right) + \frac{1}{\alpha} \bar{A}^{-1} \left(J_0^{(1,\alpha)} - J_n^{(1,\alpha)} \right). \quad (202)$$

This identity is the bridge between the (vector-valued) PR-average of $J^{(1)}$ and the centered bilinear sum bounded in Lemma 30.

Lemma 12. (Stationary first-order misadjustment bound.) Assume the stationary augmented-chain convention above, **UGE 1**, $\pi(\bar{A}) = 0$, $\pi(\epsilon) = 0$, $\|\epsilon\|_\infty < \infty$, and $0 < \alpha$, $2\alpha \leq \alpha_\infty$, and $2\alpha \leq \alpha_{\text{inv}}$. Set

$$\Phi_+(p, \alpha) := 1 + p^{3/2} t_{\text{mix}}^{1/2}/a + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha/a}. \quad (203)$$

There exists a constant $C_{\text{mis},1}$ depending on $\|\bar{A}\|$, $\|\bar{A}^{-1}\|$, κ_Q , C_A , $\|\epsilon\|_\infty$, t_{mix} , the universals $c_{W,1}, c_{W,2}$ of Lemma 30, and the constant of the centered bound in Lemma 4, such that for every $p \geq 2$ and every $n \geq 2$,

$$\begin{aligned} \|T_n^{(1)}\|_{L_p} &\leq C_{\text{mis},1} \sqrt{n} \alpha^2 + C_{\text{mis},1} p^{3/2} \sqrt{\alpha} \\ &\quad + C_{\text{mis},1} p^3 (\alpha n)^{-1/2} \log^{1/2}(1/(\alpha a)) + C_{\text{mis},1} \Phi_+(p, \alpha) n^{-1/2}. \end{aligned} \quad (204)$$

Proof. Decompose $T_n^{(1)} = T_n^{(1,b)} + T_n^{(1,c)}$ via the bias-fluctuation split

$$T_n^{(1,b)} := \sqrt{n} \left(2 \mathbb{E}_\pi [J_\infty^{(1,\alpha)}] - \mathbb{E}_\pi [J_\infty^{(1,2\alpha)}] \right), \quad (205)$$

$$T_n^{(1,c)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \sum_{w \in \{\alpha, 2\alpha\}} c_w \left(J_k^{(1,w)} - \mathbb{E}_\pi [J_\infty^{(1,w)}] \right), \quad c_\alpha = 2, c_{2\alpha} = -1. \quad (206)$$

The bias is deterministic; the centered piece carries all the L_p fluctuation.

Bias. By Lemma 29 applied at $w \in \{\alpha, 2\alpha\}$, $\mathbb{E}_\pi [J_\infty^{(1,w)}] = w \Delta + R(w)$ with $\|R(w)\| \leq 12 \|\bar{A}^{-1}\| C_A^2 t_{\text{mix}}^2 w^2 \|\epsilon\|_\infty$. The leading $w \Delta$ cancels in the RR combination, $2\alpha \Delta - 2\alpha \Delta = 0$, leaving

$$\|2 \mathbb{E}_\pi [J_\infty^{(1,\alpha)}] - \mathbb{E}_\pi [J_\infty^{(1,2\alpha)}]\| \leq 2 \|R(\alpha)\| + \|R(2\alpha)\| \leq 72 \|\bar{A}^{-1}\| C_A^2 t_{\text{mix}}^2 \alpha^2 \|\epsilon\|_\infty. \quad (207)$$

Multiplying by $\sqrt{n}$ produces the first term of the bound.

Centered. Apply Eq. 202 at $w \in \{\alpha, 2\alpha\}$, take L_p norms, and combine via the triangle inequality:

$$\begin{aligned} \|T_n^{(1,c)}\|_{L_p} &\leq \frac{1}{\sqrt{n}} \sum_{w \in \{\alpha, 2\alpha\}} |c_w| \left(\|\bar{A}^{-1}\| \cdot \left\| \sum_{k=1}^n \bar{\psi}_w \left(J_{k-1}^{(0,w)}, Z_k \right) \right\|_{L_p} \right. \\ &\quad \left. + \frac{1}{w} \|\bar{A}^{-1}\| \cdot \|J_0^{(1,w)} - J_n^{(1,w)}\|_{L_p} \right). \end{aligned} \quad (208)$$

For the bilinear sum apply Lemma 30 with $r = n$,

$$\left\| \sum_{k=1}^n \bar{\psi}_w \right\|_{L_p} \leq c_{W,1} p^{3/2} \sqrt{wn} + c_{W,2} p^3 w^{-1/2} \log^{1/2}(1/(wa)), \quad (209)$$

and divide by $\sqrt{n}$, using $w \in [\alpha, 2\alpha]$ to upper-bound by the α -form,

$$\frac{1}{\sqrt{n}} \left\| \sum \bar{\psi}_w \right\|_{L_p} \leq \sqrt{2} c_{W,1} p^{3/2} \sqrt{\alpha} + \sqrt{2} c_{W,2} p^3 (\alpha n)^{-1/2} \log^{1/2}(1/(\alpha a)). \quad (210)$$

For the boundary term, apply the stationary-limit transfer Lemma 11 to $T_n^{(1,w)} = (I - w\bar{A}) J_n^{(1,w)}$ with $w \leq \alpha_{\text{inv}}$. This gives

$$\| J_n^{(1,w)} - \mathbb{E} J_n^{(1,w)} \|_{L_p} \leq Cw \Phi_+(p, w), \quad (211)$$

and $\| \mathbb{E} J_n^{(1,w)} \| \leq w \| \Delta \| + \| R(w) \| \leq Cw$ by Lemma 29; combining,

$$\| J_n^{(1,w)} \|_{L_p} \leq Cw \Phi_+(p, w). \quad (212)$$

The same bound holds for $J_0^{(1,w)}$ under stationarity, so $w^{-1} \| J_0^{(1,w)} - J_n^{(1,w)} \|_{L_p} \leq C \Phi_+(p, w) \leq \sqrt{2} C \Phi_+(p, \alpha)$ after absorbing the factor 2 into C . Dividing by $\sqrt{n}$ produces the last term. Summing the three RR-combined pieces and absorbing universal factors into a single $C_{\text{mis},1}$ completes the proof. $\square$

Lemma 13. (Stationary second-order and depth-two remainder bound.) Under the assumptions of the previous lemma, for every $p \geq 2$ and every $q \geq 2$ satisfying $p \leq q/2$ and $2\alpha \leq \alpha_{\text{stat}}(q)$,

$$\| T_n^{(2)} \|_{L_p} + \| T_n^{(H)} \|_{L_p} \leq C_{\text{mis},2} (1 + d^{1/q}) p^{7/2} t_{\text{mix}}^{5/2} \sqrt{n} \alpha^{3/2} \log^{3/2}(1/(\alpha a)), \quad (213)$$

$$\text{with } C_{\text{mis},2} := 6\sqrt{2}^3 (D_J + D_H).$$

Proof. Triangle inequality on the n summands of $T_n^{(2)}$, then Lemma 31 and Lemma 32 applied uniformly in k and $w \in \{\alpha, 2\alpha\}$:

$$\begin{aligned} \| T_n^{(2)} \|_{L_p} &\leq \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \sum_{w \in \{\alpha, 2\alpha\}} |c_w| \| J_k^{(2,w)} \|_{L_p} \\ &\leq 3\sqrt{n} \sup_{w \in \{\alpha, 2\alpha\}} \| J_k^{(2,w)} \|_{L_p} \\ &\leq 3D_J (2\alpha)^{3/2} t_{\text{mix}}^{5/2} p^{7/2} \sqrt{n} \log^{3/2}(1/(\alpha a)), \end{aligned} \quad (214)$$

where $|c_\alpha| + |c_{2\alpha}| = 3$ absorbs the RR combination and $(2\alpha)^{3/2}$ the worse bound at the larger step. Identical argument for $T_n^{(H)}$, with the additional $d^{1/q}$ factor of Lemma 32. Adding the two bounds gives the lemma. $\square$

Theorem 2. (Stationary PR-averaged RR misadjustment bound.) Assume **UGE 1**, $\pi(\tilde{A}) = 0$, $\pi(\epsilon) = 0$, $\| \epsilon \|_\infty < \infty$, $0 < \alpha$, and the stationary augmented-chain convention above. There exists a constant C depending on the universal and problem constants of the previous two lemmas such that for every $p \geq 2$, every $q \geq 2$ satisfying $p \leq q/2$ and $2\alpha \leq \alpha_{\text{stat}}(q)$, and every $n \geq 2$,

$$\begin{aligned} \| R_n^{\text{mis, RR}} \|_{L_p} &\leq C\sqrt{n} \alpha^2 + C(1 + d^{1/q}) p^{7/2} t_{\text{mix}}^{5/2} \sqrt{n} \alpha^{3/2} \log^{3/2}(1/(\alpha a)) \\ &\quad + C p^{3/2} \sqrt{\alpha} + C p^3 (\alpha n)^{-1/2} \log^{1/2}(1/(\alpha a)) + C \Phi_+(p, \alpha) n^{-1/2}. \end{aligned} \quad (215)$$

Proof. Triangle inequality on Eq. 177: $\| R_n^{\text{mis, RR}} \|_{L_p} \leq \| T_n^{(1)} \|_{L_p} + \| T_n^{(2)} \|_{L_p} + \| T_n^{(H)} \|_{L_p}$. Combine the centered $T^{(1)}$ bound and the raw $T^{(2)} + T^{(H)}$ bound. $\square$

Corollary 4. (Stationary balanced-scale misadjustment rate.) At the working scale $\alpha = c n^{-1/2}$, with $p = \max(2, \lceil \log n \rceil)$ and $q = \max(2p, \lceil \log(ed) \rceil, 2)$, and for n large enough that $2\alpha \leq \alpha_{\text{stat}}(q)$,

$$\| R_n^{\text{mis, RR}} \|_{L_p} \leq C \text{polylog}(n) n^{-1/4}, \quad (216)$$

of the same polynomial order as the leading martingale Berry–Esseen rate, up to logarithmic factors.

Proof. At $\alpha = c n^{-1/2}$: $\sqrt{n} \alpha^2 = c^2 n^{-1/2}$, $\sqrt{n} \alpha^{3/2} = c^{3/2} n^{-1/4}$, $\sqrt{\alpha} = c^{1/2} n^{-1/4}$, $(\alpha n)^{-1/2} = c^{-1/2} n^{-1/4}$, and $\Phi_+(p, \alpha) n^{-1/2} = O(p^{3/2} n^{-1/2})$. With $p \asymp \log n$ and $q \geq \log(ed)$, the factor $d^{1/q}$ is bounded by a universal constant, and the dominant order in the stationary misadjustment bound is $\text{polylog}(n) n^{-1/4}$. $\square$

The preceding misadjustment bound is a stationary augmented-chain statement at $n_0 = 0$. A finite-start theorem with burn-in requires the burned-in weights $Q_{l,n_0}^{(\alpha)}$ and is handled in the burn-in transfer chapter.

4.11. Smoothing Assembly

The previous sections produced the four ingredients of the stationary $n_0 = 0$ Berry–Esseen program for the PR-averaged Richardson–Romberg statistic:

1. the **Poisson decomposition** $W^{\text{RR}} = -n^{-1/2}M_n^{\text{RR}} + D_{2,n}^{\text{RR}}$ with deterministic sup-norm control of $D_{2,n}^{\text{RR}}$ at the order $t_{\text{mix}}/(a^2\sqrt{n})$;
2. the **predictable quadratic variation concentration** of $u^\top \langle M^{\text{RR}} \rangle_n u$ around $n\sigma_n^{2,\text{RR}}(u)$ in L_p , obtained from the Markov concentration scale $\sqrt{p t_{\text{mix}} \sum c_i^2}$ and then written as $B(u)\sqrt{p n}$;
3. the **martingale Berry–Esseen** for $u^\top M_n^{\text{RR}}/(\sqrt{n}\sigma_n^{\text{RR}}(u))$ at rate $\log^{3/4}(n)n^{-1/4}$;
4. the **Levin depth-two misadjustment bound** on $R_n^{\text{mis,RR}}$ controlling the non-martingale residual at the same $\log^{c(n)}n^{-1/4}$ rate.

The smoothing inequality of Bobkov–Götze (Samsonov et al. 2025, Proposition 12) assembles these four into a single Kolmogorov bound on the stationary $n_0 = 0$ augmented-chain RR statistic.

Stationary assembled statistic. Combining the depth-one identity in Section 4.1 with the Poisson decomposition of Section 4.6 gives a finite-start identity. Since the transfer from a zero-start recursion to the stationary augmented chain is not proved here, the bound below is stated for the stationary $n_0 = 0$ assembled scalar statistic

$$S_{n,\text{stat}}^{\text{RR}}(u) := -\frac{u^\top M_n^{\text{RR}}}{\sqrt{n}} + u^\top \mathcal{R}_{n,\text{stat}}^{\text{RR}}, \quad (217)$$

$$\mathcal{R}_{n,\text{stat}}^{\text{RR}} := D_{2,n}^{\text{RR}} + R_n^{\text{mis,RR}}, \quad (218)$$

where $D_{2,n}^{\text{RR}}$ is the Poisson boundary/Abel remainder of Section 4.6 and $R_n^{\text{mis,RR}}$ is the Levin depth-two misadjustment defined in Eq. 172, both read under the stationary augmented-chain convention. The deterministic-start transient $D_{\text{tr}}^{\text{RR}} := 2D_{\text{tr}}^{(\alpha)} - D_{\text{tr}}^{(2\alpha)}$ and the random initial-product discrepancy $2D_{\text{init}}^{(\alpha)} - D_{\text{init}}^{(2\alpha)}$ are not part of this stationary result; handling them together with the accumulated startup error belongs to the finite-start/burn-in transfer theorem. Dividing by $\sigma_n^{\text{RR}}(u)$,

$$\frac{S_{n,\text{stat}}^{\text{RR}}(u)}{\sigma_n^{\text{RR}}(u)} = X_n + Y_n, \quad (219)$$

$$X_n := -\frac{u^\top M_n^{\text{RR}}}{\sqrt{n}\sigma_n^{\text{RR}}(u)}, \quad Y_n := \frac{u^\top \mathcal{R}_{n,\text{stat}}^{\text{RR}}}{\sigma_n^{\text{RR}}(u)}. \quad (220)$$

The martingale Berry–Esseen bound controls $d_{K(X_n, \mathcal{N}(0,1))}$ by applying Theorem 1 to the signed direction $-u$. Equivalently, use the martingale differences $-u^\top \Delta M_l^{\text{RR}}$; their bounded-increment constant and predictable variance are the same as for u , because $\kappa(-u) = \kappa(u)$ and $\sigma_n^{2,\text{RR}}(-u) = \sigma_n^{2,\text{RR}}(u)$.

Smoothing inequality. For random variables X, Y on the same probability space and any $t > 0$,

$$d_K(X + Y, \mathcal{N}(0, 1)) \leq d_K(X, \mathcal{N}(0, 1)) + \mathbb{P}(|Y| > t) + \frac{t}{\sqrt{2\pi}}, \quad (221)$$

where the third term uses the $1/\sqrt{2\pi}$ Lipschitz constant of the standard normal cdf (Bobkov–Götze; see Samsonov et al. 2025, Proposition 12 for an LSA-tailored statement). Bounding $\mathbb{P}(|Y| > t) \leq \|Y\|_{L_p}^p / t^p$ via Markov’s inequality and choosing $t = e \|Y\|_{L_p}$ gives the **working form**

$$d_K(X + Y, \mathcal{N}(0, 1)) \leq d_K(X, \mathcal{N}(0, 1)) + \frac{e \|Y\|_{L_p}}{\sqrt{2\pi}} + e^{-p}, \quad (222)$$

in which the trailing tail probability e^{-p} is absorbed into $O(1/n)$ as soon as $p \geq \log n$.

L_p -bound on the composite remainder. The two pieces of $\mathcal{R}_{n, \text{stat}}^{\text{RR}}$ contribute additively. The finite-start deterministic transient displayed in Section 4.1 is deliberately excluded: setting $\theta_0 = \theta^*$ would remove only that deterministic term, but not the startup discrepancy between zero-start perturbation variables and the stationary augmented chain.

Lemma 14. (Stationary composite remainder bound.) Assume the hypotheses and step-size restrictions of the stationary PR-averaged RR misadjustment bound above. Then for every $u \in \mathbb{R}^d$, every $p \geq 2$, and every $q \geq 2$ satisfying $p \leq q/2$ and $2\alpha \leq \alpha_{\text{stat}}(q)$,

$$\begin{aligned} \|u^\top \mathcal{R}_{n, \text{stat}}^{\text{RR}}\|_{L_p} &\leq \frac{C_{\text{D2}} \|u\|}{\sqrt{n}} + \|u\| C_{\text{mis}} \left(\sqrt{n} \alpha^2 + (1 + d^{1/q}) p^{7/2} t_{\text{mix}}^{5/2} \sqrt{n} \alpha^{3/2} \log^{3/2}(1/(\alpha a)) \right. \\ &\quad \left. + p^{3/2} \sqrt{\alpha} + p^3 (\alpha n)^{-1/2} \log^{1/2}(1/(\alpha a)) + \Phi_+(p, \alpha) n^{-1/2} \right), \end{aligned} \quad (223)$$

with constants

$$C_{\text{D2}} := 3 t_{\text{mix}} \|\epsilon\|_\infty (C_Q + C_2/a^2), \quad (224)$$

and C_{mis} the constant in the misadjustment theorem.

Proof. Triangle inequality on $u^\top \mathcal{R}_{n, \text{stat}}^{\text{RR}}$:

(a) The deterministic sup-norm bound for $D_{2,n}^{\text{RR}}$ yields $\|u^\top D_{2,n}^{\text{RR}}\|_{L_p} \leq \|u\| \|D_{2,n}^{\text{RR}}\|_\infty \leq C_{\text{D2}} \|u\| / \sqrt{n}$ for every $p \geq 1$.

(b) The misadjustment theorem gives the remaining summands directly. $\square$

Theorem 3. (Stationary augmented-chain Berry–Esseen assembly for the RR comparison statistic.) Assume **UGE 1**, $\pi(\tilde{A}) = 0$, $\pi(\epsilon) = 0$, $\|\epsilon\|_\infty < \infty$, $\sigma^2(u) > 0$, $0 < \alpha$, and the stationary augmented-chain convention above. Use the imported inputs summarized in Section 7. Set $p = \max(2, \lceil \log n \rceil)$ and $q = \max(2p, \lceil \log(e d) \rceil, 2)$. Then $p \geq 2$, $q \geq 2$, and $p \leq q/2$. If $n \geq 3$ satisfies the small-step and variance lower-bound conditions

$$2\alpha \leq \alpha_{\text{stat}}(q), \quad n \alpha a \geq \frac{2 C_3 \|u\|^2}{\sigma^2(u)}, \quad (225)$$

where the second inequality is Eq. 160, then

$$\begin{aligned} d_K\left(\frac{S_{n, \text{stat}}^{\text{RR}}(u)}{\sigma_n^{\text{RR}}(u)}, \mathcal{N}(0, 1)\right) &\leq \frac{C_{K,1}(u) \log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u) \log n}{\sqrt{n}} \\ &\quad + \frac{e \|u^\top \mathcal{R}_{n, \text{stat}}^{\text{RR}}\|_{L_p}}{\sqrt{2\pi} \sigma_n^{\text{RR}}(u)} + \frac{e}{n}, \end{aligned} \quad (226)$$

with $\|u^\top \mathcal{R}_{n, \text{stat}}^{\text{RR}}\|_{L_p}$ bounded by the preceding lemma, and $C_{K,1}(u), C_{K,2}(u)$ the constants of the martingale Berry–Esseen theorem.

Proof. Apply the smoothing inequality with $X = X_n$ and $Y = Y_n$ from the displayed decomposition above. The first two terms come from Theorem 1 applied to the signed scalar martingale with increments $-u^\top \Delta M_t^{\text{RR}}$; the constants are unchanged since the theorem’s direction-dependent quantities are invariant

under $u \rightarrow -u$. The remainder term is $e \|Y_n\|_{L_p} / \sqrt{2\pi} = e \|u^\top \mathcal{R}_{n, \text{stat}}^{\text{RR}}\|_{L_p} / (\sqrt{2\pi} \sigma_n^{\text{RR}}(u))$. Since $p = \max(2, \lceil \log n \rceil) \geq \log n$, the trailing e^{-p} is at most n^{-1} . $\square$

Corollary 5. (Stationary triangular-array admissible-step CLT.) Assume the stationary augmented-chain hypotheses of Theorem 3, including $\sigma^2(u) > 0$. Let $\alpha = \alpha_n$ be a step-size sequence and set $p_n := \max(2, \lceil \log n \rceil)$, $q_n := \max(2p_n, \lceil \log(ed) \rceil, 2)$, and $\Lambda_n := \log(1/(\alpha_n a))$. Then $p_n \leq q_n/2$. Assume that, for all sufficiently large n ,

$$2\alpha_n \leq \alpha_{\text{stat}}(q_n), \quad n \alpha_n a \geq \frac{2C_3 \|u\|^2}{\sigma^2(u)}. \quad (227)$$

If

$$p_n^3 (n\alpha_n)^{-1/2} \Lambda_n^{1/p_n} \rightarrow 0, \quad p_n^{7/2} \sqrt{n} \alpha_n^{3/2} \Lambda_n^{3/2} \rightarrow 0, \quad (228)$$

then

$$d_K \left(\frac{S_{n, \text{stat}}^{\text{RR}}(u)}{\sigma_n^{\text{RR}}(u)}, \mathcal{N}(0, 1) \right) \rightarrow 0. \quad (229)$$

In particular, for a power scale $\alpha_n = c n^{-\gamma}$, up to logarithmic factors, the admissible window is

$$\frac{1}{3} < \gamma < 1. \quad (230)$$

Proof. Apply the stationary $n_0 = 0$ Berry–Esseen bound and the preceding remainder lemma with $p = p_n$, $q = q_n$, and $\alpha = \alpha_n$. The Poisson boundary term is $O(n^{-1/2})$. The bias term $\sqrt{n}\alpha_n^2$ is smaller than $\sqrt{n}\alpha_n^{3/2}$ for all sufficiently large n , because the admissibility conditions imply $\alpha_n \rightarrow 0$. The terms $p_n^{3/2}\sqrt{\alpha_n}$ and $\Phi_+(p_n, \alpha_n)n^{-1/2}$ are also negligible under the second admissibility condition. The remaining nontrivial terms are exactly the two quantities displayed in the statement. The martingale Berry–Esseen terms $\log^{3/4}(n)n^{-1/4}$ and $\log(n)n^{-1/2}$ vanish independently of α_n . For $\alpha_n = cn^{-\gamma}$, the two displayed conditions reduce, modulo logarithms, to $n^{-(1-\gamma)/2} \rightarrow 0$ and $n^{1/2-3\gamma/2} \rightarrow 0$, i.e. $\gamma < 1$ and $\gamma > 1/3$. $\square$

Corollary 6. (Stationary balanced-scale augmented-chain Berry–Esseen bound.) At the balanced scale $\alpha = c n^{-1/2}$ with $c > 0$, put $p = \max(2, \lceil \log n \rceil)$ and $q = \max(2p, \lceil \log(ed) \rceil, 2)$. Then $p \leq q/2$. Assume the stationary augmented-chain hypotheses of Theorem 3, including $\sigma^2(u) > 0$. If $n \geq 3$ satisfies

$$2\alpha \leq \alpha_{\text{stat}}(q), \quad n \alpha a \geq \frac{2C_3 \|u\|^2}{\sigma^2(u)}, \quad (231)$$

the bound Eq. 226 reduces to

$$d_K \left(\frac{S_{n, \text{stat}}^{\text{RR}}(u)}{\sigma_n^{\text{RR}}(u)}, \mathcal{N}(0, 1) \right) \leq \frac{C(u) \text{polylog}(n)}{n^{1/4}}, \quad (232)$$

where $C(u)$ depends on $\|u\|$, $\sigma(u)$, C_Q , $\|\bar{A}\|$, $\|\bar{A}^{-1}\|$, κ_Q , C_A , $\|\epsilon\|_\infty$, $\|\Sigma_\epsilon^{(\text{M})}\|$, t_{mix} , a , α_∞ , c , and the universal constants of the smoothing inequality, Eq. 398, and Eq. 397.

Proof. The scale $\alpha = c n^{-1/2}$ satisfies Eq. 228, but gives the sharper balanced rate. Indeed, $C_{\text{D2}} \|u\| / \sqrt{n} = O(n^{-1/2})$; $\sqrt{n}\alpha^2 = c^2 n^{-1/2}$; $p^{7/2} \sqrt{n} \alpha^{3/2} \log^{3/2}(1/(\alpha a)) = O(\text{polylog}(n)n^{-1/4})$; $p^{3/2} \sqrt{\alpha} = O(\text{polylog}(n)n^{-1/4})$; $p^3 (\alpha n)^{-1/2} \log^{1/2}(1/(\alpha a)) = O(\text{polylog}(n)n^{-1/4})$; and $\Phi_+(p, \alpha)n^{-1/2} = O(\text{polylog}(n)n^{-1/2})$. After division by $\sigma_n^{\text{RR}}(u) \geq \sigma(u)/\sqrt{2}$, these terms are dominated by $\text{polylog}(n)n^{-1/4}$. Combining with the martingale Berry–Esseen term proves the claim. $\square$

Corollary 7. (Stationary asymptotic-normalization version.) Assume the stationary augmented-chain hypotheses of Theorem 3 for the same n, p, q, α, u . In particular, assume $p \leq q/2$, $\sigma^2(u) > 0$, and

$$2\alpha \leq \alpha_{\text{stat}}(q), \quad n \alpha a \geq \frac{2 C_3 \|u\|^2}{\sigma^2(u)}. \quad (233)$$

Then the same finite- n bound with asymptotic normalization contains one additional variance-comparison term:

$$\begin{aligned} d_K \left(\frac{S_{n, \text{stat}}^{\text{RR}}(u)}{\sigma(u)}, \mathcal{N}(0, 1) \right) &\leq \frac{C_{K,1}(u) \log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u) \log n}{\sqrt{n}} \\ &\quad + \frac{e \|u^\top \mathcal{R}_{n, \text{stat}}^{\text{RR}}\|_{L_p}}{\sqrt{2\pi} \sigma_n^{\text{RR}}(u)} + \frac{e}{n} + \frac{C \|u\|^2}{n \alpha a \sigma^2(u)}. \end{aligned} \quad (234)$$

Consequently, under the balanced-scale hypotheses above,

$$d_K \left(\frac{S_{n, \text{stat}}^{\text{RR}}(u)}{\sigma(u)}, \mathcal{N}(0, 1) \right) \leq \frac{C'(u) \text{polylog}(n)}{n^{1/4}}. \quad (235)$$

Proof. Set $r := \sigma_n^{\text{RR}}(u)/\sigma(u)$ and write $W := S_{n, \text{stat}}^{\text{RR}}(u)/\sigma_n^{\text{RR}}(u)$, so $Wr = S_{n, \text{stat}}^{\text{RR}}(u)/\sigma(u)$. Under the variance lower-bound condition Eq. 160, $r \in [1/\sqrt{2}, r_{\max}]$ for some finite $r_{\max}$ (from the trivial upper bound $\sigma_n^{2, \text{RR}}(u) \leq C_Q^2 \|\Sigma\| \|u\|^2$). For r in this compact interval,

$$\sup_x |\Phi(x/r) - \Phi(x)| \leq C_\Phi |r - 1|, \quad C_\Phi := \sqrt{2}/\sqrt{\pi e}, \quad (236)$$

because $\sup_x |x \varphi(x)| = 1/\sqrt{2\pi e}$. Hence

$$d_K(Wr, \mathcal{N}(0, 1)) \leq d_K(W, \mathcal{N}(0, 1)) + C_\Phi |r - 1|. \quad (237)$$

The variance comparison of Section 4.5 gives

$$|r - 1| = \frac{|\sigma_n^{2, \text{RR}}(u) - \sigma^2(u)|}{\sigma(u) (\sigma_n^{\text{RR}}(u) + \sigma(u))} \leq \frac{C_3 \|u\|^2}{n \alpha a \sigma^2(u)}, \quad (238)$$

using $\sigma_n^{\text{RR}}(u) + \sigma(u) \geq \sigma(u)$. This proves the displayed asymptotic-normalization bound. At $\alpha = cn^{-1/2}$ the final term is $O(n^{-1/2})$, hence it is absorbed into the balanced-scale bound of the preceding corollary. $\square$

The bound above is therefore an $n_0 = 0$ stationary augmented-chain statement for $S_{n, \text{stat}}^{\text{RR}}(u)$, not the deterministic-start RR average itself. A deterministic-start theorem with mixing-scale burn-in and logarithmic factors requires a separate transfer theorem with $Q_{l, n_0}^{(\alpha)}$ and the corresponding Poisson, variance-comparison, and misadjustment bounds.

5. Burn-in Transfer for Deterministic Starts

The previous chapter proves a stationary $n_0 = 0$ Berry–Esseen bound for $S_{n, \text{stat}}^{\text{RR}}(u)$. This chapter transfers that bound to the deterministic-start Richardson–Romberg average after burn-in. Burn-in changes the deterministic PR kernels, so the result is not obtained by simply inserting $n_0 > 0$ into the stationary theorem.

Throughout this chapter, constants with a "burn" subscript may depend on fixed problem parameters and the external/local startup constants of Section 7, and may absorb fixed powers of a^{-1} . They never hide dependence on n , n_0 , m , α , p , or q unless these variables appear as arguments of the constant or named quantity. Powers of t_{mix} , p , q , d , and any rate-relevant powers of a^{-1} are displayed explicitly in the surrounding bound. Unless explicitly stated otherwise, all burn-in estimates are uniform over the initial law $\xi = \mathcal{L}(Z_0)$ of the base Markov chain. Constants may depend on the deterministic initial point θ_0 only when that dependence is displayed, but they do not depend on ξ .

5.1. Target Statistic and Normalization

Let $\xi = \mathcal{L}(Z_0)$ be the initial law of the base Markov chain, let $0 \leq n_0 < n$, and set $m := n - n_0$. The burned-in PR average is

$$\bar{\theta}_{n, n_0}^{(\alpha)} := \frac{1}{m} \sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}, \quad (239)$$

and the two-level RR average is

$$\bar{\theta}_{n, n_0}^{(\text{RR}, \alpha)} := 2\bar{\theta}_{n, n_0}^{(\alpha)} - \bar{\theta}_{n, n_0}^{(2\alpha)}. \quad (240)$$

Both stepsizes are run from the same deterministic initial point θ_0 and on the same Markov trajectory. This coupling is part of the RR statistic; if the two levels use different paths, the deterministic-weight decomposition below is a different object.

The finite-start burned-in vector statistic is

$$\mathcal{J}_{n, n_0}^{\text{RR}} := \sqrt{m} \left(\bar{\theta}_{n, n_0}^{(\text{RR}, \alpha)} - \theta^* \right). \quad (241)$$

Its scalar projection in direction $u \in \mathbb{R}^d$ is

$$T_{n, n_0}^{\text{RR}}(u) := u^\top \mathcal{J}_{n, n_0}^{\text{RR}} = \sqrt{m} u^\top \left(\bar{\theta}_{n, n_0}^{(\text{RR}, \alpha)} - \theta^* \right). \quad (242)$$

The stationary augmented-chain statistic from the previous chapter is used only as a comparison object after the startup transfer.

There are two normalizations. The finite-window normalization is

$$\sigma_{n, n_0}^{\text{bRR}}(u) := \sqrt{\sigma_{n, n_0}^{2, \text{bRR}}(u)}, \quad \Xi_{n, n_0}^{\text{bRR}}(u) := \frac{T_{n, n_0}^{\text{RR}}(u)}{\sigma_{n, n_0}^{\text{bRR}}(u)}, \quad (243)$$

where $\sigma_{n, n_0}^{2, \text{bRR}}(u)$ is the deterministic variance proxy defined in Eq. 282. The asymptotic normalization is

$$\sigma(u) := \sqrt{u^\top \Sigma_\infty u}, \quad \Xi_{n, n_0}^{\text{asy, RR}}(u) := \frac{T_{n, n_0}^{\text{RR}}(u)}{\sigma(u)}. \quad (244)$$

We first control $\Xi_{n, n_0}^{\text{bRR}}(u)$ and then pass to $\Xi_{n, n_0}^{\text{asy, RR}}(u)$. A final corollary converts the $\sqrt{m}$ statistic to the final $\sqrt{n}$ statistic when the burn-in is at the mixing scale, of order α^{-1} times logarithmic factors.

5.2. Burned-in Deterministic Weights

With the $\sqrt{m}$ normalization, the depth-zero linearized term has the weights

$$Q_{l;n_0,n}^{(\alpha)} := \alpha \sum_{k=\max(n_0,l)}^{n-1} B_\alpha^{k-l}, \quad 1 \leq l \leq n-1, \quad (245)$$

and $Q_{l;n_0,n}^{(\alpha)} = 0$ for $l \geq n$. The RR weight is

$$Q_{l;n_0,n}^{\text{RR}} := 2Q_{l;n_0,n}^{(\alpha)} - Q_{l;n_0,n}^{(2\alpha)}. \quad (246)$$

Thus the leading burned-in sum is

$$W_{n,n_0}^{\text{RR}} := -\frac{1}{\sqrt{m}} \sum_{l=1}^{n-1} Q_{l;n_0,n}^{\text{RR}} \epsilon(Z_l). \quad (247)$$

For $l \geq n_0$ these weights are full-window weights with horizon $n-l$. For $l < n_0$, the lower summation limit is n_0 rather than l , so the weight comparison, Poisson decomposition, and variance proxy must be restated for $Q_{l;n_0,n}^{\text{RR}}$.

5.3. Closed Forms and Burned-in Weight Bounds

Write $B_w := I - w\bar{A}$ for $w \in \{\alpha, 2\alpha\}$ and recall $m = n - n_0$. The burned-in weights have two closed forms. If $l \geq n_0$, then

$$Q_{l;n_0,n}^{(w)} = w \sum_{k=l}^{n-1} B_w^{k-l} = \bar{A}^{-1} (I - B_w^{n-l}). \quad (248)$$

If $l < n_0$, then

$$Q_{l;n_0,n}^{(w)} = w \sum_{k=n_0}^{n-1} B_w^{k-l} = \bar{A}^{-1} (B_w^{n_0-l} - B_w^{n-l}). \quad (249)$$

Thus only the post-burn-in weights are close to the asymptotic kernel $\bar{A}^{-1}$. The pre-burn-in weights are instead exponentially small as one moves backward from n_0 . Empty sums below are interpreted as zero.

Lemma 15. (Burned-in RR weight bounds.) Assume $0 < \alpha, 2\alpha \leq \alpha_\infty$, and the Lyapunov contraction Eq. 13. Let

$$\rho_\alpha := \sqrt{1 - \alpha a}, \quad C_Q := \kappa_Q^{1/2} \|\bar{A}^{-1}\|, \quad \tilde{C}_A := \kappa_Q \|\bar{A}\|. \quad (250)$$

(i) **Post-burn-in comparison.** If $l \geq n_0$ and $k := n - l$, then

$$\|Q_{l;n_0,n}^{\text{RR}} - \bar{A}^{-1}\| \leq 3C_Q \rho_\alpha^k. \quad (251)$$

(ii) **Pre-burn-in energy.** If $1 \leq l < n_0$ and $r := n_0 - l$, then

$$\|Q_{l;n_0,n}^{\text{RR}}\| \leq 6C_Q \rho_\alpha^r. \quad (252)$$

Consequently, there are constants $C_{\text{burn,E}}$ and $C_{\text{burn,V}}$, depending only on $\kappa_Q, \|\bar{A}^{-1}\|$, and $\|\bar{A}\|$, such that, uniformly in n and n_0 ,

$$\sum_{l=1}^{n_0-1} \|Q_{l;n_0,n}^{\text{RR}}\|^2 + \sum_{l=\max(1,n_0)}^{n-1} \|Q_{l;n_0,n}^{\text{RR}} - \bar{A}^{-1}\|^2 \leq \frac{C_{\text{burn,E}}}{\alpha a}, \quad (253)$$

and

$$\sum_{l=1}^{n-2} \|Q_{l+1;n_0,n}^{\text{RR}} - Q_{l;n_0,n}^{\text{RR}}\| \leq \frac{C_{\text{burn,V}}}{a^2}. \quad (254)$$

Proof. For $l \geq n_0$, Eq. 248 gives

$$Q_{l;n_0,n}^{\text{RR}} - \bar{A}^{-1} = -\bar{A}^{-1}(2B_\alpha^k - B_{2\alpha}^k). \quad (255)$$

The same contraction argument as in the full-window estimate gives Eq. 251.

For $l < n_0$, set $r := n_0 - l$. By Eq. 249,

$$Q_{l;n_0,n}^{\text{RR}} = \bar{A}^{-1}(2(B_\alpha^r - B_\alpha^{r+m}) - (B_{2\alpha}^r - B_{2\alpha}^{r+m})). \quad (256)$$

Taking norms and using $\|B_{2\alpha}^j\|_Q \leq \rho_\alpha^j$ gives Eq. 252. Squaring Eq. 251 and Eq. 252 and summing the resulting geometric series gives Eq. 253.

It remains to bound the total variation. In the post-burn-in region $l, l+1 \geq n_0$, the full-window identity gives

$$Q_{l+1;n_0,n}^{\text{RR}} - Q_{l;n_0,n}^{\text{RR}} = -2\alpha(B_\alpha^s - B_{2\alpha}^s), \quad s := n - l - 1. \quad (257)$$

The elementary identity $X^s - Y^s = (X - Y) \sum_{i=1}^s X^{i-1} Y^{s-i}$ with $X = B_\alpha$ and $Y = B_{2\alpha}$ yields, for $s \geq 1$,

$$\|B_\alpha^s - B_{2\alpha}^s\| \leq \alpha \tilde{C}_A s \rho_\alpha^{s-1}. \quad (258)$$

The left-factored form is legitimate because B_α and $B_{2\alpha}$ are polynomials in the same matrix $\bar{A}$, hence commute with each other and with $\bar{A}$. Thus the post-burn-in contribution is bounded by a constant times a^{-2} , as in the full-window proof.

For $l, l+1 < n_0$, put $s := n_0 - l - 1$. The pre-burn-in closed form gives

$$Q_{l+1;n_0,n}^{\text{RR}} - Q_{l;n_0,n}^{\text{RR}} = 2\alpha(B_\alpha^s - B_{2\alpha}^s - B_\alpha^{s+m} + B_{2\alpha}^{s+m}). \quad (259)$$

Here $s \geq 1$. Applying Eq. 258 to the powers s and $s+m$ and summing over $s \geq 1$ again gives a constant times a^{-2} . If the boundary $l = n_0 - 1$ exists, then

$$Q_{n_0;n_0,n}^{\text{RR}} - Q_{n_0-1;n_0,n}^{\text{RR}} = 2\alpha(B_{2\alpha}^m - B_\alpha^m), \quad (260)$$

and Eq. 258 with $s = m$ bounds this term by the same a^{-2} scale. Combining the pre-burn-in, boundary, and post-burn-in contributions proves Eq. 254. $\square$

5.4. Deterministic Transient After Burn-in

For a generic step size $w \in \{\alpha, 2\alpha\}$ define the deterministic transient

$$D_{\text{tr},n,n_0}^{(w)} := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} B_w^k(\theta_0 - \theta^*), \quad B_w := I - w\bar{A}. \quad (261)$$

The Richardson–Romberg transient entering Eq. 242 is

$$D_{\text{tr},n,n_0}^{\text{RR}}(u) := u^\top \left(2D_{\text{tr},n,n_0}^{(\alpha)} - D_{\text{tr},n,n_0}^{(2\alpha)} \right). \quad (262)$$

Lemma 16. (Burned-in deterministic transient.) Assume $0 < \alpha$, $2\alpha \leq \alpha_\infty$, and the Lyapunov contraction Eq. 13. Then, for each $w \in \{\alpha, 2\alpha\}$,

$$\| D_{\text{tr},n,n_0}^{(w)} \| \leq \frac{2\sqrt{\kappa_Q}}{wa\sqrt{m}}(1-wa)^{n_0/2} \| \theta_0 - \theta^* \|. \quad (263)$$

Consequently,

$$| D_{\text{tr},n,n_0}^{\text{RR}}(u) | \leq \frac{5\sqrt{\kappa_Q} \| u \| \| \theta_0 - \theta^* \|}{\alpha a \sqrt{m}} (1 - \alpha a)^{n_0/2}. \quad (264)$$

Proof. Let $r_w := \sqrt{1 - wa}$. By Eq. 13 and equivalence of the Euclidean and Q -norms,

$$\| B_w^k(\theta_0 - \theta^*) \| \leq \sqrt{\kappa_Q} r_w^k \| \theta_0 - \theta^* \|. \quad (265)$$

Therefore

$$\| D_{\text{tr},n,n_0}^{(w)} \| \leq \frac{\sqrt{\kappa_Q} \| \theta_0 - \theta^* \|}{\sqrt{m}} \sum_{k=n_0}^{n-1} r_w^k \leq \frac{\sqrt{\kappa_Q} \| \theta_0 - \theta^* \|}{\sqrt{m}} \frac{r_w^{n_0}}{1 - r_w}. \quad (266)$$

Since $1 - \sqrt{1 - x} \geq x/2$ for $0 \leq x \leq 1$, the last display gives Eq. 263. Applying this estimate at $w = \alpha$ and $w = 2\alpha$, using $1 - 2\alpha a \leq 1 - \alpha a$, and taking the triangle inequality in Eq. 262 gives

$$| D_{\text{tr},n,n_0}^{\text{RR}}(u) | \leq \| u \| \left(2 \| D_{\text{tr},n,n_0}^{(\alpha)} \| + \| D_{\text{tr},n,n_0}^{(2\alpha)} \| \right) \leq \frac{5\sqrt{\kappa_Q} \| u \| \| \theta_0 - \theta^* \|}{\alpha a \sqrt{m}} (1 - \alpha a)^{n_0/2}. \quad (267)$$

□

Corollary 8. (Mixing-scale burn-in removes the deterministic transient.) If, for some $\beta > 0$,

$$n_0 \geq \frac{2\beta}{\alpha a} \log n, \quad (268)$$

then

$$| D_{\text{tr},n,n_0}^{\text{RR}}(u) | \leq \frac{5\sqrt{\kappa_Q} \| u \| \| \theta_0 - \theta^* \|}{\alpha a \sqrt{m}} n^{-\beta}. \quad (269)$$

In particular, at the balanced scale $\alpha = cn^{-1/2}$ and $m \geq n/2$, this is $O(n^{-\beta})$. Taking $\beta = 1$ makes the deterministic transient negligible relative to the stationary $n^{-1/4}$ Berry–Esseen rate.

Proof. The elementary inequality $(1 - \alpha a)^{n_0/2} \leq \exp(-\alpha a n_0/2)$ and Eq. 268 imply $(1 - \alpha a)^{n_0/2} \leq n^{-\beta}$. Substitute this into Eq. 264. If $\alpha = cn^{-1/2}$ and $m \geq n/2$, then $(\alpha \sqrt{m})^{-1} \leq \sqrt{2}/c$, so the displayed $O(n^{-\beta})$ bound follows. □

The deterministic transient is not the full initial-condition contribution: the exact recursion contains the random product $\Gamma_{1:k}^{(w)} := \prod_{j=1}^k (I - wA(Z_j))$. The difference between this product and B_w^k is another finite-start term. We use the following product-stability input, recorded here in the exact deterministic and conditional forms needed later. Let $\alpha_{\text{st}}(p)$ be the local threshold defined in Eq. 89; the startup section below uses the same threshold.

Lemma 17. (Technical input: deterministic and conditional product stability.) Under the stability and bounded-noise assumptions of the depth-two setup, there are product-stability ceilings $\alpha_{\text{prod}}(r)$, $r \geq 2$, and constants $C_{\text{prod}} < \infty$ and $c_{\text{prod}} > 0$, independent of p, w, s, k and of the current base-chain state, with the following property. If $p \geq 2$, $2\alpha \leq \alpha_{\text{st}}(p)$, and $w \in \{\alpha, 2\alpha\}$, then the product estimates below hold for every $0 \leq s \leq k$. The empty product is $\Gamma_{k+1:k}^{(w)} = I$.

First, for every deterministic vector v and uniformly over the current base-chain state at time s ,

$$\mathbb{E}^{1/p} \left[\left\| \Gamma_{s+1:k}^{(w)} v \right\|^p \text{ mid } Z_s \right] \leq C_{\text{prod}} \exp(-c_{\text{prod}} w a(k-s)/p) \|v\|. \quad (270)$$

Equivalently, on any coupling extension with a joint filtration $(\mathcal{G}_r)$ and a $\mathcal{G}_s$ -measurable vector W_s to which the future product is adapted,

$$\mathbb{E}^{1/p} \left[\left\| \Gamma_{s+1:k}^{(w)} W_s \right\|^p \text{ mid } \mathcal{G}_s \right] \leq C_{\text{prod}} \exp(-c_{\text{prod}} w a(k-s)/p) \|W_s\|. \quad (271)$$

Consequently, for every such $\mathcal{G}_s$ -measurable vector V_s ,

$$\left\| \Gamma_{s+1:k}^{(w)} V_s \right\|_{L_p} \leq C_{\text{prod}} \exp(-c_{\text{prod}} w a(k-s)/p) \|V_s\|_{L_p} \leq C_{\text{prod}} \exp(-c_{\text{prod}} w a(k-s)/p) \|V_s\|_{L_{2p}}. \quad (272)$$

In particular, with the natural base-chain filtration,

$$\left\| \Gamma_{s+1:k}^{(w)} V_s \right\|_{L_p} \leq C_{\text{prod}} \exp(-c_{\text{prod}} w a(k-s)/p) \|V_s\|_{L_{2p}}. \quad (273)$$

This lemma is treated as a technical input of the burn-in transfer. The thesis does not use any unrecorded intermediate version of product stability: the conditional display Eq. 271 is the form invoked at random coupling times, and Eq. 273 is its unconditional corollary for deterministic times.

Define the accumulated RR random initial-product discrepancy by

$$\mathcal{J}_{n,n_0}^{\text{init,RR}}(u) := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} u^\top \left[2 \left(\Gamma_{1:k}^{(\alpha)} - B_\alpha^k \right) - \left(\Gamma_{1:k}^{(2\alpha)} - B_{2\alpha}^k \right) \right] (\theta_0 - \theta^*), \quad (274)$$

where the empty product at $k=0$ is the identity.

Lemma 18. (Burned-in random initial-product transient.) Assume the hypotheses of Lemma 17, $0 < \alpha$, $2\alpha \leq \alpha_\infty$, $\alpha a \leq 1/4$, and the Lyapunov contraction Eq. 13. Then, for every $p \geq 2$,

$$\left\| \mathcal{J}_{n,n_0}^{\text{init,RR}}(u) \right\|_{L_p} \leq \frac{C_{\text{init,RR}} \|u\| \|\theta_0 - \theta^*\| p}{\alpha a \sqrt{m}} \exp(-c_{\text{init}} \alpha a n_0 / p), \quad (275)$$

where $C_{\text{init,RR}} < \infty$ and $c_{\text{init}} > 0$ depend only on C_{prod} , c_{prod} , and the Lyapunov norm-equivalence constant.

Proof. Let $e_0 := \theta_0 - \theta^*$. By Lemma 17 with $s=0$ and $V_0 = e_0$,

$$\left\| \Gamma_{1:k}^{(w)} e_0 \right\|_{L_p} \leq C_{\text{prod}} \exp(-c_{\text{prod}} w a k / p) \|e_0\|. \quad (276)$$

The deterministic Lyapunov contraction gives

$$\left\| B_w^k e_0 \right\| \leq \sqrt{\kappa_Q} (1 - w a)^{k/2} \|e_0\| \leq \sqrt{\kappa_Q} \exp(-w a k / (2p)) \|e_0\|, \quad (277)$$

since $p \geq 1$. Hence, after decreasing the exponential constant,

$$\left\| \left(\Gamma_{1:k}^{(w)} - B_w^k \right) e_0 \right\|_{L_p} \leq C \exp(-c_{\text{init}} w a k / p) \|e_0\|. \quad (278)$$

Apply this estimate at $w = \alpha$ and $w = 2\alpha$, use the triangle inequality in Eq. 274, and extend the geometric sum to infinity:

$$\| \mathcal{J}_{n,n_0}^{\text{init,RR}}(u) \|_{L_p} \leq \frac{C \|u\| \|e_0\|}{\sqrt{m}} \sum_{k=n_0}^{\infty} \exp(-c_{\text{init}} \alpha a k / p). \quad (279)$$

Because $\alpha a \leq 1/4$ and $p \geq 2$, $1 - \exp(-c_{\text{init}} \alpha a / p) \geq C^{-1} \alpha a / p$, which gives Eq. 275. $\square$

Corollary 9. (Mixing-scale burn-in removes the random initial-product discrepancy.) If, for some $\beta > 0$,

$$n_0 \geq \frac{\beta p}{c_{\text{init}} \alpha a} \log n, \quad (280)$$

then

$$\| \mathcal{J}_{n,n_0}^{\text{init,RR}}(u) \|_{L_p} \leq \frac{C_{\text{init,RR}} \|u\| \|\theta_0 - \theta^*\| p}{\alpha a \sqrt{m}} n^{-\beta}. \quad (281)$$

At the balanced scale $\alpha = cn^{-1/2}$, with $m \geq n/2$ and p logarithmic in n , this is $\text{polylog}(n) n^{-\beta}$.

Proof. Substitute Eq. 280 into Eq. 275. At the balanced scale, $(\alpha \sqrt{m})^{-1} = O(1)$, so the remaining factor is logarithmic. $\square$

5.5. Burned-in Variance Proxy

For the stochastic depth-zero term, write $Q_l^{\text{bRR}} := Q_{l;n_0,n}^{\text{RR}}$ and $\Sigma := \Sigma_\epsilon^{(\text{M})}$. We take $n \geq 2$ in this subsection. The martingale part produced by the Poisson decomposition below starts at $l = 2$, so the deterministic variance proxy is

$$\Sigma_{n,n_0}^{\text{bRR}} := \frac{1}{m} \sum_{l=2}^{n-1} Q_l^{\text{bRR}} \Sigma (Q_l^{\text{bRR}})^\top, \quad \sigma_{n,n_0}^{2, \text{bRR}}(u) := u^\top \Sigma_{n,n_0}^{\text{bRR}} u. \quad (282)$$

Lemma 19. (Burned-in variance comparison.) Assume the conditions of Lemma 15 and $\|\Sigma\| < \infty$. There exists a constant $C_{\text{burn},3}$, depending only on κ_Q , $\|\bar{A}^{-1}\|$, $\|\bar{A}\|$, and $\|\Sigma\|$, such that

$$\|\Sigma_{n,n_0}^{\text{bRR}} - \Sigma_\infty\| \leq \frac{C_{\text{burn},3}}{m \alpha a}. \quad (283)$$

Consequently,

$$|\sigma_{n,n_0}^{2, \text{bRR}}(u) - \sigma^2(u)| \leq \frac{C_{\text{burn},3} \|u\|^2}{m \alpha a}. \quad (284)$$

Proof. For post-burn-in indices put $\Delta_l := Q_l^{\text{bRR}} - \bar{A}^{-1}$. Then

$$Q_l^{\text{bRR}} \Sigma (Q_l^{\text{bRR}})^\top - \Sigma_\infty = \Delta_l \Sigma \bar{A}^{-\top} + \bar{A}^{-1} \Sigma \Delta_l^\top + \Delta_l \Sigma \Delta_l^\top. \quad (285)$$

By Eq. 251,

$$\sum_{l \geq \max(2, n_0)} \|\Delta_l\| \leq \frac{C}{\alpha a}, \quad \sum_{l \geq \max(2, n_0)} \|\Delta_l\|^2 \leq \frac{C}{\alpha a}. \quad (286)$$

For pre-burn-in indices there is no comparison with $\bar{A}^{-1}$; instead Eq. 253 gives

$$\sum_{2 \leq l < n_0} \|Q_l^{\text{bRR}}\|^2 \leq \frac{C}{\alpha a}. \quad (287)$$

Replacing all post-burn-in weights by $\bar{A}^{-1}$ gives at most m copies of Σ_∞ ; the martingale index set starts at $l = 2$, so the possible finite-index mismatch is bounded by two copies of Σ_∞ . Combining these three estimates and dividing by m yields Eq. 283, after absorbing the harmless $2 \|\Sigma_\infty\|_m$ term into $\frac{C_{\text{burn},3}}{m\alpha a}$ using $\alpha a \leq 1$. The scalar bound follows from $|u^\top H u| \leq \|H\| \|u\|^2$. $\square$

Assume $\sigma^2(u) > 0$ and impose the burned-in variance lower-bound condition

$$m \alpha a \geq \frac{2C_{\text{burn},3} \|u\|^2}{\sigma^2(u)}. \quad (288)$$

Then Eq. 284 gives $\sigma_{n,n_0}^{2,\text{bRR}}(u) \geq \sigma^2(u)/2$, hence $\sigma_{n,n_0}^{\text{bRR}}(u) \geq \sigma(u)/\sqrt{2}$. At the balanced scale $\alpha = c n^{-1/2}$ with $m \geq n/2$, this condition holds for all sufficiently large n depending on u , c , and the problem constants.

We say that $(n, n_0, p, q, \alpha, u)$ is in the **admissible burn-in regime** if $m := n - n_0$ and

$$n \geq 3, \quad 0 \leq n_0 < n, \quad p \geq 2, \quad q \geq 2, \quad p \leq q/4, \quad \sigma^2(u) > 0, \quad (289)$$

$$m \geq n/2, \quad 0 < \alpha, \quad 2\alpha \leq \alpha_{\text{burn}}(p, q), \quad m \alpha a \geq \frac{2C_{\text{burn},3} \|u\|^2}{\sigma^2(u)}. \quad (290)$$

The last inequality is the variance lower-bound condition Eq. 288. This regime is the standing finite-window domain for the burned-in Berry–Esseen assembly below. Every later conversion of an m -scale rate into an n -scale rate is made only after invoking $m \geq n/2$ from this regime.

5.6. Burned-in Poisson Martingale Approximation

Let $\hat{\epsilon} := \sum_{j=0}^{\infty} Q^j \epsilon$ be the Poisson solution, so

$$\hat{\epsilon} - Q\hat{\epsilon} = \epsilon, \quad \|\hat{\epsilon}\|_\infty \leq 3t_{\text{mix}} \|\epsilon\|_\infty. \quad (291)$$

The stationary Poisson identity applies because the coefficients Q_l^{bRR} are deterministic.

Lemma 20. (Burned-in Poisson martingale decomposition.) Assume **UGE 1** and $\pi(\epsilon) = 0$. Define, for $2 \leq l \leq n-1$,

$$\Delta M_l^{\text{bRR}} := Q_l^{\text{bRR}} (\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})), \quad (292)$$

and set $M_{n,n_0}^{\text{bRR}} := \sum_{l=2}^{n-1} \Delta M_l^{\text{bRR}}$. Then $\{\Delta M_l^{\text{bRR}}\}_{l=2}^{n-1}$ is a sequence of $\mathcal{F}_l$ -martingale differences and

$$W_{n,n_0}^{\text{RR}} = -\frac{1}{\sqrt{m}} M_{n,n_0}^{\text{bRR}} + D_{2,n,n_0}^{\text{bRR}}, \quad (293)$$

where

$$D_{2,n,n_0}^{\text{bRR}} := -\frac{1}{\sqrt{m}} \left[Q_1^{\text{bRR}} \hat{\epsilon}(Z_1) + \sum_{l=1}^{n-2} (Q_{l+1}^{\text{bRR}} - Q_l^{\text{bRR}}) Q\hat{\epsilon}(Z_l) \right]. \quad (294)$$

Moreover, with $C_{\text{burn},Q} := \|\bar{A}^{-1}\| + 6C_Q$,

$$\|D_{2,n,n_0}^{\text{bRR}}\|_\infty \leq \frac{3t_{\text{mix}} \|\epsilon\|_\infty}{\sqrt{m}} \left(C_{\text{burn},Q} + \frac{C_{\text{burn},V}}{a^2} \right). \quad (295)$$

Proof. The martingale-difference property follows from the Markov property: $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$.

Substitute the Poisson equation in Eq. 247. The $l = 1$ term is kept as a left boundary term, and for $l \geq 2$ we add and subtract $Q\hat{\epsilon}(Z_{l-1})$. Abel summation of the telescope gives exactly Eq. 293. The right boundary vanishes because $Q_{n-1,n_0,n}^{\text{RR}} = 2\alpha I - 2\alpha I = 0$.

For the sup-norm bound, use $\|Q\hat{\epsilon}\|_\infty \leq \|\hat{\epsilon}\|_\infty$, the uniform weight bound $\|Q_l^{\text{bRR}}\| \leq C_{\text{burn,Q}}$ from Eq. 251 and Eq. 252, and the total-variation estimate Eq. 254. $\square$

5.7. Burned-in Predictable Variance Comparison

For the martingale in Lemma 20 define

$$\mathcal{V}_\epsilon(z) := Q(\hat{\epsilon}\hat{\epsilon}^\top)(z) - (Q\hat{\epsilon})(z)(Q\hat{\epsilon})(z)^\top. \quad (296)$$

This is the same matrix-valued conditional covariance function as in the stationary chapter; it is not the vector noise ϵ . By the Poisson covariance identity Lemma 8, $\pi(\mathcal{V}_\epsilon) = \Sigma$ and

$$\|\mathcal{V}_\epsilon\|_\infty \leq 18 t_{\text{mix}}^2 \|\epsilon\|_\infty^2. \quad (297)$$

The predictable quadratic variation is

$$\langle M^{\text{bRR}} \rangle_{n,n_0} := \sum_{l=2}^{n-1} \mathbb{E} \left[\Delta M_l^{\text{bRR}} (\Delta M_l^{\text{bRR}})^\top \mid \mathcal{F}_{l-1} \right] = \sum_{l=2}^{n-1} Q_l^{\text{bRR}} \mathcal{V}_\epsilon(Z_{l-1}) (Q_l^{\text{bRR}})^\top. \quad (298)$$

Lemma 21. (Burned-in predictable-variation concentration.) Assume **UGE 1**, $\pi(\epsilon) = 0$, and $\|\epsilon\|_\infty < \infty$. There exists a universal constant $C_4 > 0$ such that, for every $u \in \mathbb{R}^d$, every $p \geq 2$, every initial distribution ξ , every $n \geq 2$, and every $0 \leq n_0 < n$,

$$\mathbb{E}_\xi^{1/p} \left[|u^\top \langle M^{\text{bRR}} \rangle_{n,n_0} u - m \sigma_{n,n_0}^2(u)|^p \right] \leq C_4 C_{\text{burn,Q}}^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2} \sqrt{pn}. \quad (299)$$

Proof. Set

$$h_{l(z)} := u^\top Q_l^{\text{bRR}} \mathcal{V}_\epsilon(z) (Q_l^{\text{bRR}})^\top u, \quad g_{l(z)} := h_{l(z)} - \pi(h_l). \quad (300)$$

By Eq. 297 and $\|Q_l^{\text{bRR}}\| \leq C_{\text{burn,Q}}$,

$$\|g_l\|_\infty \leq 36 C_{\text{burn,Q}}^2 \|u\|^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2. \quad (301)$$

Moreover, Eq. 298 and Eq. 282 imply

$$u^\top \langle M^{\text{bRR}} \rangle_{n,n_0} u - m \sigma_{n,n_0}^2(u) = \sum_{l=2}^{n-1} g_{l(Z_{l-1})}. \quad (302)$$

The scalar Markov concentration lemma used in the stationary chapter is valid for arbitrary initial law in the form Eq. 397. Applied to the time-inhomogeneous centered functions g_l with

$$c_l := 36 C_{\text{burn,Q}}^2 \|u\|^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2, \quad (303)$$

it gives

$$\left\| \sum_{l=2}^{n-1} g_{l(Z_{l-1})} \right\|_{L_p(\xi)} \leq C_{\text{MC}} \sqrt{p t_{\text{mix}} \sum_{l=2}^{n-1} c_l^2} \leq C C_{\text{burn,Q}}^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2} \sqrt{pn}, \quad (304)$$

which is Eq. 299 after increasing C_4 . $\square$

Corollary 10. (Burned-in asymptotic-variance bracket comparison.) Under the assumptions of Lemma 21,

$$\mathbb{E}_\xi^{1/p} \left[|u^\top \langle M^{\text{bRR}} \rangle_{n,n_0} u - m \sigma^2(u)|^p \right] \leq C_4 C_{\text{burn,Q}}^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2} \sqrt{pn} + \frac{C_{\text{burn,3}} \|u\|^2}{\alpha a}. \quad (305)$$

Proof. Use the triangle inequality and Eq. 284:

$$m |\sigma_{n,n_0}^{2, \text{bRR}}(u) - \sigma^2(u)| \leq \frac{C_{\text{burn},3} \|u\|^2}{\alpha a}. \quad (306)$$

The concentration sum still runs over the ambient indices $2, \dots, n-1$, including pre-burn-in weights, which is why the display keeps $\sqrt{pn}$. In the final theorem this is converted to the effective window scale using $m \geq n/2$. $\square$

5.8. Startup Transfer for Augmented-Chain Remainders

After the deterministic transient is removed, a startup discrepancy remains: the finite-start perturbation variables $J^{(\ell,w)}$ and $H^{(2,w)}$ start from zero, whereas the stationary theorem uses the invariant law of the augmented chain. This discrepancy is summed over the post-burn-in window.

For $w \in \{\alpha, 2\alpha\}$ write

$$Y_k^{(w)} := \left(Z_{k+1}, J_k^{(0,w)}, J_k^{(1,w)}, J_k^{(2,w)} \right). \quad (307)$$

The direct J -coordinate contraction, its cost Eq. 407, and the invariant J law are recorded in Lemma 33 and Lemma 34. The $H^{(2,w)}$ one-trajectory bound and representation Eq. 405 are recorded in Lemma 32. The invariant law for the full augmented state is the finite-past limit constructed in Lemma 10. In particular, for every $w \in \{\alpha, 2\alpha\}$ satisfying the admissibility condition, the stationary copy

$$\left(Z_{k+1}, J_k^{(0,w)}, J_k^{(1,w)}, J_k^{(2,w)}, H_k^{(2,w)} \right)_{k \in \mathbb{Z}} \quad (308)$$

exists, the $H^{(2,w)}$ coordinate is obtained as an L_p limit of finite-past truncations, and the limiting coordinate inherits Lemma 32. This is the full augmented-chain invariant law used below.

Lemma 22. (Conditional product stability at a coupling time.) Work on a coupling space with joint filtration $(\mathcal{G}_s)_{s \geq 0}$ for two copies of the base chain and their perturbation variables. Let T be an integer-valued $\mathcal{G}_s$ -stopping time such that the two base chains are exactly coupled from time T onward and, uniformly over the initial states used below,

$$\mathbb{P}(T > r) \leq C_T \exp(-c_T r), \quad r \geq 0. \quad (309)$$

Products are taken along the post-coupling chain; we use the convention $\Gamma_{k+1:k}^{(w)} = I$ and empty sums are zero. Assume the hypotheses of Lemma 17. Then, after increasing constants and decreasing the exponential constant, the following two estimates hold for every $p \geq 2$, every $k \geq 0$, and every $w \in \{\alpha, 2\alpha\}$ for which the product-stability input is admissible and $wa \leq 1$. In the applications below this small-step condition follows from $2\alpha \leq \alpha_{\text{burn}}(p, q)$. The constants depend only on the product-stability constants and on C_T, c_T (and on c_0 in part (ii)), but not on k, p, w, B .

(i) **Random restart.** If $(V_s)_{s \geq 0}$ is $\mathcal{G}_s$ -adapted and $\sup_s \|V_s\|_{L_{2p}} \leq B$, then $V_T 1_{T \leq k}$ is well-defined by $\sum_{s=0}^k V_s 1_{T=s}$ and

$$\| \Gamma_{T+1:k}^{(w)} V_T 1_{T \leq k} \|_{L_p} \leq C \frac{p}{wa} B \exp(-cwa k/p). \quad (310)$$

(ii) **Stable convolution.** If $c_0 > 0$, $(U_l)_{l \geq 0}$ is $\mathcal{G}_l$ -adapted, and $\|U_l\|_{L_{2p}} \leq B \exp(-c_0 wal/p)$ for all l , then

$$w \sum_{l=1}^k \| \Gamma_{l+1:k}^{(w)} U_l \|_{L_p} \leq C \frac{p}{a} B \exp(-cwa k/p). \quad (311)$$

Proof. The conditional form needed here is part of the technical input Lemma 17. Fix deterministic times $s \leq k$ and a $\mathcal{G}_s$ -measurable vector W_s . Conditional on $\mathcal{G}_s$, the future of either coupled base chain starts from its current state, and the product-stability constants are uniform over that state. Applying Eq. 271 gives

$$\mathbb{E} \left[\left\| \Gamma_{s+1:k}^{(w)} W_s \right\|_{L_p}^p \mid \mathcal{G}_s \right]^{1/p} \leq C_{\text{prod}} \exp(-c_{\text{prod}} wa(k-s)/p) \|W_s\|. \quad (312)$$

Integrating this conditional inequality gives

$$\left\| \Gamma_{s+1:k}^{(w)} W_s \right\|_{L_p} \leq C_{\text{prod}} \exp(-c_{\text{prod}} wa(k-s)/p) \|W_s\|_{L_p}. \quad (313)$$

For $s = k$ this is interpreted with the empty product $\Gamma_{k+1:k}^{(w)} = I$; after increasing C_{prod} the same display is trivial.

Since T is a stopping time, $1_{T=s}$ is $\mathcal{G}_s$ -measurable. Thus $V_s 1_{T=s}$ is an admissible input in Eq. 313. By Minkowski's inequality and the decomposition $\Gamma_{T+1:k}^{(w)} V_T 1_{T \leq k} = \sum_{s=0}^k \Gamma_{s+1:k}^{(w)} V_s 1_{T=s}$,

$$\left\| \Gamma_{T+1:k}^{(w)} V_T 1_{T \leq k} \right\|_{L_p} \leq C \sum_{s=0}^k \exp(-cwa(k-s)/p) \|V_s 1_{T=s}\|_{L_p}. \quad (314)$$

Holder's inequality gives

$$\|V_s 1_{T=s}\|_{L_p} \leq \|V_s\|_{L_{2p}} \mathbb{P}(T=s)^{1/(2p)} \leq B \mathbb{P}(T=s)^{1/(2p)}. \quad (315)$$

The tail bound Eq. 309 implies, for $s \geq 1$, $\mathbb{P}(T=s) \leq \mathbb{P}(T > s-1) \leq C_T e^{-c_T(s-1)}$, while the case $s=0$ is absorbed into the same bound after changing the constant. Since $p \geq 2$, $C_T^{1/(2p)}$ is bounded by a constant depending only on C_T , and therefore

$$\mathbb{P}(T=s)^{1/(2p)} \leq C \exp(-c_T s/(4p)). \quad (316)$$

The small-step condition $wa \leq 1$ gives, after decreasing c_T , $\exp(-c_T s/(4p)) \leq \exp(-c_T was/(4p))$. Hence

$$\sum_{s=0}^k \exp(-cwa(k-s)/p) \exp(-c_T was/(4p)) \leq C \frac{p}{wa} \exp(-c' wak/p). \quad (317)$$

Combining the last three displays proves Eq. 310 after renaming c' as c .

For the second estimate, each U_l is $\mathcal{G}_l$ -measurable, so Eq. 313 gives, including the empty-product case $l = k$,

$$\left\| \Gamma_{l+1:k}^{(w)} U_l \right\|_{L_p} \leq C \exp(-cwa(k-l)/p) \|U_l\|_{L_p} \leq C \exp(-cwa(k-l)/p) B \exp(-c_0 wal/p). \quad (318)$$

Therefore

$$w \sum_{l=1}^k \left\| \Gamma_{l+1:k}^{(w)} U_l \right\|_{L_p} \leq CwB \sum_{l=1}^k \exp(-cwa(k-l)/p) \exp(-c_0 wal/p). \quad (319)$$

The geometric convolution is bounded by $C(p/(wa)) \exp(-c' wak/p)$, with C, c' depending only on c, c_0 . Multiplication by w gives the factor p/a . Renaming c' as c proves Eq. 311. $\square$

For startup transfer, however, one must compare two copies of this display. The following lemma records the resulting full-state contraction. It is a local technical extension of Lemma 33: the $J^{(0)}$, $J^{(1)}$, and $J^{(2)}$ coordinates use the imported J contraction, while the $H^{(2)}$ coordinate is controlled by applying the local random-product stability estimate Lemma 22 to the difference of two representations Eq. 405.

The threshold $\alpha_{\text{st}}(p)$ was fixed above as the common ceiling for the local depth-two startup contraction and the random-product stability estimate. For $w \in \{\alpha, 2\alpha\}$ write

$$R_{k,\text{fin}}^{(w)} := J_{k,\text{fin}}^{(1,w)} + J_{k,\text{fin}}^{(2,w)} + H_{k,\text{fin}}^{(2,w)} \quad (320)$$

for the finite-start depth-two remainder, and define $R_{k,\text{aug}}^{(w)}$ analogously for a copy initialized from the invariant law of the full augmented chain $(Z_{k+1}, J_k^{(0,w)}, J_k^{(1,w)}, J_k^{(2,w)}, H_k^{(2,w)})$.

Lemma 23. (Full-state startup contraction for the depth-two augmented remainder.) Assume **UGE 1**, $\pi(\epsilon) = 0$, $\|\epsilon\|_\infty < \infty$, and $0 < \alpha$. There exist constants $c_{\text{st}} > 0$ and $C_{\text{st}} < \infty$, depending only on the problem constants and on the constants in Eq. 408 and Lemma 32, such that for every $p \geq 2$, every $q \geq 2$ satisfying $p \leq q/4$ and $2\alpha \leq \alpha_{\text{burn}}(p, q)$, every initial distribution ξ of the base chain with finite-start perturbation coordinates initialized at zero, every $w \in \{\alpha, 2\alpha\}$, and every $k \geq 0$, the finite-start and stationary augmented remainders can be coupled so that

$$\|R_{k,\text{fin}}^{(w)} - R_{k,\text{aug}}^{(w)}\|_{L_p} \leq A_{\text{st}}(p, q, w) \exp(-c_{\text{st}} w a k / p), \quad (321)$$

where

$$A_{\text{st}}(p, q, w) := C_{\text{st}}(1 + d^{1/q})p^7 + \frac{p^8}{a} t_{\text{mix}}^5 \sqrt{w/a} \log^3(1/(wa)). \quad (322)$$

Proof. Couple a finite-start copy, whose base chain has initial law ξ and whose perturbation coordinates are zero, with an augmented stationary copy drawn from the full invariant law constructed in Lemma 10. Let

$$\Delta J_k^{(\ell,w)} := J_{k,\text{fin}}^{(\ell,w)} - J_{k,\text{aug}}^{(\ell,w)}, \quad \Delta H_k^{(2,w)} := H_{k,\text{fin}}^{(2,w)} - H_{k,\text{aug}}^{(2,w)}. \quad (323)$$

Use the exact-coupling probability space behind Lemma 33 for the two base chains, and let T be their coupling time. UGE gives the uniform tail $\mathbb{P}(T > r) \leq C_T \exp(-c_T r)$, with constants independent of ξ . All constants below are uniform over the initial law because Eq. 408 and Lemma 17 are uniform over the current base state.

The $J^{(1)} + J^{(2)}$ part. Conditional on the initial augmented states, Lemma 33 and the coordinate projection of the cost Eq. 407 give, for $\ell \in \{0, 1, 2\}$,

$$\|J_{k,\text{fin}}^{(\ell,w)} - J_{k,\text{aug}}^{(\ell,w)}\|_{L_p} \leq C p^{7/2} t_{\text{mix}}^{5/2} \log^{3/2}(1/(wa)) e^{-c w a k / p} \mathbb{E}^{1/p} \left[c_{J,2}^{(w)} (Y_0^{\text{fin}}, Y_0^{\text{aug}})^p \right]. \quad (324)$$

The finite-start J coordinates are zero. For the augmented stationary copy, the finite-past construction of Lemma 10, the elementary stationary bounds for $J^{(0,w)}$ and $J^{(1,w)}$, and Lemma 31 for $J^{(2,w)}$ give

$$\mathbb{E}^{1/p} \left[c_{J,2}^{(w)} (Y_0^{\text{fin}}, Y_0^{\text{aug}})^p \right] \leq C(1 + d^{1/q}) p^{7/2} t_{\text{mix}}^{5/2} \sqrt{w/a} \log^{3/2}(1/(wa)). \quad (325)$$

Here the factor $1 + d^{1/q}$ also covers the indicator term in Eq. 407, and the estimate is uniform over the initial law ξ because only the stationary copy contributes to the initial cost. Combining Eq. 324 and Eq. 325 gives

$$A_J(p, q, w) := C(1 + d^{1/q}) p^{7/2} t_{\text{mix}}^{5/2} \sqrt{w/a} \log^3(1/(wa)). \quad (326)$$

Thus

$$\|\Delta J_k^{(1,w)}\|_{L_p} + \|\Delta J_k^{(2,w)}\|_{L_p} \leq A_J(p, q, w) \exp(-c w a k / p). \quad (327)$$

The p^7 summand in A_{st} is exactly this pure J -coordinate scale.

The $H^{(2)}$ part. This is the local extension beyond Lemma 33. The proof uses L_{2p} estimates; this is why the statement assumes $p \leq q/4$. With moment order $2p$, Lemma 32 and Fatou's lemma for the stationary finite-past limit give, for both finite-start and augmented stationary copies,

$$\|H_s^{(2,w)}\|_{L_{2p}} \leq B_{H(p,q,w)} := C(1 + d^{1/q}) p^{7/2} t_{\text{mix}}^{5/2} w^{3/2} \log^{3/2}(1/(wa)), \quad (328)$$

where the replacement of p by $2p$ is absorbed into C and the displayed power of p . We record the scale comparisons used below. Since $wa \leq 1$, $p \geq 2$, $t_{\text{mix}} \geq 1$, and $\log(1/(wa)) \geq \log 2$, the constants convention for fixed powers of a^{-1} gives

$$\begin{aligned} B_{H(p,q,w)} &\leq CA_J(p, q, w), \\ \frac{p}{wa} B_{H(p,q,w)} &\leq CA_J(p, q, w), \\ \frac{p}{a} A_J(2p, q, w) &\leq C(1 + d^{1/q}) \frac{p^8}{a} t_{\text{mix}}^5 \sqrt{w/a} \log^3(1/(wa)). \end{aligned} \quad (329)$$

Thus A_{st} is the sum of the J scale A_J and the convolution scale $(p/a)A_J(2p, q, w)$; the bad-coupling and random restart $H^{(2)}$ terms do not introduce additional powers of p , t_{mix} , w , or a^{-1} .

On the event $T \leq k$, the two base chains are identical after time T . Restarting the representation Eq. 405 at time T gives for each copy

$$H_k^{(2,w)} = \Gamma_{T+1:k}^{(w)} H_T^{(2,w)} - w \sum_{l=T+1}^k \Gamma_{l+1:k}^{(w)} \tilde{A}(Z_l) J_{l-1}^{(2,w)}, \quad (330)$$

where the product is the identity and the sum is empty when $T = k$. Subtracting the two restarted representations gives

$$\Delta H_k^{(2,w)} = \Gamma_{T+1:k}^{(w)} \Delta H_T^{(2,w)} - w \sum_{l=T+1}^k \Gamma_{l+1:k}^{(w)} \tilde{A}(Z_l) \Delta J_{l-1}^{(2,w)}, \quad (331)$$

on $\{T \leq k\}$.

On the bad event $\{T > k\}$, Holder's inequality gives

$$\|\Delta H_k^{(2,w)} 1_{T>k}\|_{L_p} \leq \left(\|H_{k,\text{fin}}^{(2,w)}\|_{L_{2p}} + \|H_{k,\text{aug}}^{(2,w)}\|_{L_{2p}} \right) \mathbb{P}(T > k)^{1/(2p)}. \quad (332)$$

The bound Eq. 328 controls both one-trajectory terms uniformly in k , while Eq. 309 gives $\mathbb{P}(T > k)^{1/(2p)} \leq C e^{-ck/p}$. Therefore

$$\|\Delta H_k^{(2,w)} 1_{T>k}\|_{L_p} \leq C B_{H(p,q,w)} e^{-ck/p}. \quad (333)$$

Since $wa \leq 1$, $e^{-ck/p} \leq e^{-cwak/p}$, and Eq. 329 puts this term at the target scale.

For the restart term on $\{T \leq k\}$, apply the random-time product estimate Lemma 22 with the adapted process $V_s = \Delta H_s^{(2,w)}$. The required input is exactly where the L_{2p} norm is used:

$$\sup_s \|V_s\|_{L_{2p}} \leq \|H_{s,\text{fin}}^{(2,w)}\|_{L_{2p}} + \|H_{s,\text{aug}}^{(2,w)}\|_{L_{2p}} \leq C B_{H(p,q,w)}. \quad (334)$$

Thus

$$\|\Gamma_{T+1:k}^{(w)} \Delta H_T^{(2,w)} 1_{T \leq k}\|_{L_p} \leq C \frac{p}{wa} B_{H(p,q,w)} e^{-cwak/p} \leq C A_{\text{st}}(p, q, w) e^{-cwak/p}. \quad (335)$$

For the convolution term in Eq. 331, set $U_l := \tilde{A}(Z_l) \Delta J_{l-1}^{(2,w)}$. This process is adapted to the joint filtration at time l . Applying Eq. 324 with moment order $2p$ is allowed by $p \leq q/4$ and gives

$$\|U_l\|_{L_{2p}} \leq C A_J(2p, q, w) \exp(-cwal/p), \quad (336)$$

where replacing $l-1$ by l only changes the constant. The stable convolution estimate Eq. 311 yields

$$\begin{aligned}
& w \sum_{l=1}^k \|\Gamma_{l+1:k}^{(w)} \tilde{A}(Z_l) \Delta J_{l-1}^{(2,w)}\|_{L_p} \\
& \leq C \frac{p}{a} A_J(2p, q, w) e^{-c w a k / p} \\
& \leq C A_{\text{st}}(p, q, w) e^{-c w a k / p}.
\end{aligned} \tag{337}$$

The last line is the only place where the p^8/a summand in A_{st} is needed. Combining the bad-event, restart, and convolution estimates gives

$$\|\Delta H_k^{(2,w)}\|_{L_p} \leq C A_{\text{st}}(p, q, w) e^{-c w a k / p}. \tag{338}$$

Finally,

$$R_{k, \text{fin}}^{(w)} - R_{k, \text{aug}}^{(w)} = \Delta J_k^{(1,w)} + \Delta J_k^{(2,w)} + \Delta H_k^{(2,w)}. \tag{339}$$

The triangle inequality, Eq. 327, and Eq. 338 prove Eq. 321 after renaming the constant and the exponential rate. The displayed A_{st} is the only allowed dependence on p, q, w ; all other factors are fixed problem constants, uniformly over ξ . $\square$

Define the RR startup discrepancy accumulated over the burned-in window by

$$\mathcal{U}_{n, n_0}^{\text{start,RR}} := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \left[2 \left(R_{k, \text{fin}}^{(\alpha)} - R_{k, \text{aug}}^{(\alpha)} \right) - \left(R_{k, \text{fin}}^{(2\alpha)} - R_{k, \text{aug}}^{(2\alpha)} \right) \right]. \tag{340}$$

Lemma 24. (Accumulated startup transfer.) Under Lemma 23, if $0 < \alpha$, then for every $p \geq 2$ and every $q \geq 2$ satisfying $p \leq q/4$ and $2\alpha \leq \alpha_{\text{burn}}(p, q)$,

$$\|\mathcal{U}_{n, n_0}^{\text{start,RR}}\|_{L_p} \leq \frac{C_{\text{start,RR}} p A_{\text{st}}(p, q, \alpha)}{\alpha a \sqrt{m}} \exp(-c_{\text{st}} \alpha a n_0 / p). \tag{341}$$

Proof. Apply Eq. 321 at $w = \alpha$ and $w = 2\alpha$, then use the triangle inequality:

$$\|\mathcal{U}_{n, n_0}^{\text{start,RR}}\|_{L_p} \leq \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} (2A_{\text{st}}(p, q, \alpha) e^{-c_{\text{st}} \alpha a k / p} + A_{\text{st}}(p, q, 2\alpha) e^{-2c_{\text{st}} \alpha a k / p}). \tag{342}$$

Since $\alpha a \leq 1/4$, $A_{\text{st}}(p, q, 2\alpha) \leq C A_{\text{st}}(p, q, \alpha)$ and $1 - \exp(-c_{\text{st}} \alpha a / p) \geq C^{-1} \alpha a / p$. Extending the geometric sum to infinity gives Eq. 341. $\square$

Corollary 11. (Mixing-scale burn-in removes the startup discrepancy.) If, for some $\beta > 0$,

$$n_0 \geq \frac{\beta p}{c_{\text{st}} \alpha a} \log n, \tag{343}$$

then

$$\|\mathcal{U}_{n, n_0}^{\text{start,RR}}\|_{L_p} \leq \frac{C_{\text{start,RR}} p A_{\text{st}}(p, q, \alpha)}{\alpha a \sqrt{m}} n^{-\beta}. \tag{344}$$

At the balanced scale $\alpha = c n^{-1/2}$, with $m \geq n/2$ and p, q logarithmic in n , this is $\text{polylog}(n) n^{-1/4-\beta}$. Thus the Berry–Esseen choice $p \asymp \log n$ requires a mixing-scale burn-in with logarithmic-square factor, n_0 of order $(\alpha a)^{-1} \log^2 n$, under this L_p contraction.

Proof. Substitute Eq. 343 into Eq. 341. For the balanced scale, $A_{\text{st}}(p, q, \alpha) = \text{polylog}(n) \alpha^{1/2}$, hence $A_{\text{st}}(p, q, \alpha) / (\alpha \sqrt{m}) = O(\text{polylog}(n) n^{-1/4})$. $\square$

5.9. Burned-in Depth-Two Misadjustment Bound

Define the finite-start burned-in RR misadjustment by

$$R_{n,n_0,\text{fin}}^{\text{mis,RR}} := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \left(2R_{k,\text{fin}}^{(\alpha)} - R_{k,\text{fin}}^{(2\alpha)} \right). \quad (345)$$

For comparison, let

$$R_{m,\text{aug}}^{\text{mis,RR}} := \frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} \left(2R_{j,\text{aug}}^{(\alpha)} - R_{j,\text{aug}}^{(2\alpha)} \right) \quad (346)$$

be the stationary augmented-chain depth-two misadjustment over a window of length m . By stationarity, the same distribution is obtained if the sum in Eq. 346 is taken over $j = n_0, \dots, n-1$.

Theorem 4. (Burned-in PR-averaged RR misadjustment bound.) Assume **UGE 1**, $\pi(\epsilon) = 0$, $\|\epsilon\|_\infty < \infty$, and $0 < \alpha$. Set

$$\Phi_+(p, \alpha) := 1 + p^{3/2} t_{\text{mix}}^{1/2}/a + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha/a}. \quad (347)$$

There exists a constant $C_{\text{burn,mis}}$ depending only on the stationary misadjustment constants, the startup-contraction constants, and the problem constants such that, for every $p \geq 2$, every $q \geq 2$ satisfying $p \leq q/4$ and $2\alpha \leq \alpha_{\text{burn}}(p, q)$, and every $m \geq 2$,

$$\begin{aligned} \| R_{n,n_0,\text{fin}}^{\text{mis,RR}} \|_{L_p} &\leq C_{\text{burn,mis}} \sqrt{m} \alpha^2 \\ &\quad + C_{\text{burn,mis}} (1 + d^{1/q}) p^{7/2} t_{\text{mix}}^{5/2} \sqrt{m} \alpha^{3/2} \log^{3/2}(1/(\alpha a)) \\ &\quad + C_{\text{burn,mis}} p^{3/2} \sqrt{\alpha} \\ &\quad + C_{\text{burn,mis}} p^3 (\alpha m)^{-1/2} \log^{1/2}(1/(\alpha a)) \\ &\quad + C_{\text{burn,mis}} \Phi_+(p, \alpha) m^{-1/2} \\ &\quad + \frac{C_{\text{burn,mis}} p A_{\text{st}}(p, q, \alpha)}{\alpha a \sqrt{m}} \exp(-c_{\text{st}} \alpha a n_0 / p). \end{aligned} \quad (348)$$

Proof. Couple the finite-start and stationary augmented-chain remainders as in Lemma 23, and define the stationary window on the same indices by

$$\tilde{R}_{n,n_0,\text{aug}}^{\text{mis,RR}} := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \left(2R_{k,\text{aug}}^{(\alpha)} - R_{k,\text{aug}}^{(2\alpha)} \right). \quad (349)$$

Then, with the notation of Eq. 340,

$$R_{n,n_0,\text{fin}}^{\text{mis,RR}} = \tilde{R}_{n,n_0,\text{aug}}^{\text{mis,RR}} + \mathcal{U}_{n,n_0}^{\text{start,RR}} \quad (350)$$

under this coupling. Therefore

$$\| R_{n,n_0,\text{fin}}^{\text{mis,RR}} \|_{L_p} \leq \| \tilde{R}_{n,n_0,\text{aug}}^{\text{mis,RR}} \|_{L_p} + \| \mathcal{U}_{n,n_0}^{\text{start,RR}} \|_{L_p}. \quad (351)$$

By stationarity, $\| \tilde{R}_{n,n_0,\text{aug}}^{\text{mis,RR}} \|_{L_p} = \| R_{m,\text{aug}}^{\text{mis,RR}} \|_{L_p}$, and the latter is exactly the stationary augmented-chain misadjustment bound Theorem 2 with n replaced by m . The second term is Lemma 24. Combining the two estimates gives Eq. 348. $\square$

Corollary 12. (Balanced-scale burned-in misadjustment rate.) Assume the hypotheses of Theorem 4.

Let $\alpha = c n^{-1/2}$, $m \geq n/2$, $p = \max(2, \lceil \log n \rceil)$, and $q = \max(4p, \lceil \log(e d) \rceil, 2)$. If, for some fixed $\beta > 0$,

$$n_0 \geq \frac{\beta p}{c_{\text{st}} \alpha a} \log n, \quad (352)$$

and n is large enough that $2\alpha \leq \alpha_{\text{burn}}(p, q)$, then

$$\| R_{n, n_0, \text{fin}}^{\text{mis, RR}} \|_{L_p} \leq C_{\text{burn, mis}} \text{polylog}(n) n^{-1/4}. \quad (353)$$

Proof. The stationary part of Eq. 348 is the bound of Corollary 4 with n replaced by m , and $m \geq n/2$ keeps the rate unchanged. The startup term is controlled by Corollary 11 and is $\text{polylog}(n) n^{-1/4-\beta}$ at this scale. $\square$

5.10. Burned-in Martingale Berry–Esseen

The depth-zero martingale in Lemma 20 has deterministic coefficients, but its normalization is based on the effective sample size $m = n - n_0$. We assume $m \geq n/2$ and impose the mixing-scale burn-in lower bounds with logarithmic factors, so the inhomogeneous concentration term obtained from $\sqrt{p t_{\text{mix}} \sum c_i^2}$ and then written as $C(u) \sqrt{p n}$ can be transferred to the m scale.

Fix $u \in \mathbb{R}^d$ and put $X_l^{\text{bRR}} := u^\top \Delta M_l^{\text{bRR}}$ for $2 \leq l \leq n - 1$. Then

$$|X_l^{\text{bRR}}| \leq \kappa_{\text{burn}}(u), \quad \kappa_{\text{burn}}(u) := 6 t_{\text{mix}} C_{\text{burn, Q}} \| \epsilon \|_\infty \| u \|. \quad (354)$$

Indeed, this is the same bounded-increment estimate as in the stationary chapter, with $C_{\text{burn, Q}}$ replacing the full-window weight bound. Set

$$s_{n, n_0}^2(u) := m \sigma_{n, n_0}^2, \text{bRR}(u). \quad (355)$$

Under Eq. 288, $s_{n, n_0}^2(u) \geq m \sigma^2(u)/2$. Also, $m \geq n/2$ and the uniform bound $\|Q_l^{\text{bRR}}\| \leq C_{\text{burn, Q}}$ imply the deterministic upper bound

$$s_{n, n_0}^2(u) \leq 2m C_{\text{burn, Q}}^2 \| \Sigma_\epsilon^{(\text{M})} \| \| u \|^2. \quad (356)$$

Theorem 5. (Burned-in martingale Berry–Esseen bound.) Assume **UGE 1**, $\pi(\epsilon) = 0$, $\| \epsilon \|_\infty < \infty$, $\sigma^2(u) > 0$, $0 < \alpha$, $2\alpha \leq \alpha_\infty$, $m \geq n/2$, and the burned-in variance lower-bound condition Eq. 288. There exist constants $C_{\text{bK}, 1}(u), C_{\text{bK}, 2}(u) > 0$, depending only on $\|u\|$, $\sigma(u)$, $C_{\text{burn, Q}}$, t_{mix} , $\|\epsilon\|_\infty$, $\|\Sigma_\epsilon^{(\text{M})}\|$, and the universal constants in Eq. 398, such that for every $n \geq 3$,

$$d_K \left(\frac{u^\top M_{n, n_0}^{\text{bRR}}}{\sqrt{m} \sigma_{n, n_0}^{\text{bRR}}(u)}, \mathcal{N}(0, 1) \right) \leq \frac{C_{\text{bK}, 1}(u) \log^{3/4} n}{m^{1/4}} + \frac{C_{\text{bK}, 2}(u) \log n}{\sqrt{m}}. \quad (357)$$

Proof. Apply Eq. 398 to the martingale differences X_l^{bRR} . The bounded-increment input is Eq. 354, and the target variance is $s_{n, n_0}^2(u)$. The first and third Bolthausen–Fan terms are bounded exactly as before, with n replaced by m in the denominator and the harmless factor $n/m \leq 2$ absorbed into the constants. Explicitly, from $s_{n, n_0}^2(u) \geq m \sigma^2(u)/2$ and $m \geq n/2$,

$$\frac{(2n+1) \log(2n+1)}{s_{n, n_0}^3(u)} \leq C(u) \frac{\log n}{\sqrt{m}}. \quad (358)$$

The bounded-increment Lindeberg term is controlled in the same way, using Eq. 356 and $s_{n, n_0}^2(u) \geq m \sigma^2(u)/2$. For the conditional-variance term use Lemma 21:

$$\mathbb{E}^{1/p} \left[| u^\top \langle M^{\text{bRR}} \rangle_{n, n_0} u - s_{n, n_0}^2(u) |^p \right] \leq C \sqrt{p n}. \quad (359)$$

Here the constant contains the explicit bracket coefficient $C_{\text{burn},Q}^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2}$ from Lemma 21; the underlying concentration input is Eq. 397. Together with $m \geq n/2$, Eq. 288, and Eq. 356, this gives

$$\text{Term II} \leq C(u) p^{(3p+1)/(2(2p+1))} m^{-p/(2(2p+1))}. \quad (360)$$

Taking $p = \lceil \log n \rceil$ gives the displayed $\log^{3/4}(n)m^{-1/4}$ bound. The classical Bolthausen and Lindeberg terms give the second displayed term. $\square$

5.11. Finite-Window Smoothing Assembly

The finite-start scalar statistic decomposes as

$$T_{n,n_0}^{\text{RR}}(u) = -\frac{u^\top M_{n,n_0}^{\text{bRR}}}{\sqrt{m}} + \mathcal{R}_{n,n_0,\text{fin}}^{\text{bRR}}(u), \quad (361)$$

where the composite non-Gaussian remainder is

$$\mathcal{R}_{n,n_0,\text{fin}}^{\text{bRR}}(u) := D_{\text{tr},n,n_0}^{\text{RR}}(u) + \mathcal{J}_{n,n_0}^{\text{init,RR}}(u) + u^\top D_{2,n,n_0}^{\text{bRR}} + u^\top R_{n,n_0,\text{fin}}^{\text{mis,RR}}. \quad (362)$$

The terms are controlled by Lemma 16, Lemma 18, Lemma 20, and Theorem 4, respectively.

For reference, the finite-window assembly uses the following components:

Component	Input bound	Role in smoothing
$D_{\text{tr},n,n_0}^{\text{RR}}(u)$	Eq. 264	Deterministic transient from $\theta_0 - \theta^*$.
$\mathcal{J}_{n,n_0}^{\text{init,RR}}(u)$	Eq. 275	Random initial-product discrepancy.
$u^\top D_{2,n,n_0}^{\text{bRR}}$	Eq. 295	Poisson Abel boundary remainder.
$u^\top R_{n,n_0,\text{fin}}^{\text{mis,RR}}$	Eq. 348	Depth-two RR misadjustment plus startup transfer.
$u^\top M_{n,n_0}^{\text{bRR}}$	Theorem 5 and Corollary 10	Leading martingale Berry–Esseen term.

Let $\mathcal{B}_{\text{mis}}(m, n_0, p, q, \alpha)$ denote the right-hand side of Eq. 348.

Lemma 25. (Burned-in composite remainder bound.) Assume the hypotheses of Lemma 16, Lemma 18, Lemma 20, and Theorem 4. Then, for every $p \geq 2$ and every $q \geq 2$ satisfying $p \leq q/4$ and

$$2\alpha \leq \alpha_{\text{burn}}(p, q),$$

$$\begin{aligned} \|\mathcal{R}_{n,n_0,\text{fin}}^{\text{bRR}}(u)\|_{L_p} &\leq \frac{5\sqrt{\kappa_Q} \|u\| \|\theta_0 - \theta^*\|}{\alpha a \sqrt{m}} (1 - \alpha a)^{n_0/2} \\ &\quad + \frac{C_{\text{init,RR}} \|u\| \|\theta_0 - \theta^*\| p}{\alpha a \sqrt{m}} \exp(-c_{\text{init}} \alpha a n_0 / p) \\ &\quad + \frac{C_{\text{burn,D2}} \|u\|}{\sqrt{m}} + \|u\| \mathcal{B}_{\text{mis}}(m, n_0, p, q, \alpha), \end{aligned} \quad (363)$$

where

$$C_{\text{burn,D2}} := 3 t_{\text{mix}} \|\epsilon\|_\infty \left(C_{\text{burn,Q}} + \frac{C_{\text{burn,V}}}{a^2} \right). \quad (364)$$

Proof. Apply the triangle inequality in Eq. 362. The deterministic part is Eq. 264. The random initial-product term is Eq. 275. The Poisson boundary term is bounded by

$$\|u^\top D_{2,n,n_0}^{\text{bRR}}\|_{L_p} \leq \|u\| \|D_{2,n,n_0}^{\text{bRR}}\|_\infty \leq \frac{C_{\text{burn,D2}} \|u\|}{\sqrt{m}} \quad (365)$$

using Eq. 295. The last term is exactly Eq. 348. $\square$

Dividing Eq. 361 by the finite-window normalization gives

$$\Xi_{n,n_0}^{\text{bRR}}(u) = X_{n,n_0}^{\text{bRR}} + Y_{n,n_0}^{\text{bRR}}, \quad (366)$$

where

$$X_{n,n_0}^{\text{bRR}} := -\frac{u^\top M_{n,n_0}^{\text{bRR}}}{\sqrt{m} \sigma_{n,n_0}^{\text{bRR}}(u)}, \quad Y_{n,n_0}^{\text{bRR}} := \frac{\mathcal{R}_{n,n_0,\text{fin}}^{\text{bRR}}(u)}{\sigma_{n,n_0}^{\text{bRR}}(u)}. \quad (367)$$

Theorem 6. (Finite-window deterministic-start burned-in PR-averaged RR Berry–Esseen bound.) Assume the hypotheses of Theorem 5, Lemma 18, and Theorem 4. Let $m := n - n_0$ and $p = \max(2, \lceil \log n \rceil)$ and $q = \max(4p, \lceil \log(e d) \rceil, 2)$. Then $p \geq 2$, $q \geq 2$, and $p \leq q/4$. If $(n, n_0, p, q, \alpha, u)$ is in the admissible burn-in regime Eq. 290, then

$$\begin{aligned} d_K(\Xi_{n,n_0}^{\text{bRR}}(u), \mathcal{N}(0, 1)) &\leq \frac{C_{\text{bK},1}(u) \log^{3/4} n}{m^{1/4}} + \frac{C_{\text{bK},2}(u) \log n}{\sqrt{m}} \\ &\quad + \frac{e \|\mathcal{R}_{n,n_0,\text{fin}}^{\text{bRR}}(u)\|_{L_p}}{\sqrt{2\pi} \sigma_{n,n_0}^{\text{bRR}}(u)} + \frac{e}{n}, \end{aligned} \quad (368)$$

with the composite remainder bounded by Lemma 25. The bound is uniform over the initial law $\xi = \mathcal{L}(Z_0)$ of the base Markov chain.

Proof. Apply the smoothing inequality Eq. 222 to the split Eq. 366. The martingale Berry–Esseen term is Theorem 5; the minus sign in X_{n,n_0}^{bRR} is handled by applying that theorem to the signed increments $-u^\top \Delta M_t^{\text{bRR}}$. The bounded-increment constant and predictable quadratic variation are unchanged. Since $p = \max(2, \lceil \log n \rceil)$, the smoothing tail e^{-p} is at most e/n . The L_p norm of the perturbation is

$$\|Y_{n,n_0}^{\text{bRR}}\|_{L_p} = \frac{\|\mathcal{R}_{n,n_0,\text{fin}}^{\text{bRR}}(u)\|_{L_p}}{\sigma_{n,n_0}^{\text{bRR}}(u)}, \quad (369)$$

which gives Eq. 368. $\square$

5.12. Balanced Burn-in Berry–Esseen Bound

The finite-window bound uses $\sigma_{n,n_0}^{\text{bRR}}(u)$. The final inference statement uses $\sigma(u)$, via the following burned-in analogue of Corollary 7.

Lemma 26. (Burned-in normalization transfer.) Assume $\sigma^2(u) > 0$ and the burned-in variance lower-bound condition Eq. 288. Put

$$r_{n,n_0}(u) := \frac{\sigma_{n,n_0}^{\text{bRR}}(u)}{\sigma(u)}. \quad (370)$$

Then

$$\frac{1}{\sqrt{2}} \leq r_{n,n_0}(u) \leq \sqrt{3/2}, \quad (371)$$

and, for any real random variable W ,

$$d_K(r_{n,n_0}(u)W, \mathcal{N}(0, 1)) \leq d_K(W, \mathcal{N}(0, 1)) + \frac{C_{\text{norm}} C_{\text{burn},3} \|u\|^2}{m \alpha a \sigma^2(u)}, \quad (372)$$

where C_{norm} is a universal constant.

Proof. The variance lower-bound condition and Eq. 284 give

$$|\sigma_{n,n_0}^{2, \text{bRR}}(u) - \sigma^2(u)| \leq \sigma^2(u)/2. \quad (373)$$

Hence $r_{n,n_0}(u) \in [1/\sqrt{2}, \sqrt{3/2}]$. For r in this compact interval,

$$\sup_x |\Phi(x/r) - \Phi(x)| \leq C_{\text{norm}} |r - 1|, \quad (374)$$

because $\sup_x |x\varphi(x)| < \infty$. Therefore

$$d_K(rW, \mathcal{N}(0, 1)) \leq d_K(W, \mathcal{N}(0, 1)) + C_{\text{norm}} |r - 1|. \quad (375)$$

Finally,

$$|r_{n,n_0}(u) - 1| = \frac{|\sigma_{n,n_0}^{2, \text{bRR}}(u) - \sigma^2(u)|}{\sigma(u) (\sigma_{n,n_0}^{\text{bRR}}(u) + \sigma(u))} \leq \frac{C_{\text{burn},3} \|u\|^2}{m \alpha a \sigma^2(u)}, \quad (376)$$

using Eq. 284 and $\sigma_{n,n_0}^{\text{bRR}}(u) + \sigma(u) \geq \sigma(u)$. $\square$

Theorem 7. (Balanced-scale deterministic-start burned-in PR-averaged RR Berry–Esseen bound.) Assume Assumptions 1–3 from Section 1.5. Assume also the Lyapunov contraction Eq. 13 for the two step sizes used below and the external inputs and local extensions summarized in Section 7. Fix $c > 0$ and set

$$\alpha := c n^{-1/2}, \quad m := n - n_0, \quad p := \max(2, \lceil \log n \rceil), \quad q := \max(4p, \lceil \log(e d) \rceil, 2). \quad (377)$$

Then $p \geq 2$, $q \geq 2$, and $p \leq q/4$. Use the deterministic-start admissibility threshold $\alpha_{\text{burn}}(p, q)$ defined in Eq. 91. Suppose $(n, n_0, p, q, \alpha, u)$ is in the admissible burn-in regime Eq. 290. Assume also that the burn-in satisfies the explicit mixing-scale conditions with logarithmic factors

$$n_0 \geq \frac{2}{\alpha a} \log n, \quad n_0 \geq \frac{p}{c_{\text{init}} \alpha a} \log n, \quad n_0 \geq \frac{p}{c_{\text{st}} \alpha a} \log n. \quad (378)$$

Then there exists a finite constant $C_{\text{burn,final}}(u, c, \theta_0)$, depending only on $u, c, \|\theta_0 - \theta^*\|$, and the problem and universal constants in the preceding bounds, but not on ξ , such that

$$d_K(\Xi_{n, n_0}^{\text{bRR}}(u), \mathcal{N}(0, 1)) \leq \frac{C_{\text{burn,final}}(u, c, \theta_0) \text{polylog}(n)}{n^{1/4}}, \quad (379)$$

and

$$d_K(\Xi_{n, n_0}^{\text{asy,RR}}(u), \mathcal{N}(0, 1)) \leq \frac{C_{\text{burn,final}}(u, c, \theta_0) \text{polylog}(n)}{n^{1/4}}. \quad (380)$$

Proof. Apply the finite-window assembly theorem Theorem 6. Since $m \geq n/2$, polynomial prefactors in n may be written on the m scale up to universal constants; for instance $\sqrt{n}/m \leq \sqrt{2}/\sqrt{m}$. In particular, the martingale terms satisfy

$$\frac{\log^{3/4} n}{m^{1/4}} + \frac{\log n}{\sqrt{m}} \leq C \frac{\text{polylog}(n)}{n^{1/4}}. \quad (381)$$

It remains to bound the composite remainder in Lemma 25. The first condition in Eq. 378 is Eq. 268 with $\beta = 1$; by Corollary 8 and $\alpha = c n^{-1/2}$, $m \geq n/2$, the deterministic transient is $O(n^{-1})$. The second condition is Eq. 280 with $\beta = 1$, so Corollary 9 makes the random initial-product term $O(\text{polylog}(n)n^{-1})$. The Poisson Abel remainder is $O(m^{-1/2})$. The third condition in Eq. 378 is the startup condition Eq. 343 with $\beta = 1$, so Corollary 12 gives

$$\|R_{n, n_0, \text{fin}}^{\text{mis,RR}}\|_{L_p} \leq C \text{polylog}(n) n^{-1/4}. \quad (382)$$

Together with $\sigma_{n, n_0}^{\text{bRR}}(u) \geq \sigma(u)/\sqrt{2}$, this makes the smoothing remainder in Eq. 368 of order $\text{polylog}(n)n^{-1/4}$. The smoothing tail e/n is lower order, so Eq. 379 follows.

For the asymptotic normalization, write

$$\Xi_{n, n_0}^{\text{asy,RR}}(u) = r_{n, n_0}(u) \Xi_{n, n_0}^{\text{bRR}}(u). \quad (383)$$

By Lemma 26, the additional cost is at most

$$\frac{C \|u\|^2}{m \alpha a \sigma^2(u)} = O(n^{-1/2}), \quad (384)$$

because $m \geq n/2$ and $\alpha = c n^{-1/2}$. This is absorbed into the balanced finite-window rate, proving Eq. 380. $\square$

The admissible burn-in regime Eq. 290 collects the finite- n admissibility requirements. The inequality $2\alpha \leq \alpha_{\text{burn}}(p, q)$ enforces the Lyapunov small-step ceiling, the local inverse ceiling, the Levin depth-two admissibility threshold, the random-product stability estimate Lemma 17, and the full-state startup

contraction Lemma 23. Since $\alpha = cn^{-1/2}$ and $m \geq n/2$, the elementary step-size constraints and Eq. 288 hold automatically for all sufficiently large n . The remaining non-elementary large- n requirement is

$$2cn^{-1/2} \leq \alpha_{\text{burn}}(p, q), \quad (385)$$

with p, q as in the theorem. Under this Levin/startup admissibility condition, the large- n reading of the theorem keeps only $m \geq n/2$ and the burn-in lower bounds in Eq. 378.

Corollary 13. ($\sqrt{n}$ -normalization for the burned-in RR statistic.) Under the assumptions of Theorem 7 and the admissible burn-in regime Eq. 290, in particular $m = n - n_0 \geq n/2$, $p \leq q/4$, $\sigma^2(u) > 0$, $2\alpha \leq \alpha_{\text{burn}}(p, q)$, and the burned-in variance lower bound Eq. 288, define the final scalar statistic

$$T_{n, n_0}^{\text{RR}, n}(u) := \sqrt{n} u^\top (\bar{\theta}_{n, n_0}^{(\text{RR}, \alpha)} - \theta^*) = \sqrt{n/m} T_{n, n_0}^{\text{RR}}(u), \quad (386)$$

and its asymptotic normalization

$$\Xi_{n, n_0}^{\text{RR}}(u) := \frac{T_{n, n_0}^{\text{RR}, n}(u)}{\sigma(u)}. \quad (387)$$

For the Berry–Esseen moment choice $p = \max(2, \lceil \log n \rceil)$ in Theorem 7, the lower burn-in conditions Eq. 378 are implied by

$$n_0 \geq C_- (\alpha a)^{-1} \log^2 n \quad (388)$$

with any fixed C_- large enough. At the balanced scale $\alpha = cn^{-1/2}$ this is $n_0 = O(n^{1/2} \log^2 n)$, not a purely logarithmic burn-in in n . If, in addition, the burn-in window stays in the same mixing-scale window with logarithmic-square factor,

$$C_- (\alpha a)^{-1} \log^2 n \leq n_0 \leq C_0 (\alpha a)^{-1} \log^2 n \quad (389)$$

for some finite C_0 and such a fixed C_- , then there exists a finite constant $C_{\text{burn}, n}(u, c, \theta_0, C_0, C_-)$, independent of ξ , such that

$$d_K(\Xi_{n, n_0}^{\text{RR}}(u), \mathcal{N}(0, 1)) \leq \frac{C_{\text{burn}, n}(u, c, \theta_0, C_0, C_-) \text{polylog}(n)}{n^{1/4}}. \quad (390)$$

Proof. Put $s_{n, n_0} := \sqrt{n/m}$. Since Eq. 290 gives $m \geq n/2$,

$$0 \leq s_{n, n_0} - 1 = \frac{n_0}{m(s_{n, n_0} + 1)} \leq \frac{2n_0}{n}. \quad (391)$$

For every real random variable W and every $s \in [1, \sqrt{2}]$,

$$d_K(sW, \mathcal{N}(0, 1)) \leq d_K(W, \mathcal{N}(0, 1)) + C |s - 1|, \quad (392)$$

by the same scaling argument as in Lemma 26. Applying this with $W = \Xi_{n, n_0}^{\text{RR}}(u)$ and using Theorem 7 gives the already proved $\text{polylog}(n)n^{-1/4}$ term. The upper burn-in bound in Eq. 389 and $\alpha = cn^{-1/2}$ give

$$|s_{n, n_0} - 1| \leq \frac{2C_0 \log^2 n}{ca n^{1/2}}, \quad (393)$$

which is lower order and is absorbed into the same balanced-scale rate. $\square$

The lower side of Eq. 389 is the logarithmic-square mixing-scale burn-in needed by the current L_p startup contraction; the upper side keeps the $\sqrt{n/m}$ rescaling lower order. At $\alpha = cn^{-1/2}$ this window has order $n^{1/2} \log^2 n$.

6. Numerical Experiments and Covariance-Estimator Diagnostics

This chapter records the numerical evidence currently available for the constant-stepsize Richardson–Romberg estimator and explains how the long-run-variance estimators used in the experiments fit into the inference problem. The results in this chapter are not used in the proofs of Theorem 3 or Theorem 7. They are meant to validate the finite-sample behavior suggested by the theory and to identify the next theoretical questions.

6.1. Why OBM and lugsail variance estimation enter the experiments

The main theorems above are stated either with a deterministic finite-window variance proxy or with the asymptotic variance $\sigma^2(u) = u^\top \Sigma_\infty u$. In an actual confidence interval this quantity must be estimated from the observed dependent trajectory. The naive sample variance of the iterates estimates the marginal variance, not the long-run variance, and therefore misses the serial-correlation correction.

For a scalar projected trajectory $Y_t = u^\top \theta_t$, the overlapping batch means (OBM) estimator with block size b is

$$\hat{\sigma}_{\text{OBM}}^2(b) = \frac{b}{T-b+1} \sum_{s=0}^{T-b} (\bar{Y}_{s,b} - \bar{Y}_T)^2, \quad \bar{Y}_{s,b} = \frac{1}{b} \sum_{j=s}^{s+b-1} Y_j. \quad (394)$$

It is the batch-means analogue of a Bartlett-window spectral estimator at frequency zero. This is the appropriate object for estimating the time-average covariance in Markov chain CLTs; see Flegal and Jones (2010) and Liu, Vats, and Flegal (2022).

The price is a window bias. For Bartlett/OBM estimators the standard asymptotic template is

$$\mathbb{E} \hat{\sigma}_{\text{OBM}}^2(b) = \sigma^2(u) + \frac{c_1(u)}{b} + \text{lower order terms}, \quad (395)$$

and the MSE has the characteristic bias–variance tradeoff $c_1(u)^2/b^2 + C(u)b/T$. For positively correlated output, the leading bias is typically negative, so confidence intervals based on the raw OBM estimator can be too narrow.

Lugsail windows are designed to reduce this leading negative bias (Vats and Flegal, 2022). In the implementation used here, the lugsail version is the block-size Richardson–Romberg combination

$$\hat{\sigma}_{\text{OBM-RR}}^2(b, \lambda) = \frac{\lambda}{\lambda-1} \hat{\sigma}_{\text{OBM}}^2(\lambda b) - \frac{1}{\lambda-1} \hat{\sigma}_{\text{OBM}}^2(b), \quad \lambda > 1. \quad (396)$$

For the Bartlett kernel, $\lambda = 2$ cancels the leading $1/b$ bias term. This correction is conceptually separate from Richardson–Romberg extrapolation in the stepsize. RR in α debiases the point estimator $\bar{\theta}^{(\text{RR}, \alpha)}$; lugsail or OBM-RR in b debiases the estimated standard error. In this thesis, OBM and lugsail are investigated experimentally only. A theorem for OBM/lugsail covariance estimation along RR-averaged constant-stepsize LSA trajectories is left for future work. That future analysis should combine the OBM and lugsail theory above with finite-sample spectral-density corrections in the spirit of Ng and Perron (1996) and with recent batch-means methodology for stochastic-gradient inference such as Singh, Shukla, and Vats (2025).

6.2. Experiments already completed

The main comparison experiment follows the finite-state Markovian LSA setup used in Huo, Chen, and Xie (2024). It uses 100 independently generated problems, 100 trajectories per problem, dimension $d = 5$, 10 Markov states, and trajectory length $T = 10^6$. The constant-stepsize RR estimator is compared to single constant stepsizes, diminishing-stepsize baselines, Polyak–Ruppert averaging, OBM, multiplier sub-sample bootstrap, and the OBM-RR/lugsail variance estimator. The default PR schedule is $c_0 = 200, k_0 = 20000, \gamma = 0.65$.

Method	L2	CI width	Coverage
$\alpha = 0.2$ constant	26.67	8.36	0.5%
$\alpha = 0.02$ constant	13.93	8.27	40.5%
RR constant stepsizes	4.52	8.22	94.0%
Diminishing $0.2/\sqrt{k}$	6.80	10.34	90.0%
PR + OBM	5.35	8.04	92.0%
RR + OBM	4.52	8.23	95.0%
RR + OBM-RR	4.52	8.22	95.0%

The entries are medians over the 100 problems; L2 error and CI width are reported in units of 10^{-3} . The main empirical message is that stepsize RR removes the large constant-stepsize bias without widening the intervals. The single constant-stepsize branches have poor coverage because their point-estimator bias dominates any variance-estimator correction. OBM-RR/lugsail is essentially neutral in this long-horizon run: at $T = 10^6$, the OBM window bias is already small enough that the variance inflation from the lugsail correction brings no visible coverage improvement.

A separate lugsail bias–variance experiment isolates the covariance estimator itself. It compares OBM and OBM-RR over a grid of block sizes and uses the analytic value of $\sigma^2(u)$ as ground truth. For PR iterates, OBM-RR with $\lambda = 2$ reduces the leading window bias and shifts the useful block-size range downward. At $T = 10^5$, the best OBM point has relative bias about -2.5% and MSE 0.72, while OBM-RR with $\lambda = 2$ has relative bias about -0.7% and MSE 0.34. Across $T \in \{20000, 50000, 100000\}$, the observed MSE reduction is roughly 40–55%. For constant-stepsize and RR iterate paths, the improvement is more limited because part of the error is SA transient or point-estimator bias, not Bartlett-window bias; lugsail cannot correct that component.

These experiments support the following interpretation. Step-size RR is the primary tool for improving the center of the confidence interval. OBM is the baseline practical estimator of Σ_∞ . Lugsail/OBM-RR is useful when the long-run-variance estimator has visible negative window bias, mainly in shorter-horizon or more persistent regimes.

6.3. Additional computations needed

The current experiments are enough to motivate the method, but a complete experimental section should add the following diagnostics before making strong empirical claims.

Computation	Purpose
Theory-aligned RR stepsize pairs	Run two-level pairs $(\alpha, 2\alpha)$, not only the Huo-style pair $(0.02, 0.2)$, to match the assumptions of the main theorem and separate Vandermonde RR from the specific two-level proof.
Coverage as a function of T	Repeat RR+OBM and RR+OBM-RR for $T \in \{2 \cdot 10^4, 5 \cdot 10^4, 10^5, 3 \cdot 10^5, 10^6\}$ to identify the horizon where lugsail helps and where ordinary OBM is already sufficient.
Oracle-variance comparison	Construct intervals using the analytic $\sigma^2(u)$ and compare them with OBM/MSB/OBM-RR intervals. This separates normal-approximation error and point-estimator bias from variance-estimation error.
Burn-in and initialization sweep	Vary n_0 , θ_0 , and the initial law of Z_0 to check the deterministic-start transfer and the practical size of the $(\alpha a)^{-1} \log^2 n$ burn-in window.
Block-size sweep for coverage	For each main estimator, sweep $b = \lfloor T^\eta \rfloor$ over a grid of η and report coverage, width, variance-estimator bias, and frequency of negative or clamped lugsail estimates.

Mixing and conditioning stress test	Vary the Markov-chain mixing rate, the spectral gap of $\bar{A}$, and the noise amplitude. This checks whether the empirical behavior follows the theorem's dependence on t_{mix} , a , and bounded-noise constants.
Matrix covariance diagnostics	Move beyond one scalar direction by checking several random directions and the full estimated covariance matrix, including positive-semidefiniteness of lugsail corrections.

The most important missing run is the theory-aligned stepsize sweep. The theorem is written for the pair $\alpha, 2\alpha$ and the explicit RR combination $2\bar{\theta}^{(\alpha)} - \bar{\theta}^{(2\alpha)}$, whereas the strongest completed comparison uses a wider Huo-style pair. Both are useful, but they answer different questions. The first validates the present theorem directly; the second explores a more aggressive practical Vandermonde extrapolation.

7. Appendix: External Inputs and Local Extensions

7.1. Direct Imported Working Forms

This appendix records the external statements in the exact working forms used in the thesis. Items explicitly marked as direct citations are imported from the cited papers after the sign convention conversion described in Section 4.1. Items marked as local extensions are proved in the main text and should not be read as separate external theorems.

Lemma 27. (Direct citation: Levin et al. (2025, Lemma 11) – Markov concentration.) Under **UGE 1**, there is a universal constant C_{MC} such that, for any bounded measurable functions g_i with $\pi(g_i) = 0$ and $\|g_i\|_\infty \leq c_i$, every initial distribution ξ , and every $p \geq 2$,

$$\left\| \sum_{i=1}^N g_i(Z_i) \right\|_{L_{p(\xi)}} \leq C_{\text{MC}} \sqrt{p t_{\text{mix}} \sum_{i=1}^N c_i^2}. \quad (397)$$

This is the scalar time-inhomogeneous consequence used in the predictable-variation estimates and in the preliminary last-iterate bounds.

Lemma 28. (Direct citation: Samsonov et al. (2025, Lemma 21) – Bolthausen–Fan martingale Berry–Esseen.) For scalar martingale differences X_l with $|X_l| \leq \kappa$, partial sum $S_N = \sum_l X_l$, predictable variation $V_N^2 = \sum_l \mathbb{E}[X_l^2 \mid \mathcal{F}_{l-1}]$, and deterministic scale $s_N^2 > 0$, for every $p \geq 1$,

$$\begin{aligned} d_{K(S_N/s_N, \mathcal{N}(0,1))} &\leq L_{B(\kappa)} \frac{(2N+1) \log(2N+1)}{s_N^3} \\ &\quad + C_1 \sqrt{p} s_N^{-2p/(2p+1)} (\mathbb{E}|V_N^2 - s_N^2|^p)^{1/(2p+1)} \\ &\quad + C_2 p s_N^{-2p/(2p+1)} \kappa^{2p/(2p+1)}. \end{aligned} \quad (398)$$

Here $L_{B(\kappa)} < \infty$ and C_1, C_2 are the constants in that martingale theorem.

Lemma 29. (Direct citation: Levin et al. (2025, Proposition 2) – stationary bias of $J^{(1)}$.) Under stationarity for the augmented chain $(Z_{t+1}, J_t^{(0,w)}, J_t^{(1,w)})$ and for $w \leq \alpha_{\text{L}, \text{P2}}(q, t_{\text{mix}})$,

$$\mathbb{E}_\pi[J_\infty^{(1,w)}] = w \Delta + R(w), \quad \|R(w)\| \leq 12 \|\bar{A}^{-1}\| C_A^2 t_{\text{mix}}^2 w^2 \|\epsilon\|_\infty, \quad (399)$$

with

$$\Delta := \bar{A}^{-1} \sum_{k \geq 1} \mathbb{E}_\pi[(Q^k \tilde{A})(Z_0) \epsilon(Z_0)]. \quad (400)$$

Lemma 30. (Direct citation: Levin et al. (2025, Corollary 6) – centered bilinear L_p bound.)

Define

$$\bar{\psi}_w(j, z) := \tilde{A}(z)j - \mathbb{E}_{\Pi_{J^{(0)}, w}}[\tilde{A}(Z_1) J_0^{(0,w)}]. \quad (401)$$

For $w \leq \alpha_{\text{L}, \text{C6}}(q, t_{\text{mix}})$, every initial distribution, every $r \geq 1$, and every $p \geq 2$,

$$\left\| \sum_{t=0}^{r-1} \bar{\psi}_w(J_t^{(0,w)}, Z_{t+1}) \right\|_{L_p} \leq c_{W,1} p^{3/2} \sqrt{wr} + c_{W,2} p^3 w^{-1/2} \log^{1/2}(1/(wa)), \quad (402)$$

with $c_{W,1}, c_{W,2}$ depending only on $C_A, \kappa_Q, t_{\text{mix}}, \|\epsilon\|_\infty$. The logarithmic factor is the one displayed in Levin et al. (2025, Corollary 6, Eq. (67)); it is absorbed into $\text{polylog}(n)$ in the working-scale corollaries.

Lemma 31. (Direct citation: Levin et al. (2025, Proposition 8) – depth-two $J^{(2)}$ moment bound.) For every $q \geq 2$ and every p satisfying $2 \leq p \leq q/2$, if $w \leq \alpha_{L,P8}(q, t_{\text{mix}})$, then

$$\|J_n^{(2,w)}\|_{L_p} \leq D_J t_{\text{mix}}^{5/2} p^{7/2} w^{3/2} \log^{3/2}(1/(wa)), \quad (403)$$

uniformly in n , with D_J depending only on $C_A, \kappa_Q, \|\bar{A}^{-1}\|, \|\epsilon\|_\infty$.

Lemma 32. (Direct citation: Levin et al. (2025, Proposition 9) – one-trajectory $H^{(2)}$ moment bound.) For every $q \geq 2$ and every p satisfying $2 \leq p \leq q/2$, if $w \leq \alpha_{L,P9}(q, t_{\text{mix}})$, then

$$\|H_n^{(2,w)}\|_{L_p} \leq D_H d^{1/q} t_{\text{mix}}^{5/2} p^{7/2} w^{3/2} \log^{3/2}(1/(wa)), \quad (404)$$

uniformly in n , with D_H depending only on $C_A, \kappa_Q, \|\bar{A}^{-1}\|, \|\epsilon\|_\infty$. The proof uses the representation

$$H_k^{(2,w)} = -w \sum_{l=1}^k \Gamma_{l+1:k}^{(w)} \tilde{A}(Z_l) J_{l-1}^{(2,w)}. \quad (405)$$

Lemma 33. (Direct citation/extraction: Levin et al. (2025, Appendix B.2, Proposition 5) – depth-two startup contraction for the J coordinates.) For

$$Y_k^{(w)} := \left(Z_{k+1}, J_k^{(0,w)}, J_k^{(1,w)}, J_k^{(2,w)} \right) \quad (406)$$

and $y = (z, j_0, j_1, j_2)$, $y' = (z', j'_0, j'_1, j'_2)$, define the depth-two augmented-chain cost from Levin et al. (2025, Appendix B.2, Eq. (49)):

$$\begin{aligned} c_{J,2}^{(w)}(y, y') &:= \|j_0 - j'_0\| + \|j_1 - j'_1\| + \|j_2 - j'_2\| \\ &+ (\|j_0\| + \|j'_0\| + \|j_1\| + \|j'_1\| + \|j_2\| + \|j'_2\| + \sqrt{wa} \|\epsilon\|_\infty) 1_{z \neq z'}. \end{aligned} \quad (407)$$

Under the Proposition-5 step-size restriction $w \leq \alpha_{L,P5}(p, t_{\text{mix}})$, two copies of $Y_k^{(w)}$ started from deterministic states y, y' can be coupled so that, for $\ell = 0, 1, 2$,

$$\|J_k^{(\ell,w)}(y) - J_k^{(\ell,w)}(y')\|_{L_p} \leq C_W p^{7/2} t_{\text{mix}}^{5/2} \log^{3/2}(1/(wa)) \exp(-wak/(12p)) c_{J,2}^{(w)}(y, y'). \quad (408)$$

The displayed componentwise estimate is the coordinate projection of the Wasserstein contraction, with constants as in Levin et al. (2025, Eq. (55)).

Lemma 34. (Direct citation: Levin et al. (2025, Corollary 4) – invariant depth-two J law.)

Under the same stationary depth-two ceilings as above, the Markov chain $Y_k^{(w)}$ admits a unique invariant law $\Pi_{J,2,w}$. Its stationary version has finite moments at the orders used in Lemma 31, and the contraction in Lemma 33 gives convergence to this law from every deterministic initial augmented state.

7.2. Local Extensions Used in This Thesis

The burn-in transfer also uses one technical input:

- Lemma 17 records deterministic and conditional product-stability estimates for the random linear products. Its conditional display Eq. 271 is the only product-stability form used at random coupling times; Eq. 273 is the deterministic-time corollary used for finite-past and initial-product estimates.

The following statements are local extensions or assemblies, not direct citations from Levin et al. or Samsonov et al.:

- Lemma 10 constructs the stationary full depth-two augmented state including $H^{(2)}$ by a finite-past limit. It uses Lemma 34, Lemma 33, and Lemma 32.

- Lemma 22 proves conditional product stability at a random coupling time from the deterministic product-stability working form Lemma 17 and the coupling-time tail.
- Lemma 23 proves the full-state startup contraction for $J^{(1)} + J^{(2)} + H^{(2)}$ by combining the J contraction Lemma 33 with the local random-time product estimate.

Whenever the main proof invokes these local statements, the invoked result is the local lemma named above, not an unquoted external proposition.

8. References

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